CHAPTER 4

INTRODUCTION TO ANALYTIC GEOMETRY

EXERCISE 4.1

Q.1 Describe the location in the plane of the point P(x, y) for which

(i)	$x > 0$	**(ii)**	$x > 0$ and $y > 0$	**(iii)**	$x = 0$
(iv)	$y = 0$	**(v)**	$x < 0$ and $y \geq 0$	**(vi)**	$x = y$
(vii)	$\lvert x\rvert = \lvert y\rvert$	**(viii)**	$\lvert x\rvert \geq 3$	**(ix)**	$x > 2$ and $y = 2$
(x)	x and y have opposite signs				

Solution:

(i) $x > 0$

Right half plane

(ii) $x > 0$ **and** $y > 0$

The 1st quadrant

(iii) $x = 0$

y – axis

(iv) $y = 0$

x – axis

(v) $x < 0$ **and** $y \geq 0$

2nd quadrant and – ve x – axis

(vi) $x = y$

It is a line bisecting 1st and 3 rd quadrant

(vii) $|x| = |y|$

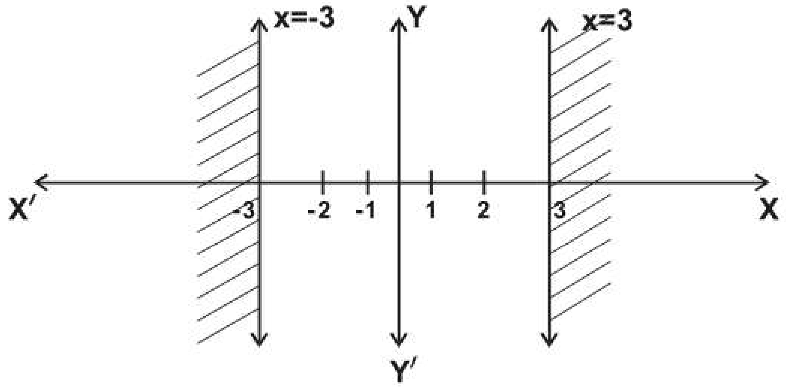
Points in the first and 3rd quadrants having both the coordinates equal or point in the second and fourth quadrant having both the coordinates equal but opposite in signs.

OR

(1) It is a line bisecting 1st and 3rd quadrant.

(2) It is a line bisecting 2nd and 4th quadrant.

(viii) $|x| \geq 3$



$\Rightarrow \quad \pm x \geq 3$

$\Rightarrow \quad x \geq 3 \quad \text{or} \quad -x \geq 3$

$\quad x \geq 3 \quad \text{or} \quad x \leq -3$

Which is the set of points lying on right side of $x = 3$ and the points lying on left side of $x = -3$

(ix) $x > 2$ **and** $y = 2$

The points lies in the first quadrant with $x > 2$ and $y = 2$

(x) **x and y have opposite signs.**

The point lies in 2^{nd} and 4^{th} quadrant.

Q.2 Find in each of the following.

(i) The distance between the two given points.

(ii) Midpoint of the line segment joining the two points.

(a) $A(3, 1)$; $B(-2, -4)$ **(b)** $A(-8, 3)$; $B(2, -1)$

(c) $A\left(-\sqrt{5}, \frac{-1}{3}\right)$; $B(-3\sqrt{5}, 5)$

Solution:

(a) $A(3, 1)$; $B(-2, -4)$ **(Lhr. Board 2007)**

(i) $|AB| = \sqrt{(-2-3)^2 + (-4-1)^2}$

$= \sqrt{(-5)^2 + (-5)^2}$

$= \sqrt{25 + 25}$

$= \sqrt{50} = 5\sqrt{2}$

(ii) Mid point $= \left(\frac{3-2}{2}, \frac{1-4}{2}\right) = \left(\frac{1}{2}, \frac{-3}{2}\right)$

(b) $A(-8, 3)$; $B(2, -1)$

(i) $|AB| = \sqrt{(2+8)^2 + (-1-3)^2}$

$= \sqrt{(10)^2 + (-4)^2}$

$$= \sqrt{100+16}$$
$$= \sqrt{116}$$
$$= \sqrt{4\times 29} = 2\sqrt{29}$$

(ii) Mid point $= \left(\dfrac{-8+2}{2}\ ,\ \dfrac{3-1}{2}\right)$

$$= \left(\frac{-6}{2}\ ,\ \frac{2}{2}\right) \quad = \ (-3, 1) \qquad \text{Ans}$$

(c) A $(-\sqrt{5}\ ,\ \frac{-1}{3})$; B$(-3\sqrt{5}\ , 5)$ (Lhr. Board 2006, Guj. Board 2008)

(i) $|AB| = \sqrt{(-3\sqrt{5}+\sqrt{5})^2 + (5+\frac{1}{3})^2}$

$$= \sqrt{(-2\sqrt{5})^2 + \left(\frac{15+1}{3}\right)^2}$$
$$= \sqrt{20 + \left(\frac{16}{3}\right)^2}$$
$$= \sqrt{20 + \frac{256}{9}}$$
$$= \sqrt{\frac{180+256}{9}}$$
$$= \sqrt{\frac{436}{9}}$$
$$= \sqrt{\frac{4\times 109}{9}}$$
$$= \frac{2\sqrt{109}}{3} \qquad \text{Ans.}$$

(ii) Mid point $= \left(\dfrac{-\sqrt{5}-3\sqrt{5}}{2},\ \dfrac{-\frac{1}{3}+5}{2}\right)$

$$= \left(\frac{-4\sqrt{5}}{2},\ \frac{\frac{-1+15}{3}}{2}\right) = \left(-2\sqrt{5}\ ,\ \frac{14}{6}\right)$$
$$= \left(-2\sqrt{5}\ ,\ \frac{7}{3}\right) \qquad \text{Ans}$$

Q.3 Which of the following points are at a distance of 15 units from origin?

(a) $(\sqrt{176}\ ,7)$ **(b)** $(10, -10)$

(c) $(1, 15)$ **(d)** $\left(\frac{15}{2}, \frac{15}{2}\right)$

Solution:

(a) **Let A $(\sqrt{176}\ ,\ 7)$, O(0, 0)**

$$|AO| = \sqrt{(0-\sqrt{176})^2 + (0-7)^2}$$
$$= \sqrt{176 + 49}$$
$$= \sqrt{225}$$
$$= 15$$

$\therefore$ A $(\sqrt{176}\ ,\ 7)$ is at a distance of 15 units from origin.

(b) **Let A(10, – 10) , O(0, 0)**

$$|AO| = \sqrt{(0-10)^2 + (0+10)^2}$$
$$= \sqrt{100 + 100}$$
$$= \sqrt{200}$$
$$= 10\sqrt{2}$$

$\therefore$ A(10, – 10) is not a distance of 15 units from origin.

(c) **Let A(1, 15) , O(0, 0)**

$$|AO| = \sqrt{(0-1)^2 + (0-15)^2}$$
$$= \sqrt{1 + 225} = \sqrt{226}$$

$\therefore$ A (1, 15) is not a distance of 15 units from origin.

(d) **Let A $\left(\frac{15}{2}, \frac{15}{2}\right)$, O(0, 0)**

$$|AO| = \sqrt{\left(0-\frac{15}{2}\right)^2 + \left(0-\frac{15}{2}\right)^2}$$

$$|AO| = \sqrt{\frac{225}{4} + \frac{225}{4}}$$

$$|AO| = \sqrt{\frac{225+225}{4}}$$

$$|AO| = \sqrt{\frac{450}{4}}$$

$$|AO| = \sqrt{\frac{225}{2}}$$

$$|AO| = \frac{15}{\sqrt{2}}$$

$\therefore$ $A\left(\frac{15}{2}, \frac{15}{2}\right)$ is not a distance of 15 units from origin.

Q.4 Show that

(i) the points A(0, 2), B $(\sqrt{3}, -1)$ and C(0, – 2) are vertices of a right triangle.

(ii) the points A(3, 1), B(–2, –3) and C(2, 2) are vertices of an isosceles triangle.

(iii) the points A(5, 2), B(– 2, 3), C(–3, –4) and D(4, –5) are vertices of a parallelogram. Is the parallelogram a square?

Solution:

(i) $A(0, 2)$, $B(\sqrt{3}, -1)$, $C(0, -2)$

$|AB| = \sqrt{(\sqrt{3}-0)^2 + (-1-2)^2}$

$|AB| = \sqrt{3 + (-3)^2} = \sqrt{3+9} = \sqrt{12}$

$|AB|^2 = 12$

$|BC| = \sqrt{(0-\sqrt{3})^2 + (-2+1)^2}$

$|BC| = \sqrt{3 + (-1)^2}$

$|BC| = \sqrt{3+1}$

$|BC| = \sqrt{4}$

$|BC|^2 = 4$

$|AC| = \sqrt{(0-0)^2 + (-2-2)^2}$

$= \sqrt{0 + (-4)^2}$

$|AC| = \sqrt{16}$

$|AC|^2 = 16$

Since

$|AC|^2 = |AB|^2 + |BC|^2$

Shows the given vertices A(0, 2), B($\sqrt{3}$, – 1) and C(0, – 2) form a right angle triangle.

(ii) **A(3, 1) , B(– 2 , – 3) , C (2, 2)**

$|AB| = \sqrt{(-2-3)^2 + (-3-1)^2}$

$|AB| = \sqrt{(-5)^2 + (-4)^2}$

$|AB| = \sqrt{25+16}$

$|AB| = \sqrt{41}$

$|BC| = \sqrt{(2+2)^2+(2+3)^2}$

$|BC| = \sqrt{(4)^2+(5)^2}$

$|BC| = \sqrt{16+25}$

$|BC| = \sqrt{41}$

$|AC| = \sqrt{(2-3)^2+(2-1)^2}$

$|AC| = \sqrt{(-1)^2+(1)^2}$

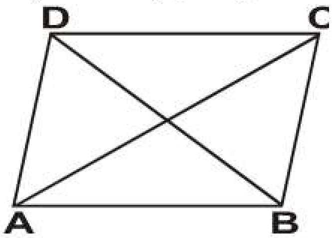
$|AC| = \sqrt{1+1}$

$|AC| = \sqrt{2}$

Since $|AB| = |BC|$

Show A (3, 1), B (–2, – 3) and C (2, 2) are vertices of an isosceles triangle.

(iii) A(5, 2), B(–2, 3), C(–3, –4) and D(4, – 5)



$|AB| = \sqrt{(-2-5)^2+(3-2)^2}$

$|AB| = \sqrt{(-7)^2+(1)^2}$

$|AB| = \sqrt{49+1}$

$|AB| = \sqrt{50}$

$|BC| = \sqrt{(-3+2)^2+(-4-3)^2}$

$|BC| = \sqrt{(-1)^2+(-7)^2}$

$|BC| = \sqrt{1+49}$

$|BC| = \sqrt{50}$

$|DC| = \sqrt{(-3-4)^2+(-4+5)^2}$

$|DC| = \sqrt{(-7)^2+(1)^2}$

$|DC| = \sqrt{49+1}$

$|DC| = \sqrt{50}$

$|AD| = \sqrt{(4-5)^2+(-5-2)^2}$

$|AD| = \sqrt{(-1)^2+(-7)^2}$

$|AD| = \sqrt{1+49} = \sqrt{50}$

$\therefore |AB| = |DC|$

and $|AD| = |BC|$

Shows that the points A (5, 2), B (–2, 3), C (– 3, – 4) and D (4, – 5) are the vertices of a parallelogram.

Let $\overline{AC}$ and $\overline{BD}$ be the diagonals of a parallelogram ABCD.

$|AC| = \sqrt{(-3-5)^2 + (-4-2)^2}$

$|AC| = \sqrt{(-8)^2 + (-6)^2}$

$|AC| = \sqrt{64 + 36}$

$|AC| = \sqrt{100}$

$|AC| = 10$

$|BD| = \sqrt{(4+2)^2 + (-5-3)^2}$

$|BD| = \sqrt{(6)^2 + (-8)^2}$

$|BD| = \sqrt{36 + 64}$

$|BD| = \sqrt{100}$

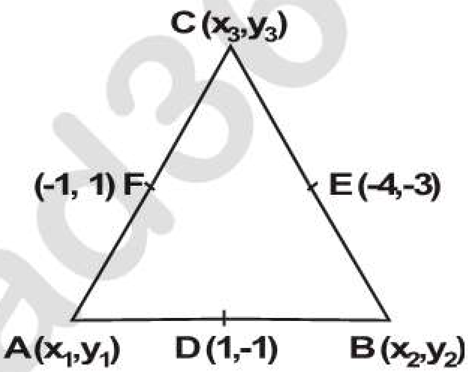
$|BD| = 10$

$\therefore \quad |AC| = |BD|$

As all sides of parallelogram are equal and diagonals are also equal so this parallelogram forms a square.

Q.5 The midpoints of the sides of a triangle are (1, –1), (–4, –3) and (–1, 1). Find the coordinates of the vertices of a triangle.

Solution:



Let D (1, – 1), E (– 4, – 3) and F (– 1, 1) be the mid points of sides $\overline{AB}$, $\overline{BC}$ and $\overline{AC}$ respectively.

Let A (x_1, y_1), B (x_2, y_2) and C(x_3, y_3) be the three vertices of a ΔABC.

Since D is the mid point of $\overline{AB}$.

$\therefore$ By ratio formula

$\frac{x_1 + x_2}{2} = 1 \qquad\qquad \frac{y_1 + y_2}{2} = -1$

$x_1 + x_2 = 2 \quad ...(1) \qquad y_1 + y_2 = -2 \quad ....(2)$

Since F be the mid point of $\overline{AC}$.

$\therefore$ By ratio formula

$\frac{x_2 + x_3}{2} = -4 \quad , \quad \frac{y_2 + y_3}{2} = -3$

$x_2 + x_3 = -8 \quad .....(3)$

$y_2 + y_3 = -6 \quad .....(4)$

$\therefore$ By ratio formula

$\frac{x_1 + x_3}{2} = -1 \qquad \frac{y_1 + y_3}{2} = 1$

$x_1 + x_3 = -2 \quad ...(5) \qquad y_1 + y_3 = 2 \quad ....(6)$

Equation (1) – equation (3), we get

$$\begin{array}{rcrcr} x_1 & + & x_2 & = & 2 \\ \underline{\ \ } x_2 & \pm & x_3 & = & \mp 8 \\ \hline x_1 & - & x_3 & = & 10 \end{array} \quad ...(7)$$

Equation (5) + Equation (7), we get

$$\begin{array}{rcrcr} x_1 & + & x_3 & = & -2 \\ x_1 & - & x_3 & = & 10 \\ \hline \end{array}$$

$$2x_1 = 8$$

$$x_1 = \frac{8}{2} = 4$$

Put $x_1 = 4$ in Equation (7)

$4 - x_3 = 10$

$4 - 10 = x_3$

$x_3 = -6$

Put $x_3 = -6$ in equation (3)

$x_2 - 6 = -8$

$x_2 = -8 + 6$

$x_2 = -2$

Equation (2) – Equation (4), we get

$$\begin{array}{rcrcr} y_1 & + & y_2 & = & -2 \\ \underline{\ \ } y_2 & \pm & y_3 & = & \mp 6 \\ \hline y_1 & - & y_3 & = & 4 \end{array} \quad ...(8)$$

Equation (6) + Equation (8), we get

$$\begin{array}{rcl} y_1 + y_3 & = & 2 \\ y_1 - y_3 & = & 4 \\ \hline 2y_1 & = & 6 \end{array}$$

$$y_1 = \frac{6}{2} = 3$$

Put $y_1 = 3$ in equation (8)

$3 - y_3 = 4$

$3 - 4 = y_3$

$y_3 = -1$

Put $y_3 = -1$ in equation (4)

$y_2 - 1 = -6$

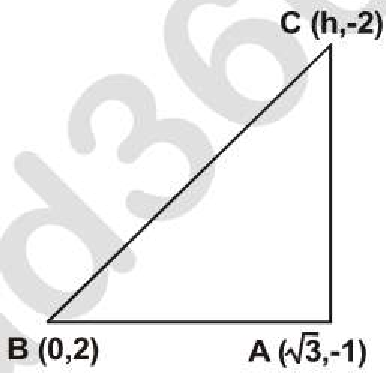
$y_2 = -6 + 1$

$y_2 = -5$

$\therefore$ A(4, 3), B(– 2, – 5), C(– 6, – 1) Ans

Q.6: Find h such that the points A ($\sqrt{3}$, – 1), B (0, 2) and C (h, – 2) are vertices of right triangle with right angle at vertex A.

Solution:



A ($\sqrt{3}$, – 1), B (0, 2) C (h, – 2)

$|AB| = \sqrt{(0-\sqrt{3})^2 + (2+1)^2}$

$|AB| = \sqrt{3 + (3)^2}$

$|AB| = \sqrt{3 + 9}$

$|AB| = \sqrt{12}$

$|BC| = \sqrt{(h-0)^2 + (-2-2)^2}$

$|BC| = \sqrt{h^2 + (-4)^2}$

$|BC| = \sqrt{h^2 + 16}$

$$|AC| = \sqrt{(h-\sqrt{3})^2 + (-2+1)^2}$$

$$|AC| = \sqrt{h^2 + 3 - 2h\sqrt{3} + (-1)^2}$$

$$|AC| = \sqrt{h^2 + 3 - 2\sqrt{3}h + 1}$$

$$|AC| = \sqrt{h^2 - 2\sqrt{3}h + 4}$$

Since ABC is a right triangle with right angle at vertex A.

$\therefore$ By Pythagrous theorem

$$|BC|^2 = |AB|^2 + |AC|^2$$

$$\left(\sqrt{h^2+16}\right)^2 = (\sqrt{12})^2 + \left(\sqrt{h^2 - 2\sqrt{3}h + 4}\right)^2$$

$$h^2 + 16 = 12 + h^2 - 2\sqrt{3}h + 4$$

$$h^2 - h^2 + 2\sqrt{3}h = 16 - 16$$

$$2\sqrt{3}h = 0$$

$$\boxed{h = 0} \text{ Ans}$$

Q.7 Find h such that A (– 1, h), B (3, 2) and C (7, 3) are collinear. (Lhr. Board 2009 (S))

Solution:

A (– 1, h), B (3, 2) and C (7, 3)

Since A, B and C are collinear.

$$\therefore \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} -1 & h & 1 \\ 3 & 2 & 1 \\ 7 & 3 & 1 \end{vmatrix} = 0$$

$$-1\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} - h\begin{vmatrix} 3 & 1 \\ 7 & 1 \end{vmatrix} + 1\begin{vmatrix} 3 & 2 \\ 7 & 3 \end{vmatrix} = 0$$

$$-1(2-3) - h(3-7) + 1(9-14) = 0$$

$$-1(-1) - h(-4) + 1(-5) = 0$$

$$1 + 4h - 5 = 0$$

$$4h - 4 = 0$$

$$4h = 4$$

$$h = \frac{4}{4}$$

$$\boxed{h = 1} \quad \text{Ans}$$

Q.8 The points A(–5, – 2) and B(5, – 4) are ends of a diameter of a circle. Find the centre and radius of the circle.

Solution:

A(– 5, – 2) , B(5, – 4)

Let C (h, k) be the center of the circle having radius r. Since C (h, k) be the mid point of $\overline{AB}$.

$\therefore$ By ratio formula

$$h = \frac{-5+5}{2} \quad , \quad k = \frac{-2-4}{2}$$

$$h = \frac{0}{2} \quad , \quad k = \frac{-6}{2} = -3$$

$$h = 0$$

$$\therefore \quad C(h, k) = C(0, -3)$$

$$r = |AC| = \sqrt{(0+5)^2 + (-3+2)^2}$$

$$= \sqrt{(5)^2 + (-1)^2}$$

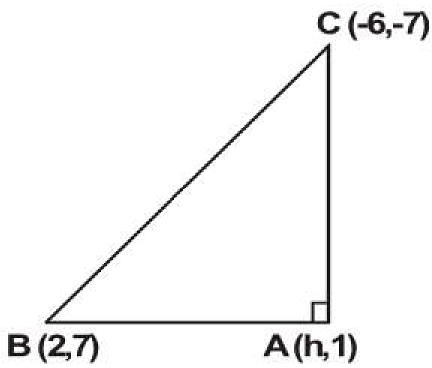
$$= \sqrt{25+1}$$

$$= \sqrt{26} \quad \text{Ans}$$

Q.9 Find h such that the points A (h, 1), B (2, 7) and C (– 6, – 7) are vertices of a right triangle with right angle at the vertex A.

Solution:

A (h, 1), B (2, 7), C (– 6, – 7)



$$|AB|^2 = (2-h)^2 + (7-1)^2$$

$$= 4 + h^2 - 4h + 36$$

$$= h^2 - 4h + 40$$

$$|BC|^2 = (-6-2)^2 + (-7-7)^2$$

$$= (-8)^2 + (-14)^2$$

$$= 64 + 196$$

$$= 260$$

$$|AC|^2 = (-6-h)^2 + (-7-1)^2$$

$$= 36 + h^2 + 12h + 64 = h^2 + 12h + 100$$

Since ABC is a right triangle with right angle at vertex A.

$\therefore$ By Pythagoras theorem

$$|BC|^2 = |AB|^2 + |AC|^2$$
$$260 = h^2 - 4h + 40 + h^2 + 12h + 100$$
$$0 = 2h^2 + 8h + 140 - 260$$
$$2h^2 + 8h - 120 = 0$$
$$2(h^2 + 4h - 60) = 0$$
$$h^2 + 4h - 60 = 0$$
$$h^2 + 10h - 6h - 60 = 0$$
$$h(h + 10) - 6(h + 10) = 0$$
$$(h + 10)(h - 6) = 0$$

Either

$h + 10 = 0$ or $h - 6 = 0$

$h = -10$ $h = 6$

$\therefore$ $h = 6, -10$ Ans

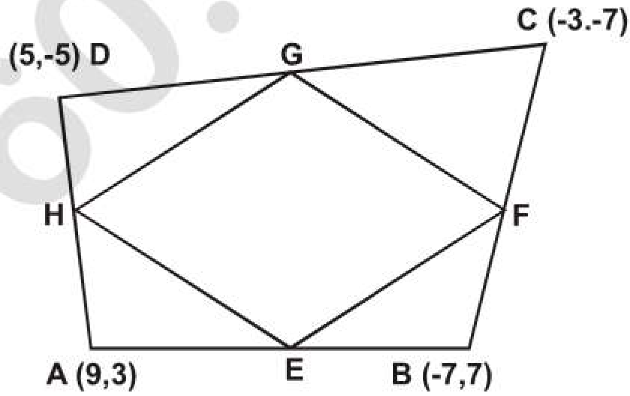
Q.10 A quadrilateral has the points A(9, 3) , B(– 7, 7), C(–3, – 7) and D(5, – 5) as its vertices. Find the midpoints of its sides. Show that the figure formed by joining the midpoints consecutively is a parallelogram.

Solution:

A (9, 3) , B(– 7, 7)

C (– 3, – 7) , D(5, – 5)

Let E, F, G and H be the mid points of sides $\overline{AB}$, $\overline{BC}$, $\overline{CD}$ and $\overline{AD}$ respectively.



$$\text{Mid point E of } \overline{AB} = \left(\frac{9-7}{2}, \frac{3+7}{2}\right)$$

$$\text{Mid point E of } \overline{AB} = \left(\frac{2}{2}, \frac{10}{2}\right)$$

$$\text{Mid point E of } \overline{AB} = (1, 5)$$

$$\text{Mid point F of } \overline{BC} = \left(\frac{-3-7}{2}, \frac{-7+7}{2}\right)$$

$$\text{Mid point F of } \overline{BC} = \left(\frac{-10}{2}, \frac{0}{2}\right)$$

Mid point F of $\overline{BC}$ $= (-5, 0)$

Mid point G of $\overline{CD}$ $= \left(\frac{5-3}{2}, \frac{-5-7}{2}\right)$

Mid point G of $\overline{CD}$ $= \left(\frac{2}{2}, \frac{-12}{2}\right)$

Mid point G of $\overline{CD}$ $= (1, -6)$

Mid point H of $\overline{AD}$ $= \left(\frac{5+9}{2}, \frac{-5+3}{2}\right)$

Mid point H of $\overline{AD}$ $= \left(\frac{14}{2}, \frac{-2}{2}\right)$

Mid point H of $\overline{AD}$ $= (7, -1)$

$|EF| = \sqrt{(-5-1)^2 + (0-5)^2}$

$|EF| = \sqrt{(-6)^2 + (-5)^2}$

$|EF| = \sqrt{36+25}$

$|EF| = \sqrt{61}$

$|FG| = \sqrt{(1+5)^2 + (-6-0)^2}$

$|FG| = \sqrt{(6)^2 + (-6)^2}$

$|FG| = \sqrt{36+36}$

$|FG| = \sqrt{72}$

$|GH| = \sqrt{(7-1)^2 + (-1+6)^2}$

$|GH| = \sqrt{(6)^2 + (5)^2}$

$|GH| = \sqrt{36+25}$

$|GH| = \sqrt{61}$

$|HE| = \sqrt{(1-7)^2 + (5+1)^2}$

$|EF| = \sqrt{(-6)^2 + (6)^2}$

$|EF| = \sqrt{36+36}$

$|EF| = \sqrt{72}$

Since $|EF| = |GH|$

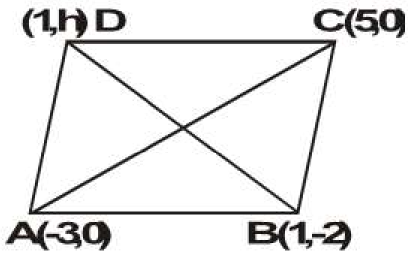
$|FG| = |HE|$

Shows the figure formed by joining the midpoints consecutively is a parallelogram.

Q.11: Find h such that the quadrilateral with vertices A(– 3, 0), B(1, – 2), C(5, 0) and D(1, h) is parallelogram. Is it square?

Solution:

A (– 3, 0) , B (1, – 2)
C (5, 0) , D (1, h)
Since ABCD is a parallelogram.
$\therefore$ $|AD| = |BC|$



$$\sqrt{(1+3)^2+(h-0)^2} = \sqrt{(5-1)^2+(0+2)^2}$$
$$\sqrt{16+h^2} = \sqrt{16+4}$$

Squaring on both sides

$$16+h^2 = 20$$
$$h^2 = 20-16$$
$$h^2 = 4$$
$$\boxed{h = \pm 2}$$

Now when $h = -2$ then D (1, h) = D (1, – 2) but we also have B (1, 2)

i.e. B an D represents the same point, which cannot be happened in quadrilateral so we cannot take $h = -2$.

$\boxed{h = 2}$ Ans

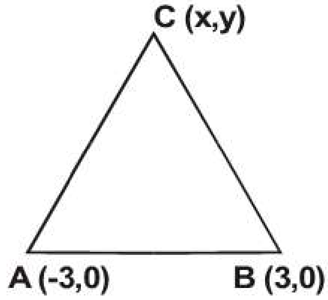
Let $\overline{AC}$ and $\overline{BC}$ be the diagonals of parallelogram ABCD.

$$|AC| = \sqrt{(5+3)^2+(0-0)^2}$$
$$|AC| = \sqrt{(8)^2+0}$$
$$|AC| = \sqrt{64+0}$$
$$|AC| = 8$$
$$|BD| = \sqrt{(1-1)^2+(2+2)^2}$$
$$|BD| = \sqrt{0+(4)^2}$$
$$|BD| = \sqrt{16}$$
$$|BD| = 4$$
$$\therefore \quad |AC| \neq |BD|$$

So the parallelogram ABCD is not a square.

Q.12 If two vertices of an equilateral triangle are A(–3, 0) and B(3, 0), find the third vertex. How many of these triangles are possible?

Solution:



A(–3, 0) , B(3, 0)

Let C (x, y) be the required vertex.

Since ABC is an equilateral triangle

$\therefore \quad |AB| = |AC| = |BC|$

$|AB| = |AC|$

$|AB|^2 = |AC|^2$

$(3+3)^2 + (0-0)^2 = (x+3)^2 + (y-0)^2$

$(6)^2 + 0 = x^2 + 9 + 6x + y^2$

$36 = x^2 + 6x + 9 + y^2$

$0 = x^2 + y^2 + 6x + 9 - 36$

$x^2 + y^2 + 6x - 27 = 0 \qquad \text{...... (1)}$

$|AC| = |BC|$

$|AC|^2 = |BC|^2$

$(x+3)^2 + (y-0)^2 = (x-3)^2 + (y-0)^2$

$x^2 + 9 + 6x + y^2 = x^2 + 9 - 6x + y^2$

$x^2 + 6x + y^2 - x^2 + 6x - y^2 = 9 - 9$

$12x = 0$

$\boxed{x = 0}$

Put $x = 0$ in equation (1)

$(0)^2 + y^2 + 6(0) - 27 = 0$

$y^2 = 27$

$\sqrt{y^2} = \sqrt{27}$

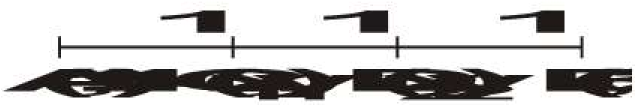
$y = \pm 3\sqrt{3}$

$\therefore \quad (0, 3\sqrt{3}) \;,\; (0, -3\sqrt{3})$

So there are two triangles are possible.

Q.13: Find the points trisecting the join of A(–1, 4) and B(6, 2). (Lhr. Board 2006, 2009, 2011) (Guj. Board 2008).

Solution:



A (–1, 4) , B (6, 2)

Let C (x_1, y_1) and D (x_2, y_2) be the required points.

Since C lies between A and B.

$\therefore$ By ratio formula

$$x_1 = \frac{1(6) + 2(-1)}{1 + 2} \quad , \quad y_1 = \frac{1(2) + 2(4)}{1 + 2}$$

$$x_1 = \frac{6 - 2}{3} \quad , \quad y_1 = \frac{2 + 8}{3}$$

$$x_1 = \frac{4}{3} \quad , \quad y_1 = \frac{10}{3}$$

$$\therefore \quad C\left(\frac{4}{3}, \frac{10}{3}\right)$$

Since D is the mid points of $\overline{CB}$.

$\therefore$ By ratio formula

$$x_2 = \frac{\frac{4}{3} + 6}{2} \quad , \quad y_2 = \frac{\frac{10}{3} + 2}{2}$$

$$x_2 = \frac{\frac{4 + 18}{3}}{2} \quad , \quad y_2 = \frac{\frac{10 + 6}{3}}{2}$$

$$x_2 = \frac{22}{6} \quad , \quad y_2 = \frac{16}{6}$$

$$x_2 = \frac{11}{3} \quad , \quad y_2 = \frac{8}{3}$$

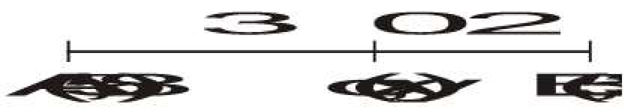
$$\therefore \quad D\left(\frac{11}{3}, \frac{8}{3}\right) \quad \text{Ans}$$

Q.14 Find the point three-fifth of the way along the line segment from A (–5, 8) to B (5, 3). (Lhr. Board 2007)

Solution:

$A(-5, 8) \quad , \quad B(5, 3)$

Let C(x, y) be the required point.



$\therefore$ By ratio formula

$$x = \frac{2(-5) + 3(5)}{3 + 2} \quad , \quad y = \frac{2(8) + 3(3)}{3 + 2}$$

$$x = \frac{-10 + 15}{5} \quad , \quad y = \frac{16 + 9}{5}$$

$$x = \frac{5}{5} \quad , \quad y = \frac{25}{5}$$

$$x = 1 \quad , \quad y = 5$$

$$\therefore \quad C(x, y) = C(1, 5) \quad \text{Ans}$$

Q.15 Find the point P on the join of A (1, 4) and B (5, 6) that is twice as far from A as B is from A and lies

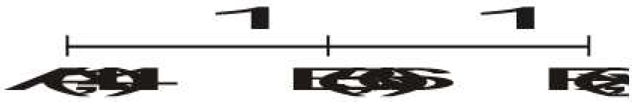
(i) On the same side of A as B does.

(ii) On the opposite side of A as B does.

Solution:

A (1, 4), B(5, 6)

Let P (x, y) be the required point.



$\therefore$ By ratio formula

$$5 = \frac{1+x}{2} \quad , \quad 6 = \frac{4+y}{2}$$

$$1 + x = 10 \quad , \quad 12 = 4 + y$$

$$x = 10 - 1 \quad , \quad y = 12 - 4$$

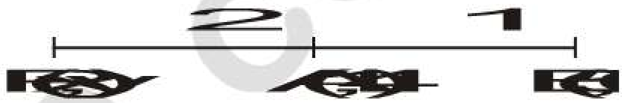
$$x = 9 \quad , \quad y = 8$$

$\therefore$ $P(x, y) = P(9, 8)$

(ii)

Let P (x, y) be the required point.

$\therefore$ By ratio formula.



$$1 = \frac{1(x) + 2(5)}{2+1} \quad , \quad 4 = \frac{1(y) + 2(6)}{2+1}$$

$$1 = \frac{x+10}{3} \quad , \quad 4 = \frac{y+12}{3}$$

$$x + 10 = 3 \quad , \quad y + 12 = 12$$

$$x = 3 - 10 \quad , \quad y = 12 - 12$$

$$x = -7 \quad , \quad y = 0$$

$\therefore$ $P(x, y) = P(-7, 0)$ Ans

Q.16 Find the point which is equidistant from the points A (5, 3), B (–2, 2) and C(4, 2). What is the radius of the circum circle of the ΔABC?

Solution:

A (5, 3), B (–2, 2), C(4, 2)

Let P (x, y) be the required point which is equidistant from A, B and C.

$$|PA| = |PB| = |PC|$$

$$|PA|^2 = |PB|^2 = |PC|^2$$

Taking $|PB|^2 = |PC|^2$

$$(-2 - x)^2 + (2 - y)^2 = (4 - x)^2 + (2 - y)^2$$

$$4 + x^2 + 4x + 4 + y^2 - 4y = 16 + x^2 - 8x + 4 + y^2 - 4y$$

$$x^2 + 4x + y^2 - 4y - x^2 + 8x - y^2 + 4y = 20 - 4 - 4$$

$$12x = 12$$
$$x = \frac{12}{12} = 1$$

and

$$|PA|^2 = |PB|^2$$
$$(5-x)^2 + (3-y)^2 = (-2-x)^2 + (2-y)^2$$
$$(5-1)^2 + 9 + y^2 - 6y = (-2-1)^2 + 4 + y^2 - 4y$$
$$16 + 9 + y^2 - 6y = 9 + 4 + y^2 - 4y$$
$$y^2 - 6y + 4y - y^2 = 13 - 16 - 9$$
$$-2y = -12$$
$$y = \frac{-12}{-2} = 6$$

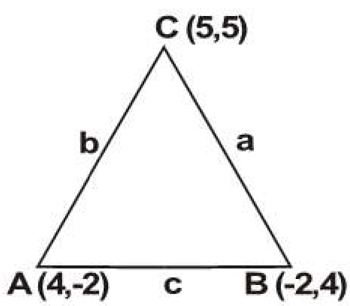
$\therefore$ $P(x, y) = P(1, 6)$

Radius of circum-circle $= |PA|$
$$= \sqrt{(1-5)^2 + (6-3)^2}$$
$$= \sqrt{(-4)^2 + (3)^2}$$
$$= \sqrt{16+9}$$
$$= \sqrt{25}$$
$$= 5 \quad \text{Ans}$$

Q.17: The points (4, – 2), (–2, 4) and (5, 5) are the vertices of a triangle. Find in-centre of the triangle.

Solution:

Let A(4, – 2), B(–2, 4) and C(5, 5) be the three given vertices of a ΔABC.



$$a = |BC| = \sqrt{(5+2)^2 + (5-4)^2}$$
$$= \sqrt{(7)^2 + (1)^2}$$
$$= \sqrt{49+1}$$
$$= \sqrt{50}$$
$$= 5\sqrt{2}$$

$$b = |AC| = \sqrt{(5-4)^2 + (5+2)^2}$$
$$= \sqrt{(1)^2 + (7)^2} = \sqrt{1+49}$$
$$= \sqrt{50} = 5\sqrt{2}$$

$$c = |AB| = \sqrt{(-2-4)^2 + (4+2)^2}$$

$$= \sqrt{(-6)^2 + (6)^2}$$
$$= \sqrt{36 + 36}$$
$$= \sqrt{72}$$
$$= 6\sqrt{2}$$

In centre of a Δ ABC is
$$\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$$

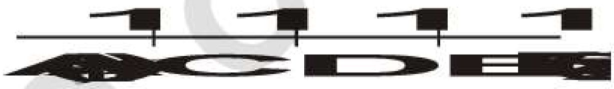
$$\Delta\ ABC = \left(\frac{5\sqrt{2}(4) + 5\sqrt{2}(-2) + 6\sqrt{2}(5)}{5\sqrt{2} + 5\sqrt{2} + 6\sqrt{2}}, \frac{5\sqrt{2}(-2) + 5\sqrt{2}(4)}{5\sqrt{2} + 5\sqrt{2} + 6\sqrt{2}}\right)$$
$$= \left(\frac{\sqrt{2}(20 - 10 + 30)}{16\sqrt{2}}, \frac{\sqrt{2}(-10 + 20 + 30)}{16\sqrt{2}}\right)$$
$$= \left(\frac{40}{16}, \frac{40}{16}\right)$$
$$= \left(\frac{5}{2}, \frac{5}{2}\right) \quad \text{Ans.}$$

Q.18: Find the points that divide the line segment joining A (x_1, y_1) and B (x_2, y_2) into four equal parts.

Solution:

A (x_1, y_1) , B (x_2, y_2)

Let C, D and E be the required points.



$$\text{Coordinates of C} = \left(\frac{1(x_2) + 3(x_1)}{1+3}, \frac{1(y_2) + 3(y_1)}{1+3}\right)$$
$$= \left(\frac{3x_1 + x_2}{4}, \frac{3y_1 + y_2}{4}\right)$$
$$\text{Coordinates of D} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$
$$\text{Coordinates of E} = \left(\frac{1(x_1) + 3(x_2)}{1+3}, \frac{1(y_1) + 3(y_2)}{1+3}\right)$$
$$= \left(\frac{x_1 + 3x_2}{4}, \frac{y_1 + 3y_2}{4}\right) \quad \text{Ans.}$$

EXERCISE 4.2

Q.1: The two points P and O' are given in xy – coordinate system. Find the XY-Coordinates of P referred to the translated axes $O'X$ and $O'Y$.

(i) P(3, 2) ; $O'(1, 3)$ (ii) P(–2, 6); $O'(–3, 2)$ (Lhr. Board 2011)

(iii) P(–6, –8); $O'(–4, –6)$ (iv) $P\left(\frac{3}{2}, \frac{5}{2}\right)$; $O'\left(-\frac{1}{2}, \frac{7}{2}\right)$

Solution:

(i) P(3, 2); $O'(1, 3)$

$x = 3, \quad y = 2, \quad h = 1, \quad k = 3$

By using

$X = x - h, \quad Y = y - k$

$X = 3 - 1, \quad Y = 2 - 3$

$X = 2, \quad Y = -1$

$\therefore \quad P(X, Y) = P(2, -1)$ Ans

(ii) **P(– 2, 6); O′(– 3, 2)**

$x = -2$, $y = 6$, $h = -3$, $k = 2$

By using

$X = x - h$, $Y = y - k$

$X = -2 + 3$, $Y = 6 - 2$

$X = 1$, $Y = 4$

$\therefore$ P(X, Y) = P (1, 4) Ans

(iii) **P(– 6, – 8); O′(– 4, – 6)**

$x = -6$, $y = -8$, $h = -4$, $k = -6$

By using

$X = x - h$, $Y = y - k$

$X = -6 + 4$, $Y = -8 + 6$

$X = -2$, $Y = -2$

$\therefore$ P(X, Y) = P (– 2, – 2) Ans

(iv) $\mathbf{P\left(\frac{3}{2}, \frac{5}{2}\right); O'\left(\frac{-1}{2}, \frac{7}{2}\right)}$

$x = \frac{3}{2}$, $y = \frac{5}{2}$, $h = \frac{-1}{2}$, $k = \frac{7}{2}$

By using

$X = x - h$, $Y = y - k$

$X = \frac{3}{2} + \frac{1}{2}$, $Y = \frac{5}{2} - \frac{7}{2}$

$X = \frac{3+1}{2}$, $Y = \frac{5-7}{2}$

$X = \frac{4}{2}$, $Y = \frac{-2}{2}$

$X = 2$, $Y = -1$

$\therefore$ P(X, Y) = P (2, – 1) Ans

Q.2: **The xy-coordinate axes are translated through the point O' whose coordinates are given in xy-coordinate system. The coordinates of P are given in the XY-coordinate system. Find the coordinates of P in xy-coordinate system.**

(i) $P(8, 10)$; $O'(3, 4)$ **(ii)** $P(-5, -3)$; $O'(-2, -6)$

(iii) $P\left(\frac{-3}{4}, \frac{-7}{6}\right)$; $O'\left(\frac{1}{4}, \frac{-1}{6}\right)$ **(iv)** $P(4, -3)$; $O'(-2, 3)$

Solution:

(i) $P(8, 10)$; $O'(3, 4)$

$X = 8, \quad Y = 10, \quad h = 3, \quad k = 4$

By using

$$\begin{aligned} X &= x - h, & Y &= y - k \\ 8 &= x - 3, & 10 &= y - 4 \\ x &= 8 + 3, & y &= 10 + 4 \\ x &= 11, & y &= 14 \end{aligned}$$

$\therefore \quad P(x, y) = P(11, 14)$ Ans

(ii) $P(-5, -3)$; $O'(-2, -6)$

$X = -5, \quad Y = -3, \quad h = -2, \quad k = -6$

By using

$$\begin{aligned} X &= x - h, & Y &= y - k \\ -5 &= x + 2, & -3 &= y + 6 \\ x &= -5 - 2, & y &= -3 - 6 \\ x &= -7, & y &= -9 \end{aligned}$$

$\therefore \quad P(x, y) = P(-7, -9)$ Ans

(iii) $P\left(\frac{-3}{4}, \frac{-7}{6}\right)$; $O'\left(\frac{1}{4}, \frac{-1}{6}\right)$

$X = \frac{-3}{4}, \quad Y = \frac{-7}{6}, \quad h = \frac{1}{4}, \quad k = \frac{-1}{6}$

By using

$$\begin{aligned} X &= x - h, & Y &= y - k \\ \frac{-3}{4} &= x - \frac{1}{4}, & \frac{-7}{6} &= y + \frac{1}{6} \\ x &= \frac{-3}{4} + \frac{1}{4}, & y &= \frac{-7}{6} - \frac{1}{6} \\ x &= \frac{-3+1}{4}, & y &= \frac{-7-1}{6} \end{aligned}$$

$x = \frac{-2}{4}$, $y = \frac{-8}{6}$

$x = \frac{-1}{2}$, $y = \frac{-4}{3}$

$\therefore$ $P(x, y) = P\left(\frac{-1}{2}, \frac{-4}{3}\right)$ Ans.

(iv) P(4, – 3) ; O/ (– 2, 3)

$X = 4$, $Y = -3$, $h = -2$, $k = 3$

By using

$X = x - h$, $Y = y - K$

$4 = x + 2$, $-3 = y - 3$

$x = 4 - 2$, $y = -3 + 3$

$x = 2$, $y = 0$

$\therefore$ $P(x, y) = P(2, 0)$ Ans

Q.3: The xy-coordinate axes are rotated about the origin through the indicated angle. The new axes are OX and OY. Find the XY-coordinates of the point P with the given xy-coordinates.

Solution:

(i) P(5, 3) ; θ = 45°

$x = 5$, $y = 3$

By using

$X = x\cos\theta + y\sin\theta$, $Y = y\cos\theta - x\sin\theta$

$X = 5\cos 45^\circ + 3\sin 45^\circ$ $Y = 3\cos 45^\circ - 5\sin 45^\circ$

$X = \frac{5}{\sqrt{2}} + \frac{3}{\sqrt{2}}$, $Y = \frac{3}{\sqrt{2}} - \frac{5}{\sqrt{2}}$

$X = \frac{5+3}{\sqrt{2}}$, $Y = \frac{3-5}{\sqrt{2}}$

$X = \frac{8}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$, $Y = \frac{-2}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$

$X = \frac{8\sqrt{2}}{2}$, $Y = \frac{-2\sqrt{2}}{2}$

$X = 4\sqrt{2}$, $Y = -\sqrt{2}$

$\therefore$ $P(X, Y) = P(4\sqrt{2}, -\sqrt{2})$ Ans

(ii) P(3, – 7) ; θ = 30°

$x = 3$, $y = -7$

By using

$X = x\cos\theta + y\sin\theta$, $Y = y\cos\theta - x\sin\theta$

$X = 3\cos 30^\circ - 7\sin 30^\circ \qquad Y = -7\cos 30^\circ - 3\sin 30^\circ$

$X = \dfrac{3\sqrt{3}}{2} - \dfrac{7}{2} \quad , \quad Y = \dfrac{-7\sqrt{3}}{2} - \dfrac{3}{2}$

$X = \dfrac{3\sqrt{3}-7}{2} \quad , \quad Y = \dfrac{-7\sqrt{3}-3}{2}$

$\therefore \quad P(X, Y) = P\left(\dfrac{3\sqrt{3}-7}{2}, \dfrac{-7\sqrt{3}-3}{2}\right)$ Ans

(iii) P (11, – 15); $\theta = 60^\circ$

$x = 11, \quad y = -15$

By using

$X = x\cos\theta + y\sin\theta \quad , \quad Y = y\cos\theta - x\sin\theta$

$X = 11\cos 60^\circ - 15\sin 60^\circ \quad , \quad Y = -15\cos 60^\circ - 11\sin 60^\circ$

$X = \dfrac{11}{2} - \dfrac{15\sqrt{3}}{2} \qquad Y = \dfrac{-15}{2} - \dfrac{11\sqrt{3}}{2}$

$X = \dfrac{11-15\sqrt{3}}{2} \qquad Y = \dfrac{-15-11\sqrt{3}}{2}$

$\therefore \quad P(X, Y) = P\left(\dfrac{11-15\sqrt{3}}{2} \quad \dfrac{-15-11\sqrt{3}}{2}\right)$ Ans

(iv) P(15, 10) ; $\theta = \arctan\dfrac{1}{3}$

$x = 15, \quad y = 10 \quad , \quad \theta = \tan^{-1}\dfrac{1}{3}$

$\tan\theta = \dfrac{1}{3}$

$\sin\theta = \dfrac{1}{\sqrt{10}} \quad , \quad \cos\theta = \dfrac{3}{\sqrt{10}}$

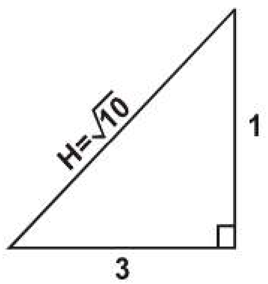
$H^2 = (3)^2 + (1)^2$

$H^2 = 9 + 1$

$H^2 = 10$

$H = \sqrt{10}$

By using



$X = x\cos\theta + y\sin\theta \quad , \quad Y = y\cos\theta - x\sin\theta$

$X = 15\left(\dfrac{3}{\sqrt{10}}\right) + 10\left(\dfrac{1}{\sqrt{10}}\right)$

$$X = \frac{45}{\sqrt{10}} + \frac{10}{\sqrt{10}}$$

$$X = \frac{45+10}{\sqrt{10}} = \frac{55}{\sqrt{10}}$$

$$Y = 10\left(\frac{3}{\sqrt{10}}\right) - 15\left(\frac{1}{\sqrt{10}}\right)$$

$$Y = \frac{30}{\sqrt{10}} - \frac{15}{\sqrt{10}}$$

$$Y = \frac{30-15}{\sqrt{10}}$$

$$Y = \frac{15}{\sqrt{10}}$$

$\therefore \quad P(X, Y) = P\left(\frac{55}{\sqrt{10}}, \frac{15}{\sqrt{10}}\right)$ Ans.

Q.4: The xy-coordinate axes are rotated about the origin through the indicated angle and the new axes are OX and OY. Find the xy-coordinates of P with the given XY-Coordinates.

(i) $P(-5, 3)$; $\theta = 30^\circ$ (ii) $P(-7\sqrt{2}, 5\sqrt{2})$; $\theta = 45^\circ$

Solution:

(i) $P(-5, 3)$; $\theta = 30^\circ$

$X = -5 \quad , \quad Y = 3$

By using $X = x\cos\theta + y\sin\theta \quad , \quad Y = y\cos\theta - x\sin\theta$

$-5 = x\cos 30^\circ + y\sin 30^\circ \quad , \quad 3 = y\cos 30^\circ - x\sin 30^\circ$

$-5 = \frac{x\sqrt{3}}{2} + \frac{y}{2} \quad , \quad 3 = \frac{y\sqrt{3}}{2} - \frac{x}{2}$

$3 = \frac{\sqrt{3}y - x}{2}$

$-5 = \frac{\sqrt{3}x + y}{2} \quad , \quad \sqrt{3}\,y - x = 6 \quad \ldots\ldots (2)$

$\sqrt{3}\,x + y = -10 \quad \ldots. (1)$

Equation (1) + Equation (2) $\times \sqrt{3}$, we get

$$\begin{aligned} \sqrt{3}\,x + y &= -10 \\ 3y - \sqrt{3}\,x &= 6\sqrt{3} \\ \hline 4y &= -10 + 6\sqrt{3} \end{aligned}$$

$$y = \frac{2(-5+3\sqrt{3})}{4}$$

$$y = \frac{-5+3\sqrt{3}}{2}$$

$$= \frac{3\sqrt{3}-5}{2}$$

Put $y = \frac{3\sqrt{3}-5}{2}$ in equation (1)

$$\sqrt{3}\,x + y = -10$$

$$\sqrt{3}\,x = -10 - y$$

$$\sqrt{3}\,x = -10 - \frac{3\sqrt{3}-5}{2}$$

$$\sqrt{3}\,x = \frac{-20-3\sqrt{3}+5}{2}$$

$$x = \frac{-15-3\sqrt{3}}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{3(-5-\sqrt{3})\sqrt{3}}{2(3)}$$

$$= \frac{-5\sqrt{3}-3}{2}$$

$$\therefore \quad P(x, y) = P\left(\frac{-5\sqrt{3}-3}{2}, \frac{3\sqrt{3}-5}{2}\right) \quad \text{Ans}$$

(ii) $\mathbf{P(-7\sqrt{2}, 5\sqrt{2})}$; $\mathbf{\theta = 45^o}$

$X = -7\sqrt{2}$, $Y = 5\sqrt{2}$

By using

$$X = x\cos\theta + y\sin\theta,$$

$$-7\sqrt{2} = x\cos 45^o + y\sin 45^o$$

$$Y = y\cos\theta - x\sin\theta$$

$$Y = y\cos 45^o - x\sin 45^o$$

$$-7\sqrt{2} = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} \quad , \quad 5\sqrt{2} = \frac{y}{\sqrt{2}} - \frac{x}{\sqrt{2}}$$

$$-7\sqrt{2} = \frac{x+y}{\sqrt{2}} \quad , \quad 5\sqrt{2} = \frac{y-x}{\sqrt{2}}$$

$x + y = -14$ ….. (1) $y - x = 10$ ….. (2)

Adding equation (1) and equation (2), we get

$$\begin{aligned} x + y &= -14 \\ \underline{y - x} &\underline{= 10} \\ 2y &= -4 \\ y &= \frac{-4}{2} = -2 \end{aligned}$$

Put $y = -2$ in equation (1)

$$\begin{aligned} x - 2 &= -14 \\ x &= -14 + 2 \\ &= -12 \end{aligned}$$

$\therefore$ $P(x, y) = P(-12, -2)$ Ans

EXERCISE 4.3

Q.1 Find the slope and inclination of the line joining the points:

(i) (– 2, 4) ; (5, 11) (ii) (3, – 2) ; (2, 7)

(iii) (4, 6) ; (4, 8)

Solution:

(i) (– 2, 4) ; (5, 11)

Let A (– 2, 4) ; B (5, 11)

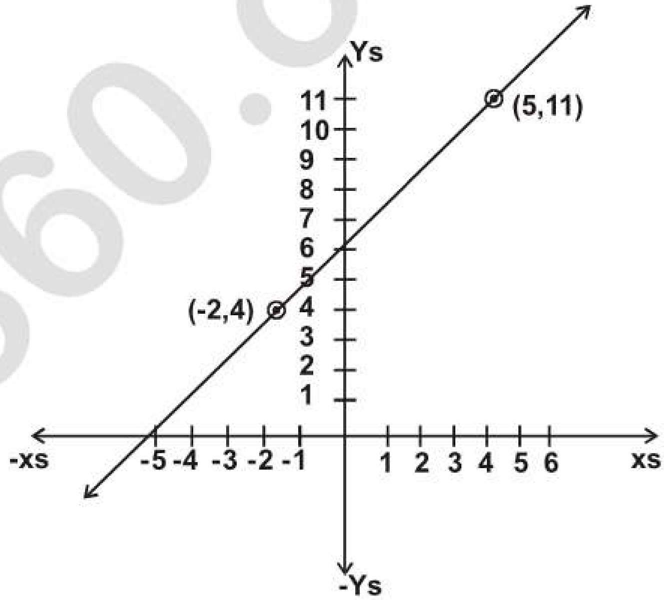
Slope of line AB = $m = \dfrac{11-4}{5+2}$

$= \dfrac{7}{7} = 1$

$\tan\alpha = m$

$\tan\alpha = 1$

$\alpha = \tan^{-1}(1) = 45^\circ$



(ii) (3, – 2) ; (2, 7)

Let A (3, – 2) , B(2, 7)

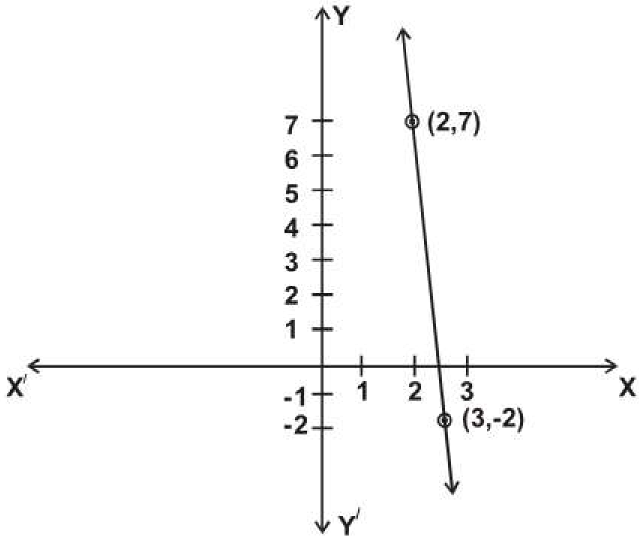
Slope of line AB $= m = \dfrac{7+2}{2-3} = \dfrac{9}{-1} = -9$

$\tan\alpha = m = -9$

$\alpha = \tan^{-1}(-9) = 180^\circ - \tan^{-1} 9$

$= 180 - 83.66^\circ$

$= 96.34^\circ$ Ans.



(iii) (4, 6) ; (4, 8)

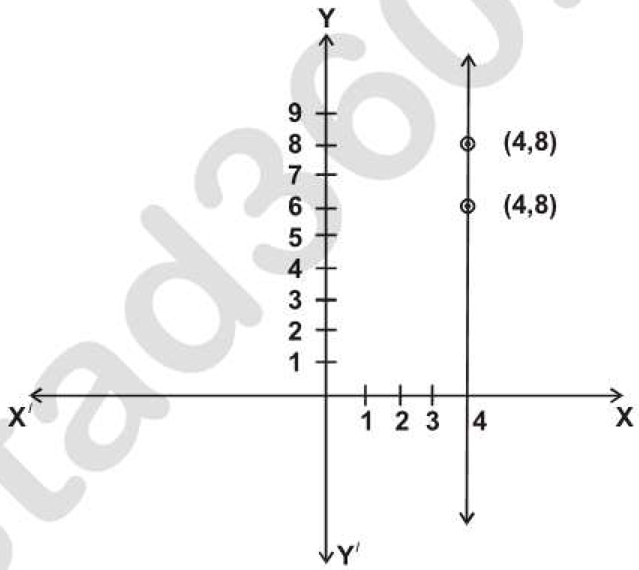
Let A(4, 6) , B(4, 8)

Slope of line AB $= \dfrac{8-6}{4-4} = \dfrac{2}{0} = \infty$ (Undefined)

$m = \tan\alpha = \infty$

$\alpha = \tan^{-1}(\infty)$

$\alpha = 90^\circ$



Q.2 In the triangle A (8, 6), B(– 4, 2), C(– 2, – 6) find the slope of

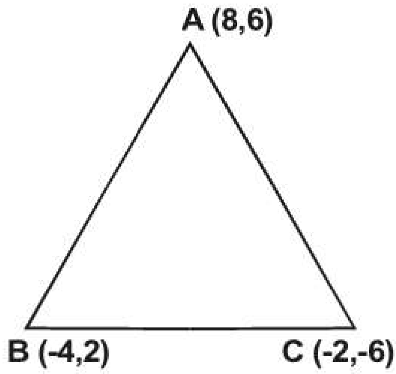
(i) each side of the triangle

(ii) each median of the triangle

(iii) each altitude of the triangle

Solution:

A (8, 6), B (– 4, 2), C (– 2, – 6)



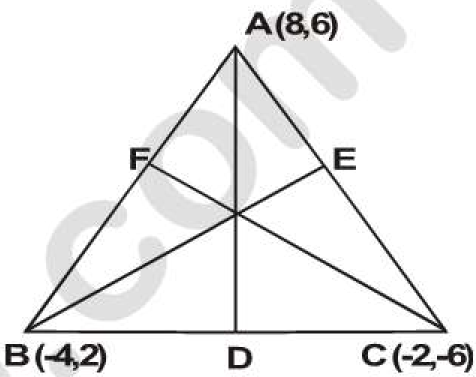
(i) Slope of side AB $= \dfrac{2-6}{-4-8}$

$= \dfrac{-4}{-12} = \dfrac{1}{3}$

Slope of side BC $= \dfrac{-6-2}{-2+4} = \dfrac{-8}{2} = -4$

Slope of side AC $= \dfrac{-6-6}{-2-8} = \dfrac{-12}{-10} = \dfrac{6}{5}$

(ii)



Let $\overline{AD}$, $\overline{BE}$ and $\overline{CF}$ be the medians of a triangle ABC and D, E, F are the mid points of sides $\overline{BC}$, $\overline{AC}$ and $\overline{AB}$ respectively.

Coordinates of D $= \left(\dfrac{-4-2}{2}, \dfrac{2-6}{2}\right)$

$= \left(\dfrac{-6}{2}, \dfrac{-4}{2}\right) = (-3, -2)$

Coordinates of E $= \left(\dfrac{8-2}{2}, \dfrac{6-6}{2}\right)$

$= \left(\dfrac{6}{2}, \dfrac{0}{2}\right) = (3, 0)$

Coordinates of F $= \left(\dfrac{8-4}{2}, \dfrac{6+2}{2}\right)$

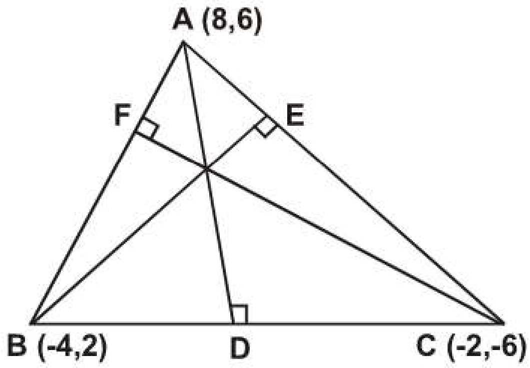
$= \left(\dfrac{4}{2}, \dfrac{8}{2}\right) = (2, 4)$

Slope of median AD $= \dfrac{-2-6}{-3-8} = \dfrac{-8}{-11} = \dfrac{8}{11}$

Slope of median BE $= \dfrac{0-2}{3+4} = \dfrac{-2}{7}$

Slope of median CF $= \dfrac{4+6}{2+2} = \dfrac{10}{4} = \dfrac{5}{2}$

(iii) Let $\overline{AD}$, $\overline{BE}$ and $\overline{CF}$ are the altitudes of ΔABC. Since altitudes are perpendicular to the sides.



$$\text{Slope of altitude } AD = \frac{-1}{\text{Slope of side } BC} = \frac{-1}{-4} = \frac{1}{4}$$

$$\text{Slope of altitude } BE = \frac{-1}{\text{Slope of side } AC} = \frac{-1}{\frac{6}{5}} = \frac{-5}{6}$$

$$\text{Slope of altitude } CF = \frac{-1}{\text{Slope of side } AB} = \frac{-1}{\frac{1}{3}} = -3$$

Q.3 By means of slopes, show that the following points lie on the same line.

(a) $(-1, -3)$; $(1, 5)$; $(2, 9)$
(b) $(4, -5)$; $(7, 5)$; $(10, 15)$
(c) $(-4, 6)$; $(3, 8)$; $(10, 10)$
(d) $(a, 2b)$; $(c, a + b)$; $(2c - a, 2a)$

Solution:

(a) $(-1, -3)$; $(1, 5)$; $(2, 9)$

Let $A(-1, -3)$, $B(1, 5)$, $C(2, 9)$

$$\text{Slope of } AB = \frac{5+3}{1+1} = \frac{8}{2} = 4$$

$$\text{Slope of } BC = \frac{9-5}{2-1} = \frac{4}{1} = 4$$

$\therefore$ Slope of AB = Slope of BC

Show A, B and C lie on the same line.

(b) $(4, -5)$; $(7, 5)$; $(10, 15)$ **(Guj. Board 2006)**

Let $A(4, -5)$, $B(7, 5)$, $C(10, 15)$

$$\text{Slope of } AB = \frac{5+5}{7-4} = \frac{10}{3}$$

Slope of BC $= \dfrac{15-5}{10-7} = \dfrac{10}{3}$

$\therefore$ Slope of AB = Slope of BC

Shows A, B and C lie on the same line.

(c) (– 4, 6) ; (3, 8); (10, 10)

Let A (– 4, 6) , B (3, 8), C(10, 10)

Slope of AB $= \dfrac{8-6}{3+4} = \dfrac{2}{7}$

Slope of BC $= \dfrac{10-8}{10-3} = \dfrac{2}{7}$

$\therefore$ Slope of AB = Slope of BC

Shows A, B and C lie on the same line.

(d) (a, 2b) ; (c, a + b); (2c – a, 2a)

Let A(a, 2b) , B(c, a + b), C(2c – a, 2a)

Slope of AB $= \dfrac{a+b-2b}{c-a} = \dfrac{a-b}{c-a}$

Slope of BC $= \dfrac{2a-(a+b)}{2c-a-c} = \dfrac{2a-a-b}{c-a} = \dfrac{a-b}{c-a}$

$\therefore$ Slope of AB = Slope of BC

Shows the points A, B and C lie on the same line.

Q.4 Find K so that the line joining A (7, 3); B (K, – 6) and the line joining C (– 4, 5); D(– 6, 4) are (i) Parallel (ii) Perpendicular

Solution:

A (7, 3) ; B(K, – 6)

Let m_1 be the slope of line AB.

$m_1 = \dfrac{-6-3}{K-7} = \dfrac{-9}{K-7}$

C (– 4, 5) ; D (– 6, 4)

Let m_2 be the slope of line CD

$m_2 = \dfrac{4-5}{-6+4} = \dfrac{-1}{-2} = \dfrac{1}{2}$

(i) Since the lines AB and CD are parallel

$\therefore\ m_1 = m_2$

$\dfrac{-9}{K-7} = \dfrac{1}{2}$

$-18 = K-7$

$-18+7 = K$

$$\boxed{K = -11} \text{ Ans}$$

(ii) Since the lines AB and CD are perpendicular.

$$\therefore \quad m_1 \times m_2 = -1$$

$$\left(\frac{-9}{K-7}\right)\left(\frac{1}{2}\right) = -1$$

$$9 = 2(K-7)$$

$$9 = 2K - 14$$

$$9 + 14 = 2K$$

$$2K = 23$$

$$\boxed{K = \frac{23}{2}} \quad \text{Ans.}$$

Q.5: Using slopes, show that the triangle with its vertices A (6, 1), B (2, 7) and C(– 6, – 7) is a right triangle.

Solution:

A (6, 1), B (2, 7), C (– 6, – 7)

Let m_1, m_2 and m_3 be the slopes of sides AB, BC and AC respectively.

$$m_1 = \frac{7-1}{2-6} = \frac{6}{-4} = \frac{-3}{2}$$

$$m_2 = \frac{-7-7}{-6-2} = \frac{-14}{-8} = \frac{7}{4}$$

$$m_3 = \frac{-7-1}{-6-6} = \frac{-8}{-12} = \frac{2}{3}$$

Since $\quad m_1 \times m_3 = \frac{-3}{2} \times \frac{2}{3}$

$$m_1 \times m_3 = -1$$

$\therefore$ The side AB and AC are perpendicular.

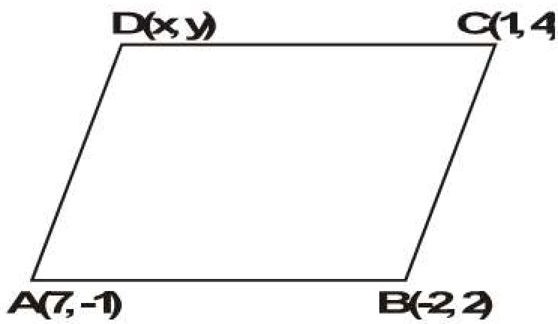
Shows the triangle ABC is a right angle triangle.

Q.6 The three points A (7, – 1), B (–2, 2) and C (1, 4) are consecutive vertices of a parallelogram. Find the fourth vertex.

Solution:

A (7, –1), B (– 2, 2), C (1, 4)

Let D (x, y)be the required vertex of a parallelogram.



$$\text{Slope of } \overline{AB} = \frac{2+1}{-2-7}$$

$$= \frac{3}{-9} = \frac{-1}{3}$$

Slope of $\overline{CD} = \dfrac{y-4}{x-1}$

Slope of $\overline{AD} = \dfrac{y+1}{x-7}$

Slope of $\overline{BC} = \dfrac{4-2}{1+2} = \dfrac{2}{3}$

Since ABCD is a parallelogram.

$\therefore$ Slope of $\overline{AB}$ = Slope of $\overline{CD}$

$$\frac{-1}{3} = \frac{y-4}{x-1}$$

$$-x+1 = 3y-12$$

$$1+12 = x+3y$$

$$x+3y = 13 \quad \text{.... (1)}$$

Slope of $\overline{AD}$ = Slope of $\overline{BC}$

$$\frac{y+1}{x-7} = \frac{2}{3}$$

$$3y+3 = 2x-14$$

$$3+14 = 2x-3y$$

$$2x-3y = 17 \quad \text{.... (2)}$$

Eq. (1) + Eq. (2), we get

$$\begin{aligned} x+3y &= 13 \\ 2x-3y &= 17 \\ \hline 3x &= 30 \\ x &= \frac{30}{3} = 10 \end{aligned}$$

Put $x = 10$ in equation (1)

$$10+3y = 13$$

$$3y = 13-10$$

$$y = \frac{3}{3} = 1$$

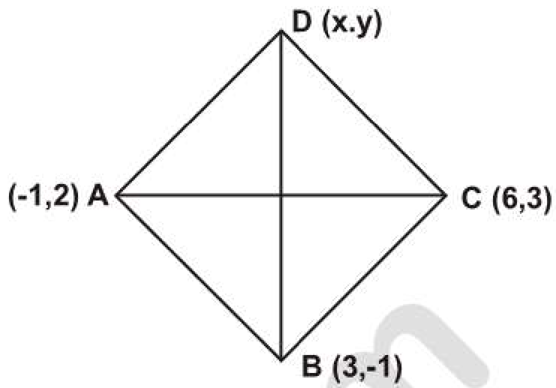
$\therefore$ D (10, 1) is the fourth vertex of parallelogram.

Q.7 The points A (– 1, 2), B (3, – 1) and C (6, 3) are consecutive vertices of a rhombus. Find the fourth vertex and show that the diagonals of the rhombus are perpendicular to each other.

Solution:

Let A (– 1, 2), B (3, – 1), C (6, 3)

Let D (x, y) by the fourth vertex of rhombus.



Slope of side AB $= \dfrac{-1-2}{3+1} = \dfrac{-3}{4}$

Slope of side BC $= \dfrac{3+1}{6-3} = \dfrac{4}{3}$

Slope of side CD $= \dfrac{y-3}{x-6}$

Slope of side DA $= \dfrac{2-y}{-1-x}$

Since ABCD is a rhombus

$\therefore$ Slope of side AB = Slope of side CD

$$\frac{-3}{4} = \frac{y-3}{x-6}$$

$$-3(x-6) = 4(y-3)$$

$$-3x+18 = 4y-12$$

$$18+12 = 3x+4y$$

$$3x+4y = 30 \quad ......(1)$$

Slope of side BC = Slope of side DA

$$\frac{4}{3} = \frac{2-y}{-1-x}$$

$$4(-1-x) = 3(2-y)$$

$$-4-4x = 6-3y$$

$$-4-6 = 4x-3y$$

$$4x-3y = -10 \quad ......(2)$$

Eq. (1) × 3 + Eq, (2) × 4, we get

$$9x + 12y = 90$$

$$16x - 12y = -40$$

$$25x = 50$$

$$x = \frac{50}{25} = 2$$

Put $x = 2$ in eq. (1)

$$3(2) + 4y = 30$$
$$6 + 4y = 30$$
$$4y = 30 - 6$$
$$y = \frac{24}{4} = 6$$

$\therefore$ D (2, 6) is the fourth vertex of rhombus.

$$\text{Slope of diagonal } \overline{AC} = \frac{3-2}{6+1} = \frac{1}{7}$$

$$\text{Slope of diagonal } \overline{BD} = \frac{y+1}{x-3}$$
$$= \frac{6+1}{2-3} = -7$$

Since (Slope of diagonal $\overline{AC}$) (Slope of diagonal $\overline{BD}$) $= -1$

Shows the diagonals $\overline{AC}$ and $\overline{BD}$ are perpendicular to each other.

Q.8 Two pairs of points are given. Find whether the two lines determined by these points are

(i) Parallel (ii) Perpendicular (iii) None

(a) (1, – 2), (2, 4) and (4, 1) , (– 8, 2)

(b) (– 3, 4), (6, 2) and (4, 5) , (– 2, – 7)

Solution:

(a) (1, – 2), (2, 4) and (4, 1) (– 8, 2)

Let A (1, – 2), B (2, 4) and C (4, 1), D (– 8, 2)

Let m_1 and m_2 be the slopes of lines AB and CD.

$$m_1 = \frac{4+2}{2-1} = \frac{6}{1} = 6$$

$$m_2 = \frac{2-1}{-8-4} = \frac{1}{-12} = \frac{-1}{12}$$

Since

$$m_1 \times m_2 = 6 \times \frac{-1}{12} \neq -1$$

and $m_1 \neq m_2$

So the given two lines neither parallel nor perpendicular.

(b) (– 3, 4), (6, 2) and (4, 5) , (– 2, – 7)

Let A (– 3, 4), B (6, 2) and C (4, 5), (– 2, – 7)

Let m_1 and m_2 be the slope of lines AB and CD.

$$m_1 = \frac{2-4}{6+3} = \frac{-2}{9}$$

$$m_2 = \frac{-7-5}{-2-4} = \frac{-12}{-6} = 2$$

Since $m_1 \times m_2 \neq -1$ and $m_1 \neq m_2$

So the given lines are neither parallel nor perpendicular.

Q.9: Find an equation of

(a) the horizontal line through (7, – 9)

(b) the vertical line through (– 5, 3)

(c) the line bisecting the first and 3rd quadrants.

(d) the line bisecting the second and 4th quadrant.

Solution:

(a) the horizontal line through (7, – 9)

Slope of horizontal line $= m = \tan 0^\circ = 0$

Equation of line is

$$y - y_1 = m(x - x_1)$$

$$y + 9 = 0(x - 7)$$

$$\boxed{y + 9 = 0}$$

$$\boxed{y = -9}$$ Ans

(b) the vertical line through (– 5, 3)

As line is vertical and passing through (–5, 3) and eq. of vertical line is $x = c$

here $\boxed{x = -5}$

(c) the line bisecting the first and 3rd quadrant passing through origin.

Slope $= m = \tan 45^\circ = 1$ (Guj. Board 2006)

Eq. of line is

$$y = mx$$

$$\boxed{y = x}$$ Ans

(d) The line bisecting the 2nd and 4th quadrant passing through origin.

Slope = m = tan 135° = – 1

Eq. of line is

$y = mx$

$y = -x$

$\boxed{x = -y}$ Ans

Q.10: Find an equation of the line

(a) through A (– 6, 5) having slope 7

(b) through (8, – 3) having slope 0

(c) through (– 8, 5) having slope undefined (Lhr. Board 2009 (S))

(d) through (–5, – 3) and (9, – 1) (Lhr. Board 2009)

(e) y – intercept = – 7 and Slope = – 5

(f) x – intercept = – 3 and y – intercept = 4

(g) x – intercept = 9 and Slope = – 4 (Lhr. Board 2008, 2011)

Solution:

(a) through A (– 6, 5) having slope 7

$y - y_1 = m(x - x_1)$

$y - 5 = 7(x + 6)$

$y - 5 = 7x + 42$

$0 = 7x + 42 - y + 5$

$7x - y + 47 = 0$ Ans

(b) through (8, – 3) having slope 0

$y - y_1 = m(x - x_1)$

$y + 3 = 0(x - 8)$

$y + 3 = 0$

$y = -3$ Ans

(c) through (– 8, 5) having slope undefined

$y - y_1 = m(x - x_1)$

$y - 5 = \infty(x + 8)$

$y - 5 = \frac{1}{0}(x + 8)$

$x + 8 = 0$

$x = -8$ Ans

(d) through (– 5, – 3) and (9, – 1)

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y+3}{-1+3} = \frac{x+5}{9+5}$$

$$\frac{y+3}{2} = \frac{x+5}{14}$$

$$14y + 42 = 2x + 10$$

$$0 = 2x + 10 - 14y - 42$$

$$2x - 14y - 32 = 0$$

$$2(x - 7y - 16) = 0$$

$$x - 7y - 16 = 0 \qquad \text{Ans}$$

(e) **y-intercept = – 7 and Slope = – 5**

$$y = mx + c$$

$$y = -5x - 7$$

$$5x + y + 7 = 0 \qquad \text{Ans}$$

(f) **x – intercept = – 3 , y – intercept = 4**

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{-3} + \frac{y}{4} = 1$$

$$\frac{-4x + 3y}{12} = 1$$

$$-4x + 3y = 12$$

$$0 = 12 + 4x - 3y$$

$$4x - 3y + 12 = 0$$

(g)

x – intercept = – 9 , Slope = – 4

Since x – intercept = – 9 mean the curve cuts the x-axis at (– 9, 0)

$$y - y_1 = m(x - x_1)$$

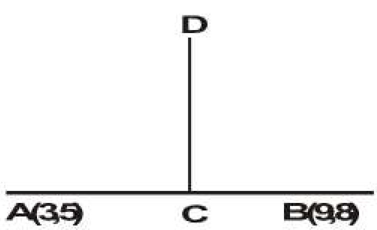
$$y - 0 = -4(x + 9)$$

$$y = -4x - 36$$

$$4x + y + 36 = 0 \qquad \text{Ans}$$

Q.11 **Find an equation of the perpendicular bisector of the line segment joining the points A (3, 5) and B (9, 8).**

Solution:



Let $\overline{CD}$ be the perpendicular bisector and C be the mid point of $\overline{AB}$.

$$\text{Coordinates of C} = \left(\frac{3+9}{2}, \frac{5+8}{2}\right)$$

Coordinates of C $= \left(\dfrac{12}{2}\ ,\ \dfrac{13}{2}\right)$

Coordinates of C $= \left(6\ ,\ \dfrac{13}{2}\right)$

Slope of AB $= \dfrac{8-5}{9-3} = \dfrac{3}{6} = \dfrac{1}{2}$

Since $\overline{CD}$ is $\perp$ to $\overline{AB}$

Slope of $\overline{CD}$ $= \dfrac{-1}{\frac{1}{2}} = -2$

Eq. of line is

$$y - y_1 = m(x - x_1)$$

$$y - \frac{13}{2} = -2(x - 6)$$

$$\frac{2y - 13}{2} = -2x + 12$$

$$2y - 13 = -4x + 24$$

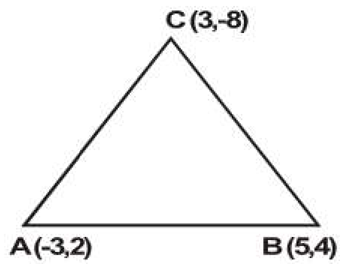
$$4x + 2y - 13 - 24 = 0$$

$$4x + 2y - 37 = 0 \quad \text{Ans}$$

Q.12: Find equations of the sides, altitudes and medians of the triangle whose vertices are A (– 3, 2), B (5, 4) and C (3, – 8). (Lhr. Board 2007, 2011)

Solution:

A (– 3, 2), B (5, 4), C (3, – 8)



<u>Equations of sides</u>

Equation of side AB

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - 2}{4 - 2} = \frac{x + 3}{5 + 3}$$

$$\frac{y - 2}{2} = \frac{x + 3}{8}$$

$$8y - 16 = 2x + 6$$

$$0 = 2x + 6 - 8y + 16$$

$$2x - 8y + 22 = 0$$

$$2(x - 4y + 11) = 0$$

$$\boxed{x - 4y + 11 = 0} \quad \text{Ans.}$$

Equation of side BC

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y-4}{-8-4} = \frac{x-5}{3-5}$$

$$\frac{y-4}{-12} = \frac{x-5}{-2}$$

$$-2y+8 = -12x+60$$

$$12x-2y+8-60 = 0$$

$$12x-2y-52 = 0$$

$$2(6x-y-26) = 0$$

$$\boxed{6x - y - 26 = 0} \quad \text{Ans}$$

Equation of side AC is

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y-2}{-8-2} = \frac{x+3}{3+3}$$

$$\frac{y-2}{-10} = \frac{x+3}{6}$$

$$6y-12 = -10x-30$$

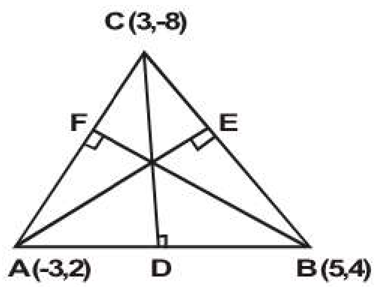
$$10x+6y-12+30 = 0$$

$$10x+6y+18 = 0$$

$$2(5x+3y+9) = 0$$

$$\boxed{5x + 3y + 9 = 0} \quad \text{Ans}$$

Equations of altitudes



$$\text{Slope of side AB} = \frac{4-2}{5+3}$$

$$= \frac{2}{8} = \frac{1}{4}$$

$$\text{Slope of side BC} = \frac{-8-4}{3-5}$$

$$= \frac{-12}{-2} = 6$$

$$\text{Slope of side AC} = \frac{-8-2}{3+3}$$

$$= \frac{-10}{6} = \frac{-5}{3}$$

Slope of altitude CD $= \dfrac{-1}{\frac{1}{4}} = -4$

Slope of altitude AE $= \dfrac{-1}{6}$

Slope of altitude BF $= \dfrac{-1}{\frac{-5}{3}} = \dfrac{3}{5}$

Equation of altitude CD passing through (3, – 8) and having slope – 4 is

$$y - y_1 = m(x - x_1)$$
$$y + 8 = -4(x - 3)$$
$$y + 8 = -4x + 12$$
$$4x + y + 8 - 12 = 0$$
$$\boxed{4x + y - 4 = 0} \quad \text{Ans}$$

Equation of altitude AE passing through (– 3, 2) and having slope $\dfrac{-1}{6}$ is

$$y - y_1 = m(x - x_1)$$
$$y - 2 = \frac{-1}{6}(x + 3)$$
$$6y - 12 = -x - 3$$
$$x + 6y - 12 + 3 = 0$$
$$\boxed{x + 6y - 9 = 0} \quad \text{Ans}$$

Equations of altitude BF passing through B (5, 4) and having slope $\dfrac{3}{5}$ is

$$y - y_1 = m(x - x_1)$$
$$y - 4 = \frac{3}{5}(x - 5)$$
$$5y - 20 = 3x - 15$$
$$0 = 3x - 15 - 5y + 20$$
$$\boxed{3x - 5y + 5 = 0} \quad \text{Ans}$$

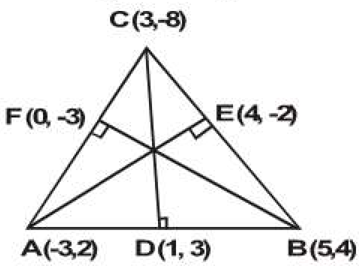
Equations of Medians

Since D, E and F be the mid points of sides AB, BC and AC respectively.

Coordinates of D $= \left(\dfrac{-3+5}{2}, \dfrac{2+4}{2}\right)$

$= \left(\dfrac{2}{2}, \dfrac{6}{2}\right) = (1, 3)$

Coordinates of E $= \left(\dfrac{3+5}{2}, \dfrac{-8+4}{2}\right)$



$$= \left(\frac{8}{2}, \frac{-4}{2}\right) = (4, -2)$$

$$\text{Coordinates of F} = \left(\frac{-3+3}{2}, \frac{2-8}{2}\right)$$

$$\text{Coordinates of F} = \left(\frac{0}{2}, \frac{-6}{2}\right) = (0, -3)$$

Equations of median CD is

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y+8}{3+8} = \frac{x-3}{1-3}$$

$$\frac{y+8}{11} = \frac{x-3}{-2}$$

$$-2y-16 = 11x-33$$

$$0 = 11x-33+2y+16$$

$$\boxed{11x+2y-17 = 0} \quad \text{Ans}$$

Equation of median AE is

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y-2}{-2-2} = \frac{x+3}{4+3}$$

$$\frac{y-2}{-4} = \frac{x+3}{7}$$

$$7y-14 = -4x-12$$

$$4x+7y-14+12 = 0$$

$$\boxed{4x+7y-2 = 0} \quad \text{Ans}$$

Equation of median BF is

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y-4}{-3-4} = \frac{x-5}{0-5}$$

$$\frac{y-4}{-7} = \frac{x-5}{-5}$$

$$-5y+20 = -7x+35$$

$$7x-5y+20-35 = 0$$

$$\boxed{7x-5y-15 = 0} \quad \text{Ans}$$

Q.13 Find an equation of the line through (– 4, – 6) and perpendicular to a line having slope $\frac{-3}{2}$. (Lhr. Board 2006, 2007).

Solution:

Slope of given line $= \frac{-3}{2}$

Slope of required line $= \frac{-1}{\frac{-3}{2}} = \frac{2}{3}$

Equation of the line passing through (– 4, – 6) and having slope $\frac{-2}{3}$ is

$$y - y_1 = m(x - x_1)$$
$$y + 6 = \frac{2}{3}(x + 4)$$
$$3y + 18 = 2x + 8$$
$$0 = 2x + 8 - 3y - 18$$
$$2x - 3y - 10 = 0 \quad \text{Ans}$$

Q.14 Find an equation of the line through (11, – 5) and parallel to line with slope – 24.

Solution:

Slope of given line $= -24$

Slope of required line $= -24$

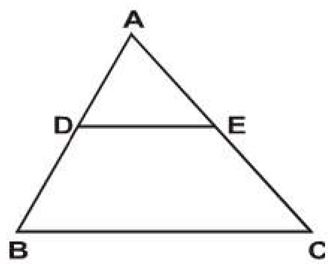
Eq. of line passing through (11, – 5) and having slope – 24 is

$$y - y_1 = m(x - x_1)$$
$$y + 5 = -24(x - 11)$$
$$y + 5 = -24x + 264$$
$$24x + y - 259 = 0 \quad \text{Ans}$$

Q.15 The points A (– 1, 2), B (6, 3) and C (2, – 4) are vertices of a triangle. Show that the line joining the midpoint D of AB and the mid point E of AC is parallel to BC and DE = $\frac{1}{2}$ BC. (Lhr. Board 2006)

Solution:

A (– 1, 2), B (6, 3), C (2, – 4)



Coordinates of D $= \left(\frac{-1+6}{2}, \frac{2+3}{2}\right)$

Coordinates of D $= \left(\frac{5}{2}, \frac{5}{2}\right)$

Coordinates of E $= \left(\frac{-1+2}{2}, \frac{2-4}{2}\right)$

Coordinates of E $= \left(\frac{1}{2}, \frac{-2}{2}\right)$

Coordinates of E $= \left(\frac{1}{2}, -1\right)$

Slope of DE $= \dfrac{-1-\frac{5}{2}}{\frac{1}{2}-\frac{5}{2}} = \dfrac{\frac{-2-5}{2}}{\frac{1-5}{2}} = \dfrac{-7}{-4} = \dfrac{7}{4}$

Slope of BC $= \dfrac{-4-3}{2-6} = \dfrac{-7}{-4} = \dfrac{7}{4}$

Since Slope of DE = Slope of BC

$\therefore$ DE is parallel to BC

$$|DE| = \sqrt{\left(\frac{1}{2}-\frac{5}{2}\right)^2 + \left(-1-\frac{5}{2}\right)^2}$$

$$|DE| = \sqrt{\left(\frac{1-5}{2}\right)^2 + \left(\frac{-2-5}{2}\right)^2}$$

$$|DE| = \sqrt{\left(\frac{-4}{2}\right)^2 + \left(\frac{-7}{2}\right)^2}$$

$$|DE| = \sqrt{4+\frac{49}{4}}$$

$$|DE| = \sqrt{\frac{16+49}{4}}$$

$$|DE| = \frac{\sqrt{65}}{2}$$

$$|BC| = \sqrt{(2-6)^2 + (-4-3)^2}$$

$$|BC| = \sqrt{16+49}$$

$$|BC| = \sqrt{65}$$

$$\therefore \quad |DE| = \frac{\sqrt{65}}{2}$$

$|DE| = \frac{1}{2}|BC|$ Hence proved.

Q.16: A milkman can sell 560 litres of milk at Rs. 12.50 per litre and 700 litres of milk at 12.00 per litre. Assuming the graph of the sale price and the milk sold to be a straight line, find the number of litres of milk that the milkman can sell 12.25 per litre.

Solution:

Let ℓ denoted the number of litres of milk and P denotes the price of milk per litre.

The $(\ell_1, P_1) = (560, 12.50)$ and $(\ell_2, P_2) = (700, 12.00)$

Equation of line is

$$\frac{\ell - \ell_1}{\ell_2 - \ell_1} = \frac{P - P_1}{P_2 - P_1}$$

$$\frac{\ell - 560}{700 - 560} = \frac{P - 12.50}{12.00 - 12.50}$$

$$\frac{\ell - 560}{140} = \frac{P - 12.50}{-0.50}$$

$$-0.50\,\ell + 280 = 140\,P - 1750$$

$$0 = 140\,P - 1750 + 0.50\ell - 280$$

$$0.50\ell + 140\,P - 2030 = 0$$

Put $P = 12.25$

$$0.50\ell + 140(12.25) - 2030 = 0$$

$$0.50\ell + 1715 - 2030 = 0$$

$$0.50\ell = 315$$

$$\ell = \frac{315}{0.50}$$

$\ell = 630$ Ans.

So the milkman can sell 630 litres at Rs. 12.25 per litre.

Q.17 The population of Pakistan to the nearest million was 60 million in 1961 and 95 million in 1981. Using t as the number of years after 1961, find an equation of the line that gives the population in terms of t. Use this equation to find the population in (a) 1947 (b) 1997

Solution:

Let P be the population and t be the number of years.

$(P_1, t_1) = (60, 1961), (P_2, t_2) = (95, 1981)$

Eq. of line is

$$\frac{P - P_1}{P_2 - P_1} = \frac{t - t_1}{t_2 - t_1}$$

$$\frac{P - 60}{95 - 60} = \frac{t - 1961}{1981 - 1961}$$

$$\frac{P - 60}{35} = \frac{t - 1961}{20}$$

$$P - 60 = \frac{35}{20}(t - 1961)$$

$$P = \frac{7}{4}(t - 1961) + 60$$

$$P = \frac{7t - 13727}{4} + 240$$

$$P = \frac{7t - 13487}{4} \quad \text{..........} (1)$$

(i) Put $t = 1947$ in equation (1)

$$P = \frac{7(1947) - 13487}{4}$$

$$P = \frac{13629 - 13487}{4}$$

$$P = 35.5 \text{ million} \qquad \text{Ans.}$$

(ii) Put $t = 1997$ in equation (1)

$$P = \frac{7(1997) - 13487}{4}$$

$$= \frac{13979 - 13487}{4} = 123 \text{ million} \qquad \text{Ans.}$$

Q.18 A house was purchased for Rs. 1 million in 1980. It is worth Rs. 4 million in 1996. Assuming that the value increased by the same amount each year, find an equation that gives the value of the house after years of the date of purchase. What was its value in 1990?

Solution:

Let V be the value of house and t be the years.

$(V_1, t_1) = (1, 1980), \quad (V_2, t_2) = (4, 1996)$

$$\frac{V - V_1}{V_2 - V_1} = \frac{t - t_1}{t_2 - t_1}$$

$$\frac{V - 1}{4 - 1} = \frac{t - 1980}{1996 - 1980}$$

$$\frac{V - 1}{3} = \frac{t - 1980}{16}$$

$$V - 1 = \frac{3}{16}(t - 1980)$$

$$V = \frac{3t - 5940}{16} + 1$$

$$V = \frac{3t - 5940 + 16}{16}$$

$$V = \frac{3t - 5924}{16}$$

Put $t = 1990$

$$V = \frac{3(1990) - 5924}{16}$$

$$V = \frac{5970 - 5924}{16}$$

$V = 2.875$ Million Ans.

Q.19: Plot the Celsius (C) and Fahrenheit (F) temperature scales on the horizontal axis and vertical axis respectively. Draw the line joining the freezing point and the boiling point of water. Find an equation giving F temperature in terms of C.

Solution:

Freezing point of water $= 0°C = 32$ F

Boiling point of water $= 100°C = 212$ F

$$(C_1, F_1) = (0, 32)$$

$$(C_2, F_2) = (100, 212)$$

$$\frac{F - F_1}{F_2 - F_1} = \frac{C - C_1}{C_2 - C_1}$$

$$\frac{F - 32}{212 - 32} = \frac{C - 0}{100 - 0}$$

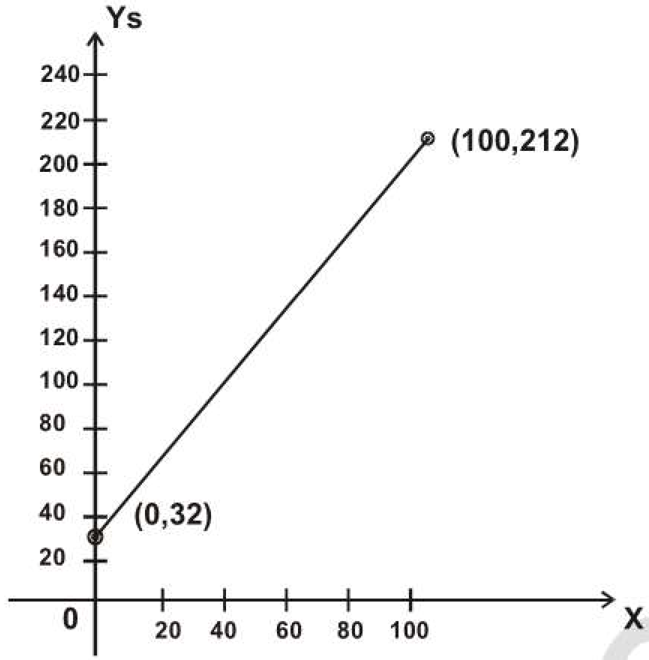
$$\frac{F - 32}{180} = \frac{C}{100}$$

$$100F - 3200 = 180C$$

$$100F = 180C + 3200$$

$$F = \frac{180}{100}C + \frac{3200}{100}$$

$$F = \frac{9}{5}C + 32 \quad \text{Ans.}$$



Q.20 The average entry test score of engineering candidates was 592 in the year 1998 while the score was 564 in 2002. Assuming that the relationship between time and score is linear, find the average score for 2006.

Solution:

Let S be the score and t be the time.

$$(S_1, t_1) = (592, 1998), \quad (S_2, t_2) = (564, 2002)$$

$$\frac{S - S_1}{S_2 - S_1} = \frac{t - t_1}{t_2 - t_1}$$

$$\frac{S - 592}{564 - 592} = \frac{t - 1998}{2002 - 1998}$$

$$\frac{S - 592}{-28} = \frac{t - 1998}{4}$$

$$\frac{S - 592}{-7} = t - 1998$$

$$S - 592 = -7t + 13986$$

$$S = -7t + 13986 + 592$$

$$S = -7t + 14578$$

Put $t = 2006$

$$S = -7(2006) + 14578$$

$$S = -14042 + 14578$$

$$\boxed{S = 536} \quad \text{Ans}$$

Q.21 Convert each of the following equation into
(i) Slope intercept form (ii) two intercepts form
(iii) Normal form
(a) $2x - 4y + 11 = 0$ (b) $4x + 7y - 2 = 0$
(c) $15y - 8x + 3 = 0$ (Lhr. Board 2006, 2009, 2009 (S))
Also find the length of perpendicular from (0, 0) to each line.

Solution:

(a) $2x - 4y + 11 = 0$

(i) Slope intercept Form

$$2x - 4y + 11 = 0$$
$$-4y = -2x - 11$$
$$y = \frac{-2x}{-4} - \frac{11}{-4}$$
$$y = \frac{x}{2} + \frac{11}{4} \quad \text{Ans}$$

(ii) Two intercepts Form

$$2x - 4y + 11 = 0$$
$$2x - 4y = -11$$

Dividing both sides by -11

$$\frac{2x}{-11} - \frac{4y}{-11} = \frac{-11}{-11}$$
$$\frac{x}{\frac{-11}{2}} + \frac{y}{\frac{11}{4}} = 1 \quad \text{Ans}$$

(iii) Normal Form

$$2x - 4y + 11 = 0$$
$$2x - 4y = -11$$
$$-2x + 4y = 11$$

Dividing by $\sqrt{(-2)^2 + (4)^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$

$$\frac{-2x}{2\sqrt{5}} + \frac{4y}{2\sqrt{5}} = \frac{11}{2\sqrt{5}}$$
$$\frac{-x}{\sqrt{5}} + \frac{2y}{\sqrt{5}} = \frac{11}{2\sqrt{5}} \quad \text{........... (1)}$$

Compare it with

$$x\cos\alpha + y\sin\alpha = P$$

$\cos\alpha = \frac{-1}{\sqrt{5}}$, $\sin\alpha = \frac{2}{\sqrt{5}}$, $P = \frac{11}{2\sqrt{5}}$

α lies in 2^{nd} quadrant

$\therefore \quad \alpha = \cos^{-1}\left(\frac{-1}{\sqrt{5}}\right) = 116.57^\circ$

Put in equation (ii)

$$x\cos 116.57^\circ + y\sin 116.57^\circ = \frac{11}{2\sqrt{5}}$$

Length of perpendicular from origin to line is $P = \frac{11}{2\sqrt{5}}$ Ans.

(b) $4x + 7y - 2 = 0$

(i) <u>Slope intercept form</u>

$4x + 7y - 2 = 0$

$7y = -4x + 2$

$y = \frac{-4}{7}x + \frac{2}{7}$ Ans

(ii) <u>Two intercepts form</u>

$4x + 7y - 2 = 0$

$4x + 7y = 2$

Dividing both sides by 2

$$\frac{4x}{2} + \frac{7y}{2} = \frac{2}{2}$$

$$2x + \frac{y}{2/7} = 1$$

$$\frac{x}{\frac{1}{2}} + \frac{y}{\frac{2}{7}} = 1 \quad \text{Ans}$$

(iii) <u>Normal Form</u>

$4x + 7y - 2 = 0$

$4x + 7y = 2$

Dividing by $\sqrt{(4)^2 + (7)^2} = \sqrt{16 + 49} = \sqrt{65}$

$$\frac{4x}{\sqrt{65}} + \frac{7y}{\sqrt{65}} = \frac{2}{\sqrt{65}} \quad \text{........ (1)}$$

Compare it with

$x\cos\alpha + y\sin\alpha = P$

$\cos\alpha = \frac{4}{\sqrt{65}}, \quad \sin\alpha = \frac{7}{\sqrt{65}}, \quad P = \frac{2}{\sqrt{65}}$

α lies in 1^{st} quadrant

$\therefore \quad \alpha = \cos^{-1}\left(\frac{4}{\sqrt{65}}\right) = 60.26^o$

Put in equation (1)

$x\cos 60.26^o + y\sin 60.26^o = \frac{2}{\sqrt{65}}$

Length of perpendicular from origin to line is $P = \frac{2}{\sqrt{65}}$

(c) $15y - 8x + 3 = 0$

(i) <u>Slope intercept form</u>

$15y - 8x + 3 = 0$

$15y = 8x - 3$

$y = \frac{8}{15}x - \frac{3}{15}$

$y = \frac{8}{15}x - \frac{1}{15}$ Ans

(ii) <u>Two intercept form</u>

$15y - 8x + 3 = 0$

$-8x + 15y = -3$

Dividing both sides by -3

$\frac{-8x}{-3} + \frac{15y}{-3} = \frac{-3}{-3}$

$\frac{x}{\frac{3}{8}} + \frac{y}{\frac{-3}{15}} = 1$

$\frac{x}{\frac{3}{8}} + \frac{x}{\frac{-1}{5}} = 1$ Ans

(iii) <u>Normal Form</u>

$15y - 8x + 3 = 0$

$-8x + 15y = -3$

$8x - 15y = 3$

Dividing by $\sqrt{(8)^2 + (-15)^2} = \sqrt{64 + 225} = \sqrt{289} = 17$

$\frac{8x}{17} - \frac{15y}{17} = \frac{3}{17}$ .... (1)

Compare it with

$x\cos\alpha + y\sin\alpha = P$

$\cos\alpha = \frac{8}{17}, \quad \sin\alpha = \frac{-15}{17}, \quad P = \frac{3}{17}$

α lies in 4^{th} quadrant

$\therefore \quad \alpha = \cos^{-1}\left(\frac{8}{17}\right) = -61.93^\circ + 360^\circ$

$= 360^\circ - 61.93^\circ$

$= 298.07^\circ$

Put in eq. (1)

$x\cos 298.07^\circ + y\sin 298.07^\circ = \frac{3}{17}$

Length of perpendicular from origin to line is $P = \frac{3}{17}$

Q.22: In each of the following check whether the two lines are

(i) Parallel (ii) Perpendicular

(iii) Neither parallel nor perpendicular

(a) $2x + y - 3 = 0 \quad ; \quad 4x + 2y + 5 = 0$

(b) $3y = 2x + 5 \quad ; \quad 3x + 2y - 8 = 0$

(c) $4y + 2x - 1 = 0 \quad ; \quad x - 2y - 7 = 0$

(d) $4x - y + 2 = 0 \quad ; \quad 12x - 3y + 1 = 0$

(e) $12x + 35y - 7 = 0 \quad ; \quad 105x - 36y + 11 = 0$

Solution:

(a) $2x + y - 3 = 0 \quad ; \quad 4x + 2y + 5 = 0$

Let m_1 be the slope of 1^{st} line and m_2 be the slope of 2^{nd} line.

$m_1 = \frac{-2}{1} = -2 \quad , \quad m_2 = \frac{-4}{2} = -2$

$\therefore \quad m_1 = m_2$

So the lines are parallel.

(b) $3y = 2x + 5 \quad ; \quad 3x + 2y - 8 = 0$

$2x - 3y + 5 = 0$

Let m_1 be the slope of 1^{st} line and m_2 be the slope of 2^{nd} line.

$m_1 = \frac{-2}{-3} = \frac{2}{3} \quad , \quad m_2 = \frac{-3}{2}$

$\therefore \quad m_1 \times m_2 = \frac{2}{3} \times \frac{-3}{2} = -1$

So the lines are perpendicular.

(c) $\mathbf{4y + 2x - 1 = 0}$; $\mathbf{x - 2y - 7 = 0}$

Let m_1 be the slope of 1^{st} line and m_2 be the slope of 2^{nd} line.

$$m_1 = \frac{-2}{4} = \frac{-1}{2} \quad , \quad m_2 = -\frac{1}{-2} = \frac{1}{2}$$

Since $m_1 \times m_2 \neq -1$, $m_1 \neq m_2$

So the given lines are neither parallel nor perpendicular.

(d) $\mathbf{4x - y + 2 = 0}$; $\mathbf{12x - 3y + 1 = 0}$

Let m_1 be the slope of 1^{st} line and m_2 be the slope of 2^{nd} line.

$$m_1 = -\frac{4}{-1} = 4 \quad , \quad m_2 = -\frac{12}{-3} = 4$$

$\therefore$ $m_1 = m_2$

So the lines are parallel.

(e) $\mathbf{12x + 35y - 7 = 0}$; $\mathbf{105x - 36y + 11 = 0}$

Let m_1 be the slope of 1^{st} line and m_2 be the slope of 2^{nd} line.

$$m_1 = -\frac{12}{35} \quad , \quad m_2 = \frac{-105}{-36} = \frac{35}{12}$$

Since $m_1 \times m_2 = \frac{-12}{35} \times \frac{35}{12} = -1$

So the lines are perpendicular.

Q.23: Find the distance between the given parallel lines. Sketch the lines. Also find an equation of the parallel line lying midway between them.

(a) $\mathbf{3x - 4y + 3 = 0}$; $\mathbf{3x - 4y + 7 = 0}$

(b) $\mathbf{12x + 5y - 6 = 0}$; $\mathbf{12x + 5y + 13 = 0}$

(c) $\mathbf{x + 2y - 5 = 0}$; $\mathbf{2x + 4y = 1}$

Solution:

(a) $\mathbf{3x - 4y + 3 = 0}$ **.... (1)**

$\mathbf{3x - 4y + 7 = 0}$ **.... (2)**

Put $x = 0$ in eq. (1)

$-4y + 3 = 0$

$-4y = -3 \qquad y = \frac{3}{4}$

$\therefore$ Point is $\left(0\ ,\ \frac{3}{4}\right)$

Now we find the distance from $(0, \frac{3}{4})$ to the line (2).

$$d = \frac{\left|3(0) - 4\left(\frac{3}{4}\right) + 7\right|}{\sqrt{(3)^2 + (-4)^2}}$$

$$d = \frac{|0 - 3 + 7|}{\sqrt{9 + 16}}$$

$$d = \frac{4}{\sqrt{25}}$$

$$d = \frac{4}{5} \quad \text{Ans.}$$

Put $y = 0$ in eq. (1)

$$3x + 3 = 0$$

$$3x = -3$$

$$x = \frac{-3}{3} = -1$$

$\therefore$ Point is $(-1, 0)$

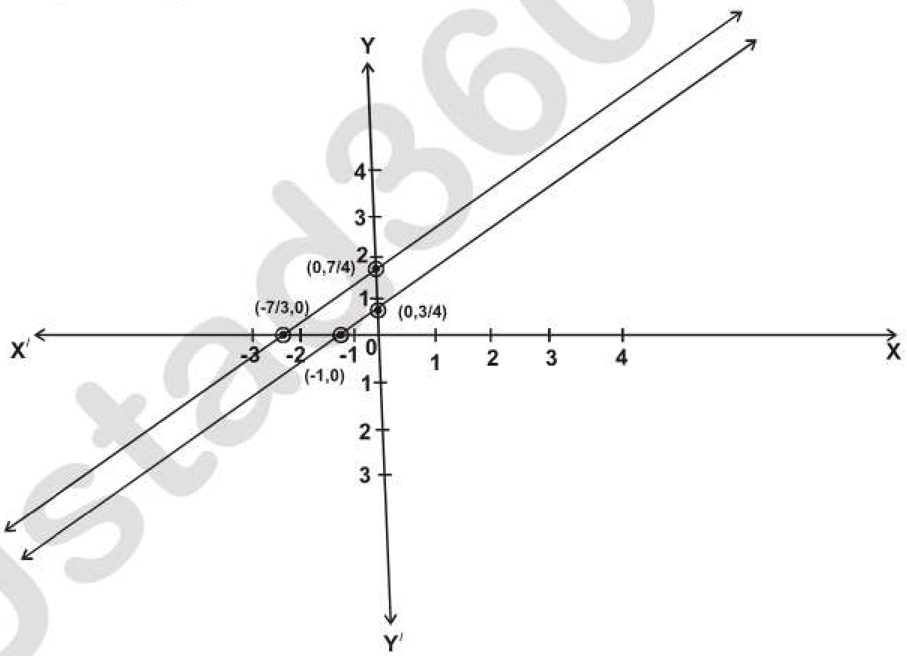
Put $y = 0$ in eq. (2)

$$3x - 4(0) + 7 = 0$$

$$3x = -7$$

$$x = \frac{-7}{3}$$

$\therefore$ Point is $\left(\frac{-7}{3}, 0\right)$



Mid point of $\left(\frac{-7}{3}, 0\right)$ and $(-1, 0)$ is

$$= \left(\frac{\frac{-7}{3}-1}{2}, \frac{0+0}{2}\right) = \left(\frac{\frac{-7-3}{3}}{2}, \frac{0}{2}\right) = \left(\frac{-10}{6}, 0\right) = \left(\frac{-5}{3}, 0\right)$$

Slope of parallel lines = $\frac{-3}{-4} = \frac{3}{4}$

Eq. of line passing through $\left(\frac{-5}{3}, 0\right)$ and having slope $\frac{3}{4}$ is

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{3}{4}\left(x + \frac{5}{3}\right)$$

$$4y = 3x + 5$$

$$\boxed{3x - 4y + 5 = 0} \quad \text{Ans}$$

(b) $\mathbf{12x + 5y - 6 = 0}$ **..... (1)**

$\mathbf{12x + 5y + 13 = 0}$ **..... (2)**

Put $x = 0$ in eq. (1)

$$5y - 6 = 0$$

$$5y = 6$$

$$5y = \frac{6}{5}$$

$\therefore$ Point is $\left(0, \frac{6}{5}\right)$

Now we find the distance from the point $\left(0, \frac{6}{5}\right)$ to the line (2) is

$$d = \frac{\left|12(0) + 5\left(\frac{6}{5}\right) + 13\right|}{\sqrt{(12)^2 + (5)^2}} = \frac{|0 + 6 + 13|}{\sqrt{144 + 25}} = \frac{|19|}{\sqrt{169}} = \frac{19}{13} \quad \text{Ans}$$

Put $y = 0$ in eq. (1)

$$12x + 0 - 6 = 0$$

$$12x = 6$$

$$x = \frac{6}{12} = \frac{1}{2}$$

$\therefore$ Point is $\left(\frac{1}{2}, 0\right)$

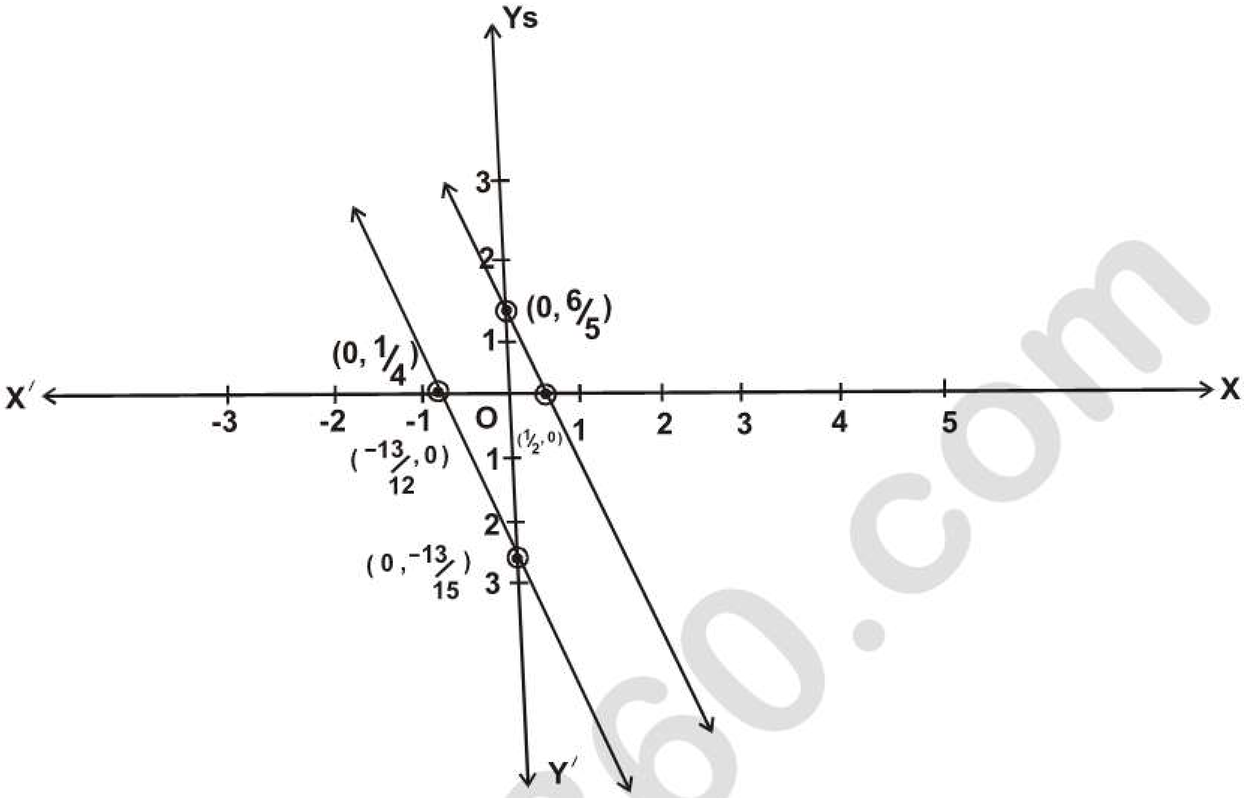
Put $y = 0$ in eq. (2)

$$12x + 0 + 13 = 0$$

$$12x = -13$$

$$x = \frac{-13}{12}$$

$\therefore$ Point is $\left(\frac{-13}{12}, 0\right)$



Mid point of $\left(\frac{-13}{12}, 0\right)$ and $\left(\frac{1}{2}, 0\right)$ is

$$= \left(\frac{\frac{-13}{12}+\frac{1}{2}}{2}, \frac{0+0}{2}\right) = \left(\frac{\frac{-13+6}{12}}{2}, \frac{0}{2}\right)$$

$$= \left(\frac{-7}{24}, 0\right)$$

Slope of parallel lines = $\frac{-12}{5}$

Eq. of line passing through $\left(\frac{-7}{24}, 0\right)$ and having slope $\frac{-12}{15}$ is

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{-12}{5}\left(x + \frac{7}{24}\right)$$

$$5y = -12x - \frac{7}{2}$$

$$5y = \frac{-24x - 7}{2}$$

$$10y = -24x - 7$$

$$24x + 10y + 7 = 0 \qquad \text{Ans}$$

(c) $\mathbf{x + 2y - 5 = 0}$ **..... (1)**

$$2x + 4y = 1$$

$$2x + 4y - 1 = 0 \qquad \text{..... (2)}$$

Put $x = 0$ in eq. (1)

$$0 + 2y - 5 = 0$$

$$2y = 5$$

$$y = \frac{5}{2}$$

$\therefore$ Point is $\left(0\ ,\ \frac{5}{2}\right)$

Now we find the distance from the point $\left(0\ ,\ \frac{5}{2}\right)$ to the line (2)

$$d = \frac{\left|2(0) + 4\left(\frac{5}{2}\right) - 1\right|}{\sqrt{(2)^2 + (4)^2}}$$

$$d = \frac{|0 + 10 - 1|}{\sqrt{4 + 16}} = \frac{9}{\sqrt{20}} = \frac{9}{2\sqrt{5}} \qquad \text{Ans}$$

Put $y = 0$ in eq. (1)

$$x + 0 - 5 = 0$$

$$x = 5$$

$\therefore$ Point is (5, 0)

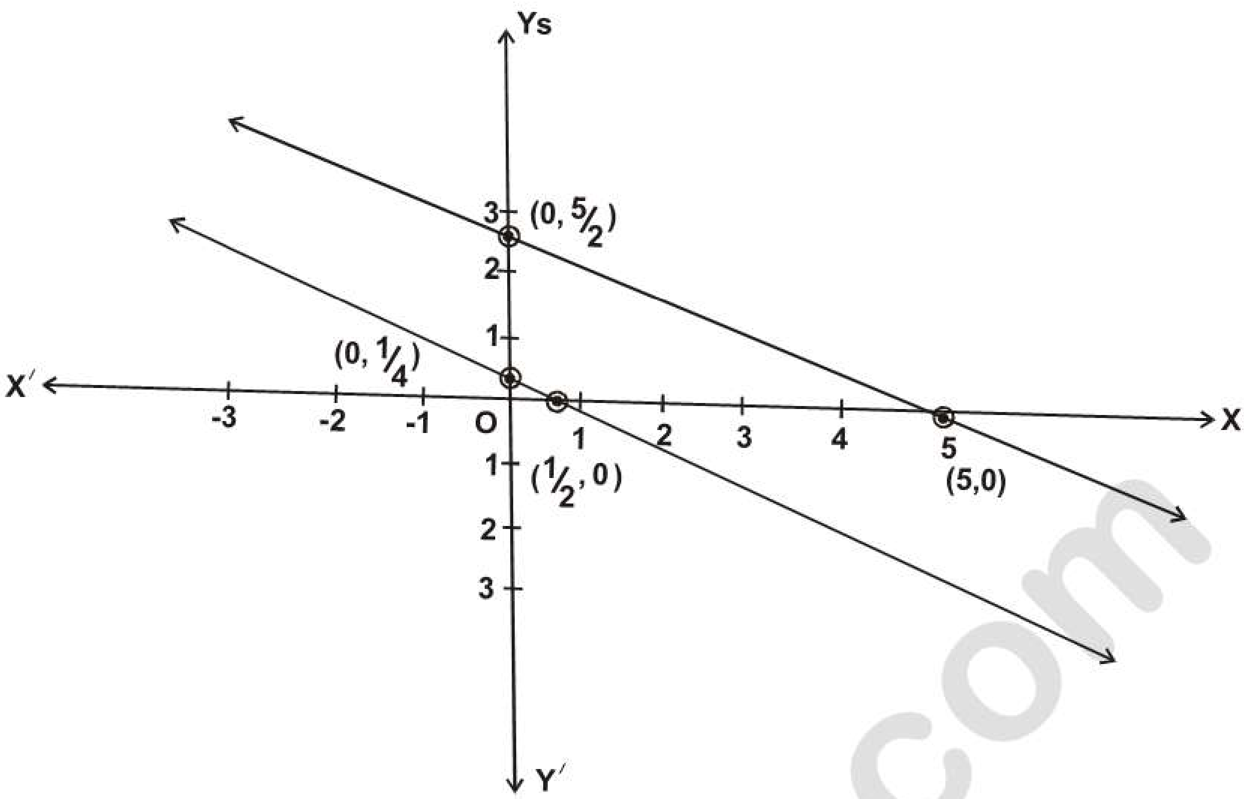
Put $y = 0$ in eq. (2)

$$2x + 0 - 1 = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

$\therefore$ Point is $(\frac{1}{2}, 0)$



Mid point of $(\frac{1}{2}, 0)$ and $(5, 0)$ is

$$= \left(\frac{\frac{1}{2}+5}{2}, \frac{0+0}{2}\right) \qquad = \left(\frac{\frac{1+10}{2}}{2}, \frac{0}{2}\right)$$

$$= \left(\frac{11}{4}, 0\right)$$

Slope of parallel lines $= \frac{-1}{2}$

Equation of line passing through $\left(\frac{11}{4}, 0\right)$ and having slope $\frac{-1}{2}$ is

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{-1}{2}\left(x - \frac{11}{4}\right)$$

$$2y = -x + \frac{11}{4}$$

$$\boxed{x + 2y - \frac{11}{4} = 0} \quad \text{Ans}$$

Q.24 Find an equation of the line through (– 4, 7) and parallel to the line 2x – 7y + 4 = 0. (Gur. Board 2007) (Lhr. Board 2007)

Solution:

$$2x - 7y + 4 = 0$$

$$\text{Slope} = \frac{-2}{-7} = \frac{2}{7}$$

$$\text{Slope of required line} = \frac{2}{7}$$

Eq. of line passing through (– 4, 7) and having slope $\frac{2}{7}$ is

$$y - y_1 = m(x - x_1)$$

$$y - 7 = \frac{2}{7}(x + 4)$$

$$7y - 49 = 2x + 8$$

$$0 = 2x + 8 - 7y + 49$$

$$\boxed{2x - 7y + 57 = 0} \quad \text{Ans.}$$

Q.25: Find an equation of the line through (5, – 8) and perpendicular to the join of A(– 15, – 8), B (10, 7). (Guj. Board 2006)

Solution:

A (– 15, – 8), B (10, 7)

$$\text{Slope of AB} = \frac{7 + 8}{10 + 15} = \frac{15}{25} = \frac{3}{5}$$

$$\text{Slope of required line} = \frac{-1}{\frac{3}{5}} = \frac{-5}{3}$$

Eq. of line passing through (5, – 8) and having slope $\frac{-5}{3}$ is

$$y - y_1 = m(x - x_1)$$

$$y + 8 = \frac{-5}{3}(x - 5)$$

$$3y + 24 = -5x + 25$$

$$5x + 3y + 24 - 25 = 0$$

$$\boxed{5x + 3y - 1 = 0} \quad \text{Ans.}$$

Q.26 Find equations of two parallel lines perpendicular to $2x - y + 3 = 0$ such that the product of the x and y – intercepts of each is 3.

Solution:

$$2x - y + 3 = 0$$

$$\text{Slope} = \frac{-2}{-1} = 2$$

$$\text{Slope of required lines} = \frac{-1}{2}$$

Equations of required lines are

$$y = mx + c$$

$$y = \frac{-1}{2}x + c$$

$$y = \frac{-x + 2c}{2}$$

$$2y = -x + 2c$$

$$x + 2y - 2c = 0 \qquad \text{..... (1)}$$

<u>x-intercept</u>

Put $y = 0$ in eq. (1)

$$x + 0 - 2c = 0$$

$$x = 2c$$

<u>y-intercept</u>

Put $x = 0$ in eq. (1)

$$2y - 2c = 0$$

$$2y = 2c$$

$$y = \frac{2c}{2} = c$$

By given condition

$$(\text{x-intercept})(\text{y-intercept}) = 3$$

$$(2c)(c) = 3$$

$$2c^2 = 3$$

$$c^2 = \frac{3}{2}$$

$$\sqrt{c^2} = \sqrt{\frac{3}{2}}$$

$$c = \pm\sqrt{\frac{3}{2}}$$

Put in eq. (1)

$$x + 2y \pm 2\sqrt{\frac{3}{2}} = 0$$

$$x + 2y \pm \sqrt{2}.\sqrt{3} = 0$$

$$x + 2y \pm \sqrt{6} = 0$$

$$\boxed{x + 2y + \sqrt{6} = 0 \quad \text{and} \quad x + 2y - \sqrt{6} = 0} \qquad \text{Ans.}$$

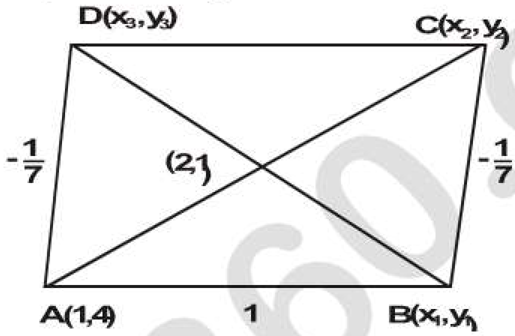
Q.27 One vertex of a parallelogram is (1, 4); the diagonals intersects at (2, 1) and the sides have slopes 1 and $\frac{-1}{7}$. Find the other three vertices.

Solution:

A (1, 4)

Let B (x_1, y_1), C (x_2, y_2) and D (x_3, y_3) be the required vertices of parallelogram.

Since (2, 1) is the mid point of diagonal $\overline{AC}$.



$$2 = \frac{1 + x_2}{2} \quad , \quad 1 = \frac{4 + y_2}{2}$$

$$1 + x_2 = 4 \quad , \quad 2 = 4 + y_2$$

$$x_2 = 4 - 1 \quad , \quad y_2 = 2 - 4$$

$$x_2 = 3 \quad , \quad y_2 = -2$$

$\therefore$ $C(x_2, y_2) = C(3, -2)$

Now slope of AB $= 1$

$$\frac{y_1 - 4}{x_1 - 1} = 1$$

$$y_1 - 4 = x_1 - 1$$

$$-4 + 1 = x_1 - y_1$$

$$x_1 - y_1 = -3 \quad \text{.... (1)}$$

and Slope of BC $= \frac{-1}{7}$

$$\frac{-2-y_1}{3-x_1} = \frac{-1}{7}$$

$$-14-7y_1 = -3+x_1$$

$$-14+3 = x_1+7y_1$$

$$x_1+7y_1 = -11 \quad \text{.... (2)}$$

Eq. (1) – Eq. (2), we get

$$\begin{array}{rcl} x_1 - y_1 & = & -3 \\ \underline{\ }x_1 \pm 7y_1 & = & \mp 11 \\ \hline -8y_1 & = & 8 \end{array}$$

$$y_1 = \frac{8}{-8} = -1$$

Put $y_1 = -1$ in eq. (1)

$$x_1+1 = -3$$

$$x_1 = -3-1$$

$$= -4$$

$\therefore$ $B(x_1, y_1) = B(-4, -1)$

Now (2, 1) is the mid points of diagonal $\overline{BD}$.

$$2 = \frac{-4+x_3}{2} \quad , \quad 1 = \frac{-1+y_3}{2}$$

$$4 = -4+x_3 \quad , \quad 2 = -1+y_3$$

$$4+4 = x_3 \quad , \quad 2+1 = y_3$$

$$x_3 = 8 \quad , \quad y_3 = 3$$

$\therefore$ $D(x_3, y_3) = D(8, 3)$

Hence B (– 4, – 1), C (3, – 2) and D (8, 3) are the required vertices of parallelogram.

Q.28 Find whether the given points lies above and below the given line.

(a) (5, 8) ; $2x-3y+6 = 0$

(b) (– 7, 6) ; $4x+3y-9 = 0$

Solution:

(a) $2x-3y+6 = 0$

To make coefficient of y positive we multiply above equation with -1

$$-2x+3y-6 = 0$$

Put (5, 8) on L.H.S

$$-2(5)+3(8)-6 = -10+24-6 = 8 > 0$$

Hence (5, 8) lies above the line.

(b) $\quad$ **$4x + 3y - 9 = 0$** $\quad$ **(Guj. Board 2008)**

Put $(-7, 6)$ on L.H.S

$4(-7) + 3(6) - 9 = -28 + 18 - 9 = -19 \quad .... (1)$

Since the coefficient of y and expression (1) have opposite signs therefore $(-7, 6)$ lies below the line.

Q.29 Check whether the given points are on the same or opposite sides of the given line.

(a) $\quad$ **$(0, 0)$ and $(-4, 7)$; $6x - 7y + 70 = 0$**

(b) $\quad$ **$(2, 3)$ and $(-2, 3)$; $3x - 5y + 8 = 0$**

Solution:

(a) $\quad$ **$6x - 7y + 70 = 0$**

To make coefficient of y positive we multiply above eq. with -1

$-6x + 7y - 70 = 0 \quad ..... (1)$

Put $(0, 0)$ on L.H.S of eq. (1)

$-6(0) + 7(0) - 70 = -70 < 0$

=> $(0, 0)$ lies below the line.

Put $(-4, 7)$ on L.H.S of eq. (1)

$-6(-4) - 7(7) + 70 = 24 - 49 + 70 = 45 > 0$

$\therefore$ $(-4, 7)$ lies above the line.

Hence $(0, 0)$ $(-4, 7)$ lies on the opposite side of line.

(b) $\quad$ $3x - 5y + 8 = 0$

To make coefficient of y positive we multiply above equation with -1

$-3x + 5y - 8 = 0 \quad .... (1)$

Put $(2, 3)$ on L.H.S. of eq. (1)

$-3(2) + 5(3) - 8 = -6 + 15 - 8 = 1 > 0$

$\therefore$ $(2, 3)$ lies above the line.

Put $(-2, 3)$ on L.H.S of eq. (1)

$-3(-2) + 5(3) + 8 = 6 + 15 - 8 = 13 > 0$

$\therefore$ $(-2, 3)$ lies above the line.

Hence $(2, 3)$ and $(-2, 3)$ lies on the same side of line.

Q.30 Find the distance from the point P $(6, -1)$ to the line $6x - 4y + 9 = 0$ (Guj. Board 2007)

Solution:

Let d be the distance from P $(6, -1)$ to the line $6x - 4y + 9 = 0$

$$d = \frac{|6(6) - 4(-1) + 9|}{\sqrt{(6)^2 + (-4)^2}}$$

$$d = \frac{|36+4+9|}{\sqrt{36+16}} = \frac{49}{\sqrt{52}}$$

$$\boxed{d = \frac{49}{2\sqrt{13}}} \quad \text{Ans.}$$

Q.31 Find the area of the triangular region whose vertices are A (5, 3), B (– 2, 2), C (4, 2). (Lhr. Board 2008)

Solution:

A (5, 3), B (– 2, 2), C (4, 2)

$$\text{Area of } \Delta ABC = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\begin{vmatrix} 5 & 3 & 1 \\ -2 & 2 & 1 \\ 4 & 2 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\left[5\begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} - 3\begin{vmatrix} -2 & 1 \\ 4 & 1 \end{vmatrix} + 1\begin{vmatrix} -2 & 2 \\ 4 & 2 \end{vmatrix}\right]$$

$$= \frac{1}{2}\left[5(2-2) - 3(-2-4) + 1(-4-8)\right]$$

$$= \frac{1}{2}\left[5(0) - 3(-6) + 1(-12)\right]$$

$$= \frac{1}{2}(0 + 18 - 12) = \frac{6}{2} = \boxed{3 \text{ Square unit}} \quad \text{Ans.}$$

Q.32 The coordinates of three points are A (2, 3), B (– 1, 1) and C (4, – 5). By computing the area bounded by ABC check whether the points are collinear.

Solution:

A (2, 3), B (– 1, 1), C (4, – 5)

$$\text{Area of } \Delta ABC = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\begin{vmatrix} 2 & 3 & 1 \\ -1 & 1 & 1 \\ 4 & -5 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\left[2\begin{vmatrix} 1 & 1 \\ -5 & 1 \end{vmatrix} - 3\begin{vmatrix} -1 & 1 \\ 4 & 1 \end{vmatrix} + 1\begin{vmatrix} -1 & 1 \\ 4 & -5 \end{vmatrix}\right]$$

$$= \frac{1}{2}\ [\,2\,(1+5) - 3\,(-1-4) + 1\,(5-4)\,]$$

$$= \frac{1}{2}\ [\,2(6) - 3\,(-5) + 1\,(1)\,]$$

$$= \frac{1}{2}\ (12 + 15 + 1)$$

$$= \frac{28}{2} = \boxed{14 \quad \text{Square unit}} \quad \text{Ans.}$$

Since Area of $\Delta ABC \neq 0$ so the points are not collinear.

EXERCISE 4.4

Q.1: Find the point of intersection of the lines.

(i) $x - 2y + 1 = 0$ and $2x - y + 2 = 0$

(ii) $3x + y + 12 = 0$ and $x + 2y - 1 = 0$

(iii) $x + 4y - 12 = 0$ and $x - 3y + 3 = 0$

Solution:

(i) $x - 2y + 1 = 0$ **(Guj. Board 2005, 2007)**

$2x - y + 2 = 0$

Let P (x, y) be the point of intersection of given lines.

$$\frac{x}{(-2)(2) - 1(-1)} = \frac{-y}{(1)(2) - 1(2)} = \frac{1}{1(-1) - (2)(-2)}$$

$$\frac{x}{-4 + 1} = \frac{-y}{2 - 2} = \frac{1}{-1 + 4}$$

$$\frac{x}{-3} = \frac{-y}{0} = \frac{1}{3}$$

$$\Rightarrow \frac{x}{-3} = \frac{1}{3} \quad \text{and} \quad \frac{-y}{0} = \frac{1}{3}$$

$3x = -3$ $\qquad$ $-3y = 0$

$x = \frac{-3}{3}$ $\qquad$ $y = \frac{0}{-3}$

$x = -1$ $\qquad$ $y = 0$

$\therefore$ P (x, y) = P (– 1, 0) Ans

(ii) $\mathbf{3x + y + 12 = 0}$

$\mathbf{x + 2y - 1 = 0}$

Let P (x, y) be the point of intersection of given lines.

$$\frac{x}{1(-1)-2(12)} = \frac{-y}{3(-1)-1(12)} = \frac{1}{3(2)-1(1)}$$

$$\frac{x}{-1-24} = \frac{-y}{-3-12} = \frac{1}{6-1}$$

$$\frac{x}{-25} = \frac{-y}{-15} = \frac{1}{5}$$

$\Rightarrow \frac{x}{-25} = \frac{1}{5}$ and $\frac{-y}{-15} = \frac{1}{5}$

$x = \frac{-25}{5}$ $\qquad$ $y = \frac{15}{5}$

$x = -5$ $\qquad$ $y = 3$

$\therefore$ P (x, y) = P (– 5, 3) Ans

(iii) $\mathbf{x + 4y - 12 = 0}$

$\mathbf{x - 3y + 3 = 0}$

Let P (x, y) be the point of intersection of given lines.

$$\frac{x}{4(3)-(-3)(-12)} = \frac{-y}{1(3)-1(-12)} = \frac{1}{1(-3)-4(1)}$$

$$\frac{x}{12-36} = \frac{-y}{3+12} = \frac{1}{-3-4}$$

$$\frac{x}{-24} = \frac{-y}{15} = \frac{1}{-7}$$

$\Rightarrow \frac{x}{-24} = \frac{1}{-7}$ and $\frac{-y}{15} = \frac{1}{-7}$

$x = \frac{-24}{-7}$ $\qquad$ $y = \frac{15}{7}$

$x = \frac{24}{7}$

$\therefore$ P (x, y) = $P\left(\frac{24}{7}, \frac{15}{7}\right)$ Ans

Q.2 Find an equation of the line through

(i) the point (2, – 9) and the intersection of the lines 2x + 5y – 8 = 0 and 3x – 4y – 6 = 0

(ii) the intersection of the lines x – y – 4 = 0 and 7x + y + 20 = 0 and

(a) Parallel (b) Perpendicular

to the line 6x + y – 14 = 0 (Lhr. Board 2009 (S)) (Guj. Board 2005)

(iii) through the intersection of lines x + 2y + 3 = 0, 3x + 4y + 7 = 0 and making equal intercepts on the axes.

Solution:

(i) $2x + 5y - 8 = 0$

$3x - 4y - 6 = 0$

Let P (x, y) be the point of intersection of given lines.

$$\frac{x}{5(-6)-(-8)(-4)} = \frac{-y}{2(-6)-3(-8)} = \frac{1}{2(-4)-3(5)}$$

$$\frac{x}{-30-32} = \frac{-y}{-12+24} = \frac{1}{-8-15}$$

$$\frac{x}{-62} = \frac{-y}{12} = \frac{1}{-23}$$

$$\Rightarrow \frac{x}{-62} = \frac{1}{-23} \quad \text{and} \quad \frac{-y}{12} = \frac{1}{-23}$$

$$x = \frac{62}{23} \qquad y = \frac{12}{23}$$

$$\therefore\ P(x, y) = P\left(\frac{62}{23}, \frac{12}{23}\right)$$

Equation of the line passing through (2, – 9) and $\left(\frac{62}{23}, \frac{12}{23}\right)$ is

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y+9}{\frac{12}{23}+9} = \frac{x-2}{\frac{62}{23}-2}$$

$$\frac{y+9}{\frac{12+207}{23}} = \frac{x-2}{\frac{62-46}{23}}$$

$$\frac{23(y+9)}{219} = \frac{23(x-2)}{16}$$

$$16(y+9) = 219(x-2)$$

$$16y + 144 = 219x - 438$$
$$0 = 219x - 438 - 16y - 144$$
$$\boxed{219x - 16y - 582 = 0} \quad \text{Ans}$$

(ii) $\mathbf{x - y - 4 = 0}$

$\mathbf{7x + y + 20 = 0}$

Let P (x, y) be the point of intersection of given lines.

$$\frac{x}{(-1)(20) - 1(-4)} = \frac{-y}{1(20) - 7(-4)} = \frac{1}{1(1) - 7(-1)}$$

$$\frac{x}{-20 + 4} = \frac{-y}{20 + 28} = \frac{1}{1 + 7}$$

$$\frac{x}{-16} = \frac{-y}{48} = \frac{1}{8}$$

$$\Rightarrow \frac{x}{-16} = \frac{1}{8} \quad \text{and} \quad \frac{-y}{48} = \frac{1}{8}$$

$$x = \frac{-16}{8} \qquad y = \frac{-48}{8}$$

$$x = -2 \qquad y = -6$$

$\therefore$ P (x, y) = P (– 2, – 6)

Given line is

$6x + y - 14 = 0$

Let m_1 be the slope of given line and m_2 be the slope of required line.

$$m_1 = -\frac{6}{1} = -6$$

(a) Since the given line and the required line is parallel.

$\therefore$ $m_1 = m_2$

$m_2 = -6$

Equation of line passing through (– 2, – 6) and having slope $m_2 = -6$ is

$$y - y_1 = m_2(x - x_1)$$
$$y + 6 = -6(x + 2)$$
$$y + 6 = -6x - 12$$
$$6x + y + 6 + 12 = 0$$
$$\boxed{6x + y = 0}$$

(b) Since the given line and the required line is perpendicular.

$\therefore$ $m_1 \times m_2 = -1$

$-6 \times m_2 = -1$

$$m_2 = \frac{-1}{-6} = \frac{1}{6}$$

Equation of line passing through (– 2, – 6) and having slope $m_2 = \frac{1}{6}$ is

$$y - y_1 = m_2 (x - x_1)$$

$$y + 6 = \frac{1}{6}(x + 2)$$

$$6y + 36 = x + 2$$

$$0 = x + 2 - 6y - 36$$

$$\boxed{x - 6y - 34 = 0} \quad \text{Ans}$$

(iii) $\mathbf{x + 2y + 3 = 0}$

$\mathbf{3x + 4y + 7 = 0}$

Let P (x, y) be the point of intersection of given lines.

$$\frac{x}{2(7) - 3(4)} = \frac{-y}{1(7) - 3(3)} = \frac{1}{1(4) - 2(3)}$$

$$\frac{x}{14 - 12} = \frac{-y}{7 - 9} = \frac{1}{4 - 6}$$

$$\frac{x}{2} = \frac{-y}{-2} = \frac{1}{-2}$$

$$\Rightarrow \frac{x}{2} = \frac{1}{-2} \quad \text{and} \quad \frac{-y}{-2} = \frac{1}{-2}$$

$$x = \frac{2}{-2} \qquad y = \frac{2}{-2}$$

$$x = -1 \qquad y = -1$$

$\therefore$ P (x, y) = P (– 1, – 1)

Equation of line passing through P (– 1, – 1) and having slope m is

$$y - y_1 = m(x - x_1)$$

$$y + 1 = m(x + 1)$$

$$y + 1 = mx + m$$

$$y - mx + 1 - m = 0 \quad \text{..... (1)}$$

x–intercept

Put $y = 0$ in equation (1)

$$0 - mx + 1 - m = 0$$

$$1 - m = mx$$

$$x = \frac{1 - m}{m}$$

y–intercept

Put $x = 0$ in equation (1)

$y + 1 - m = 0$

$y = m - 1$

Since x – intercept = y intercept

$$\frac{1-m}{m} = m - 1$$

$1 - m = m^2 - m$

$1 = m2 - m + m$

$m^2 = 1$

$m = \pm 1$

Put $m = 1$ in eq. (1)

$y - x + 1 - 1 = 0$

$y - x = 0$

$y = x$

This equation passing through origin and having no intercepts on the axes. So we neglect it.

Put $m = -1$ in eq. (1)

$y + x + 1 + 1 = 0$

$\boxed{x + y + 2 = 0}$ Ans

Q.3 Find an equation of the line through the intersection of 16x – 10y – 33 = 0; 12x + 14y + 29 = 0 and the intersection of x – y + 4 = 0 ; x – 7y + 2 = 0

Solution:

$16x - 10y - 33 = 0$

$12x + 14y + 29 = 0$

Let P(x, y) be the point of intersection of above lines.

$$\frac{x}{(-10)(29) - 14(-33)} = \frac{-y}{16(29) - 12(-33)} = \frac{1}{16(14) - 12(-10)}$$

$$\frac{x}{-290 + 462} = \frac{-y}{464 + 396} = \frac{1}{224 + 120}$$

$$\frac{x}{172} = \frac{-y}{860} = \frac{1}{344}$$

$$\Rightarrow \frac{x}{172} = \frac{1}{344}, \quad \frac{-y}{860} = \frac{1}{344}$$

$$x = \frac{172}{344} \qquad y = \frac{-860}{344}$$

$$x = \frac{1}{2} \qquad y = \frac{-5}{2}$$

$\therefore \ P(x, y) = P\left(\frac{1}{2}, \frac{-5}{2}\right)$

$x - y + 4 = 0 \qquad x - 7y + 2 = 0$

Let P_1 (x, y) be the point of intersection of above lines.

$$\frac{x}{(-1)(2) - 4(-7)} = \frac{-y}{1(2) - 1(4)} = \frac{1}{1(-7) - 1(-1)}$$

$$\frac{x}{-2 + 28} = \frac{-y}{2 - 4} = \frac{1}{-7 + 1}$$

$$\frac{x}{26} = \frac{-y}{-2} = \frac{1}{-6}$$

$$\Rightarrow \frac{x}{26} = \frac{1}{-6} \quad \text{and} \quad \frac{-y}{-2} = \frac{1}{-6}$$

$$x = \frac{-26}{6} \qquad y = \frac{2}{-6}$$

$$x = \frac{-13}{3} \qquad y = \frac{-1}{3}$$

$\therefore \ P_1(x, y) = P_1\left(\frac{-13}{3}, \frac{-1}{3}\right)$

Equation of line passing through $\left(\frac{1}{2}, \frac{-5}{2}\right)$ and $\left(\frac{-13}{3}, \frac{-1}{3}\right)$ is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y + \frac{5}{2}}{\frac{-1}{3} + \frac{5}{2}} = \frac{x - \frac{1}{2}}{\frac{-13}{3} - \frac{1}{2}}$$

$$\frac{\frac{2y + 5}{2}}{\frac{-2 + 15}{6}} = \frac{\frac{2x - 1}{2}}{\frac{-26 - 3}{6}}$$

$$\frac{6(2y + 5)}{2(13)} = \frac{6(2x - 1)}{2(-29)}$$

$$-29(2y + 5) = 13(2x - 1)$$

$$-58y - 145 = 26x - 13$$

$$0 = 26x - 13 + 58y + 145$$

$$26x + 58y + 132 = 0$$

$$2(13x + 29y + 66) = 0 \qquad \boxed{13x + 29y + 66 = 0} \qquad \text{Ans}$$

Q.4: Find the condition that the lines $y = m_1x + c_1$; $y = m_2x + c_2$ and $y = m_3x + c_3$ are concurrent. (Lhr. Board 2007)

Solution:

$$y = m_1x + c_1$$
$$y = m_2x + c_2$$
$$y = m_3x + c_3$$
$$m_1x - y + c_1 = 0$$
$$m_2x - y + c_2 = 0$$
$$m_3x - y + c_3 = 0$$

Since the lines are concurrent

$$\therefore \begin{vmatrix} x_1 & y_1 & c_1 \\ x_2 & y_2 & c_2 \\ x_3 & y_3 & c_3 \end{vmatrix} = 0$$

$$\begin{vmatrix} m_1 & -1 & c_1 \\ m_2 & -1 & c_2 \\ m_3 & -1 & c_3 \end{vmatrix} = 0$$

$$-\begin{vmatrix} m_1 & 1 & c_1 \\ m_2 & 1 & c_2 \\ m_3 & 1 & c_3 \end{vmatrix} = 0$$

$R_2 - R_1$, $R_3 - R_1$

$$\begin{vmatrix} m_1 & 1 & c_1 \\ m_2 - m_1 & 0 & c_2 - c_1 \\ m_3 - m_1 & 0 & c_3 - c_1 \end{vmatrix} = 0$$

Expanding from C_2

$$-(-1)\begin{vmatrix} m_2 - m_1 & c_2 - c_1 \\ m_3 - m_1 & c_3 - c_1 \end{vmatrix} = 0$$

$$(m_2 - m_1)(c_3 - c_1) - (m_3 - m_1)(c_2 - c_1) = 0$$

$(m_2 - m_1)(c_3 - c_1) = (m_3 - m_1)(c_2 - c_1)$ is the required condition.

Q.5 Determine the value of P such that the lines $2x - 3y - 1 = 0$, $3x - y - 5 = 0$ and $3x + py + 8 = 0$ meet at a point.

(Lhr. Board 2007, 2009) (Guj. Board 2008)

Solution:

$2x - 3y - 1 = 0$

$3x - y - 5 = 0$

$3x + Py + 8 = 0$

Since the lines are concurrent

$$\therefore \begin{vmatrix} 2 & -3 & -1 \\ 3 & -1 & -5 \\ 3 & P & 8 \end{vmatrix} = 0$$

$$2\begin{vmatrix} -1 & -5 \\ P & 8 \end{vmatrix} - (-3)\begin{vmatrix} 3 & -5 \\ 3 & 8 \end{vmatrix} + (-1)\begin{vmatrix} 3 & -1 \\ 3 & P \end{vmatrix} = 0$$

$2(-8 + 5P) + 3(24 + 15) - 1(3P + 3) = 0$

$-16 + 10P + 3(39) - 3P - 3 = 0$

$-16 + 10P + 117 - 3P - 3 = 0$

$7P + 98 = 0$

$7P = -98$

$P = \dfrac{-98}{7}$

$\boxed{P = -14}$ Ans

Q.6 Show that the lines $4x - 3y - 8 = 0$, $3x - 4y - 6 = 0$ and $x - y - 2 = 0$ are concurrent and third line bisects the angle formed by the first two lines.

(Lhr. Board 2011) (Guj. Board 2008)

Solution:

$4x - 3y - 8 = 0$ ... (1)

$3x - 4y - 6 = 0$ ... (2)

$x - y - 2 = 0$ ... (3)

Taking

$$\begin{vmatrix} 4 & -3 & -8 \\ 3 & -4 & -6 \\ 1 & -1 & -2 \end{vmatrix}$$

$$= 4\begin{vmatrix} -4 & -6 \\ -1 & -2 \end{vmatrix} - (-3)\begin{vmatrix} 3 & -6 \\ 1 & -2 \end{vmatrix} + (-8)\begin{vmatrix} 3 & -4 \\ 1 & -1 \end{vmatrix}$$

$= 4(8-6) + 3(-6+6) - 8(-3+4)$

$= 4(2) + 3(0) - 8(1) = 8 + 0 - 8 = 0$

Shows the lines are concurrent.

Let m_1, m_2 and m_3 be the slopes of lines (1), (2) and (3).

$m_1 = \frac{-4}{-3} = \frac{4}{3}$

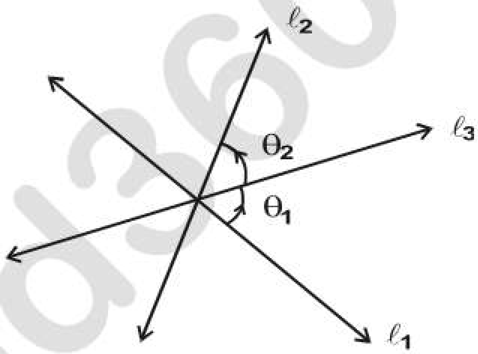
$m_2 = \frac{-3}{-4} = \frac{3}{4}$

$m_3 = \frac{-1}{-1} = 1$

Let θ_1 be an angle from ℓ_1 to ℓ_3 and θ_2 be an angle from ℓ_3 to ℓ_2.

$$\tan\theta_1 = \frac{m_3 - m_1}{1 + m_3 m_1}$$

$$= \frac{1 - \frac{4}{3}}{1 + (1)\left(\frac{4}{3}\right)} = \frac{\frac{3-4}{3}}{\frac{3+4}{3}}$$



$\tan\theta_1 = \frac{-1}{7}$

$$\tan\theta_2 = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{\frac{3}{4} - 1}{1 + \left(\frac{3}{4}\right)(1)}$$

$$= \frac{\frac{3-4}{4}}{\frac{4+3}{4}}$$

$\tan\theta_2 = \dfrac{-1}{7}$

$\therefore \tan\theta_1 = \tan\theta_2$

$\Rightarrow \theta_1 = \theta_2$

$\Rightarrow \ell_3$ bisect the angle formed by the first two lines.

Q.7 The vertices of a triangle are A (– 2, 3), B (– 4, 1) and C (3, 5). Find coordinates of the

(i) centroid (L.B 2006) (ii) Orthocentre (L.B 2009 (S))

(iii) Circumcentre of the triangle. Are these three points are collinear?

Solution:

A (– 2, 3), B (– 4, 1) C (3, 5)

(i) Centroid of a triangle is the intersection of medians is

$$= \left(\frac{x_1 + x_2 + x_3}{2}, \frac{y_1 + y_2 + y_3}{2}\right)$$

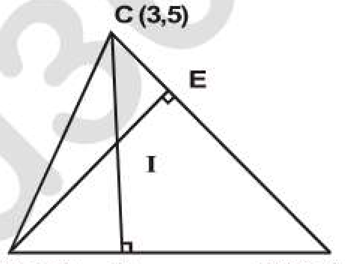
$$= \left(\frac{-2-4+3}{3}, \frac{3+1+5}{3}\right) = \left(\frac{-3}{3}, \frac{9}{3}\right)$$

$= (-1, 3)$ is the centroid of Δ ABC.

(ii) Orthocentre is the point of intersection of altitudes.

Let I be an orthocentre of a Δ ABC.

Let $\overline{CD}$ and $\overline{AE}$ be the altitudes of a Δ ABC.



Slope of side BC $= \dfrac{5-1}{3+4} = \dfrac{4}{7}$

Slope of side AB $= \dfrac{1-3}{-4+2} = \dfrac{-2}{-2} = 1$

Since altitudes are $\perp$ to sides.

$\therefore$ Slope of altitude $\overline{AE} = \dfrac{-1}{\frac{4}{7}} = \dfrac{-7}{4}$

Slope of altitude $\overline{CD}$ $= \dfrac{-1}{1} = -1$

Equation of altitude $\overline{AE}$ is

$$y - y_1 = m(x - x_1)$$
$$y - 3 = \frac{-7}{4}(x + 2)$$
$$4(y - 3) = -7x - 14$$
$$4y - 12 + 7x + 14 = 0$$
$$7x + 4y + 2 = 0 \quad \text{....... (1)}$$

Equation of altitude $\overline{CD}$ is

$$y - y_1 = m(x - x_1)$$
$$y - 5 = -1(x - 3)$$
$$y - 5 = -x + 3$$
$$x + y - 5 - 3 = 0$$
$$x + y - 8 = 0 \quad \text{....... (2)}$$

To find the point of intersection solving equation (1) and equation (2)

$$\frac{x}{4(-8) - 1(2)} = \frac{-y}{7(-8) - 1(2)} = \frac{1}{7(1) - 1(4)}$$
$$\frac{x}{-32 - 2} = \frac{-y}{-56 - 2} = \frac{1}{7 - 4}$$
$$\frac{x}{-34} = \frac{-y}{-58} = \frac{1}{3}$$
$$\Rightarrow \frac{x}{-34} = \frac{1}{3} \quad \text{and} \quad \frac{-y}{-58} = \frac{1}{3}$$
$$x = \frac{-34}{3} \qquad y = \frac{58}{3}$$

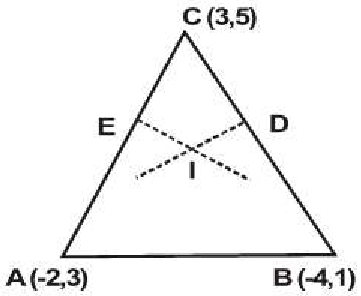
$\therefore \left(\dfrac{-34}{3}, \dfrac{58}{3}\right)$ is orthocenter of ΔABC.

(iii) Circumcentre of the triangle is the point of intersection of perpendicular bisector.

Let I be the centre of the circumcircle.

Since D and E are the mid points of sides BC and AC respectively.

Let $\overline{CE}$ and $\overline{CD}$ are the perpendicular bisectors of sides $\overline{AC}$ and $\overline{BC}$ respectively.



Coordinates of D $= \left(\dfrac{3-4}{2} \quad , \quad \dfrac{5+1}{2}\right)$

$= \left(\dfrac{-1}{2} \quad , \quad \dfrac{6}{2}\right) \quad = \left(\dfrac{-1}{2} \, , \, 3\right)$

Coordinates of E $= \left(\dfrac{3-2}{2} \quad , \quad \dfrac{5+3}{2}\right)$

$= \left(\dfrac{1}{2} \quad , \quad \dfrac{8}{2}\right) = \left(\dfrac{1}{2} \, , \, 4\right)$

Slope of side $\overline{AC} = \dfrac{5-3}{3+2} \qquad = \dfrac{2}{5}$

Slope of side $\overline{BC} = \dfrac{5-1}{3+4} \qquad = \dfrac{4}{7}$

Since perpendicular bisectors are perpendicular to the sides.

$\therefore$ Slope of $\overline{CE} \qquad = \dfrac{-1}{\frac{2}{5}} \quad = \dfrac{-5}{2}$

Slope of $\overline{CD} \qquad = \dfrac{-1}{\frac{4}{7}} \quad = \dfrac{-7}{4}$

Equation of perpendicular bisector $\overline{CE}$ is

$y - y_1 \qquad = \qquad m\,(x - x_1)$

$y - 4 \qquad = \qquad \dfrac{-5}{2}\,(x - \dfrac{1}{2})$

$2(y-4) \; = \qquad -5\left(\dfrac{2x-1}{2}\right)$

$10x + 4y - 21 = 0 \qquad .... (1)$

Equation of perpendicular bisector $\overline{CD}$ is

$y - y_1 = \qquad m\,(x - x_1)$

$$y - 3 = \frac{-7}{4}\left(x + \frac{1}{2}\right)$$

$$y - 3 = \frac{-7}{4}\left(\frac{2x+1}{2}\right)$$

$$8(y-3) = -14x - 7$$

$$8y - 24 + 14x + 7 = 0$$

$$14x + 8y - 17 = 0 \qquad .... (2)$$

For point of intersection solving equation (1) and equation (2)

$$\frac{x}{4(-17) - 8(-21)} = \frac{-y}{(-17)(10) - 14(-21)} = \frac{1}{10(8) - 4(14)}$$

$$\frac{x}{-68 + 168} = \frac{-y}{-170 + 294} = \frac{1}{80 - 56}$$

$$\frac{x}{100} = \frac{-y}{124} = \frac{1}{24}$$

$$\Rightarrow \frac{x}{100} = \frac{1}{24} \quad \text{and} \quad \frac{-y}{124} = \frac{1}{24}$$

$$x = \frac{100}{24} \qquad y = \frac{-124}{24}$$

$$x = \frac{25}{6} \qquad y = \frac{-31}{6}$$

$\therefore \left(\frac{25}{6}, \frac{-31}{6}\right)$ is circumcentre of Δ ABC.

Taking $\begin{vmatrix} -1 & 3 & 1 \\ \frac{-34}{3} & \frac{58}{3} & 1 \\ \frac{25}{6} & \frac{-31}{6} & 1 \end{vmatrix}$

$$= -1\begin{vmatrix} \frac{58}{3} & 1 \\ \frac{-31}{6} & 1 \end{vmatrix} - 3\begin{vmatrix} \frac{-34}{3} & 1 \\ \frac{25}{6} & 1 \end{vmatrix} + 1\begin{vmatrix} \frac{-34}{3} & \frac{58}{3} \\ \frac{25}{6} & \frac{-31}{6} \end{vmatrix}$$

$$= -1\left(\frac{58}{3} + \frac{31}{6}\right) - 3\left(\frac{-34}{3} - \frac{25}{6}\right) + 1\left(\frac{1054}{18} - \frac{1450}{18}\right)$$

$$= -1\left(\frac{116+31}{6}\right) - 3\left(\frac{-68-25}{6}\right) + 1\left(\frac{-396}{18}\right)$$

$$= -\frac{147}{6} - 3\left(\frac{-93}{6}\right) - 22 \qquad = -\frac{49}{2} + \frac{93}{2} - 22$$

$$= -\frac{49}{2} - 3\left(\frac{93}{2}\right) - 22 \qquad = \frac{-49+93-44}{2} = \frac{0}{2} = 0$$

Hence centroid, orthocentre and circumcentre of ΔABC are concurrent.

Q.8 Check whether the lines $4x - 3y - 8 = 0$; $3x - 4y - 6 = 0$; $x - y - 2 = 0$ are concurrent. If so, find the point where they meet.

(Lhr. Board 2005) (Guj. Board 2006)

Solution:

$$4x - 3y - 8 = 0 \qquad .... (1)$$
$$3x - 4y - 6 = 0 \qquad .... (2)$$
$$x - y - 2 = 0 \qquad .... (3)$$

Taking $\begin{vmatrix} 4 & -3 & -8 \\ 3 & -4 & -6 \\ 1 & -1 & -2 \end{vmatrix}$

$$= 4\begin{vmatrix} -4 & -6 \\ -1 & -2 \end{vmatrix} - (-3)\begin{vmatrix} 3 & -6 \\ 1 & -2 \end{vmatrix} + (-8)\begin{vmatrix} 3 & -4 \\ 1 & -1 \end{vmatrix}$$

$$= 4(8-6) + 3(-6+6) - 8(-3+4)$$

$$= 4(2) + 3(0) - 8(1) = 8 + 0 - 8 = 0$$

Hence the lines are concurrent. Now we find the point of intersection solving eq. (1) & eq. (2)

$$\frac{x}{-3(-6)-(-4)(-8)} = \frac{-y}{4(-6)-3(-8)} = \frac{1}{4(-4)-3(-3)}$$

$$\frac{x}{18-32} = \frac{-y}{-24+24} = \frac{1}{-16+9}$$

$$\frac{x}{-14} = \frac{-y}{0} = \frac{1}{-7}$$

$$\Rightarrow \frac{x}{-14} = \frac{1}{-7} \quad \text{and} \quad \frac{-y}{0} = \frac{1}{-7}$$

$$x = \frac{-14}{-7} \quad , \quad 7y = 0$$

$$x = 2 \quad , \quad y = 0$$

$\therefore$ (2, 0) is the point of concurrency of three lines.

Q.9 Find the coordinates of the vertices of the triangle formed by the lines $x - 2y - 6 = 0$; $3x - y + 3 = 0$; $2x + y - 4 = 0$. Also find measures of the angles of the triangle. (Lhr. Board 2005) (Guj. Board 2005)

Solution:

$$x - 2y - 6 = 0 \quad .... (1)$$
$$3x - y + 3 = 0 \quad .... (2)$$
$$2x + y - 4 = 0 \quad .... (3)$$

For point of intersection solving (1) and (2)

$$\frac{x}{-2(3) - (-1)(-6)} = \frac{-y}{1(3) - (3)(-6)} = \frac{1}{1(-1) - 3(-2)}$$

$$\frac{x}{-6-6} = \frac{-y}{3+18} = \frac{1}{-1+6}$$

$$\frac{x}{-12} = \frac{-y}{21} = \frac{1}{5}$$

$$\Rightarrow \frac{x}{-12} = \frac{1}{5} \quad \text{and} \quad \frac{-y}{21} = \frac{1}{5}$$

$$x = \frac{-12}{5} \qquad y = \frac{-21}{5}$$

$\therefore \left(\frac{-12}{5}, \frac{-21}{5}\right)$ is the point of intersection of (1) and (2).

For point of intersection solving equation (2) and equation (3)

$$\frac{x}{(-1)(-4) - 1(3)} = \frac{-y}{3(-4) - 2(3)} = \frac{1}{3(1) - 2(-1)}$$

$$\frac{x}{4-3} = \frac{-y}{-12-6} = \frac{1}{3+2}$$

$$\frac{x}{1} = \frac{-y}{-18} = \frac{1}{5}$$

$$\Rightarrow \frac{x}{1} = \frac{1}{5} \quad \text{and} \quad \frac{-y}{-18} = \frac{1}{5}$$

$$x = \frac{1}{5} \qquad y = \frac{18}{5}$$

$\therefore \left(\frac{1}{5}, \frac{18}{5}\right)$ is the point of intersection of (2) & (3)

For point of intersection solving equation (1) & equation (3)

$$\frac{x}{(-2)(-4) - 1(-6)} = \frac{-y}{1(-4) - 2(-6)} = \frac{1}{1(1) - 2(-2)}$$

$$\frac{x}{8+6} = \frac{-y}{-4+12} = \frac{1}{1+4}$$

$$\frac{x}{14} = \frac{-y}{8} = \frac{1}{5}$$

$$\frac{x}{14} = \frac{1}{5} \quad \text{and} \quad \frac{-y}{8} = \frac{1}{5}$$

$$x = \frac{14}{5} \qquad y = \frac{-8}{5}$$

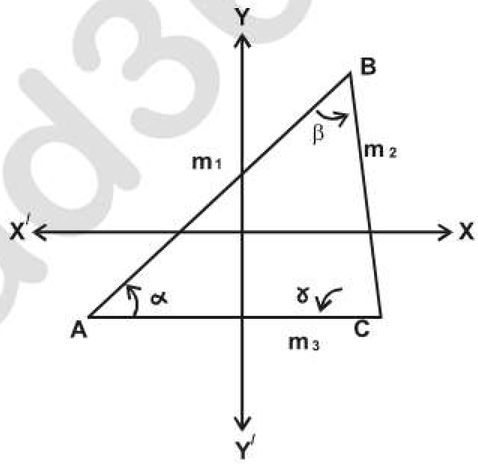
$\therefore$ $\left(\frac{14}{5}, \frac{-8}{5}\right)$ is the point of intersection of equation (1) and equation (3).

Let m_1, m_2 and m_3 be the slopes of sides AB, BC and CA respectively.

$$m_1 = \frac{\frac{18}{5}+\frac{21}{5}}{\frac{1}{5}+\frac{12}{5}} = \frac{\frac{18+21}{5}}{\frac{1+12}{5}} = \frac{39}{13} = 3$$

$$m_2 = \frac{\frac{-8}{5}+\frac{18}{5}}{\frac{14}{5}-\frac{1}{5}} = \frac{\frac{-8-18}{5}}{\frac{14-1}{5}} = \frac{-26}{13} = -2$$

$$m_3 = \frac{\frac{-21}{5}+\frac{8}{5}}{\frac{-12}{5}-\frac{14}{5}} = \frac{\frac{-21+8}{5}}{\frac{-12-14}{5}} = \frac{13}{26} = \frac{1}{2}$$



Let α, β and γ be the angles of a triangle ABC.

$$\tan\alpha = \frac{m_1 - m_3}{1 + m_1 m_3} = \frac{3 - \frac{1}{2}}{1 + (3)\left(\frac{1}{2}\right)}$$

$$= \frac{\frac{6-1}{2}}{\frac{2+3}{2}} \quad = \frac{5}{5}$$

$$\tan\alpha = 1$$

$$\alpha = \tan^{-1}(1) = 45^\circ$$

$$\tan\beta = \frac{m_2 - m_1}{1 + m_2 m_1}$$

$$= \frac{-2-3}{1+(-2)(3)}$$

$$= \frac{-5}{1-6}$$

$$= \frac{-5}{-5}$$

$$\tan\beta = 1$$

$$\beta = \tan^{-1}(1) = 45^\circ$$

$$\tan\gamma = \frac{m_3 - m_2}{1 + m_3 m_2}$$

$$= \frac{\frac{1}{2}+2}{1+\left(\frac{1}{2}\right)(-2)} = \frac{\frac{1+4}{2}}{1-1} = \frac{\frac{5}{2}}{0}$$

$$\tan\gamma = \infty$$

$$\gamma = \tan^{-1}(\infty) = 90^\circ$$

Q.10 Find the angle measured from the line ℓ_1 to the line ℓ_2 where

(a) ℓ_1 : joining (2, 7) and (7, 10)

ℓ_2 : joining (1, 1) and (– 5, 3)

(b) ℓ_1 : joining (3, – 1) and (5, 7)

ℓ_2 : joining (2, 4) and (– 8, 2)

(c) ℓ_1 : joining (1, – 7) and (6, – 4)

ℓ_2 : joining (– 1, 2) and (– 6, – 1)

(d) ℓ_1 : joining (– 9, – 1) and (3, – 5)

ℓ_2 : joining (2, 7) and (– 6, – 7)

Also find the acute angle in each case.

Solution:

(a) Let m_1 be the slope of line ℓ_1 and m_2 be the slope of line ℓ_2.

$$m_1 = \frac{10-7}{7-2} = \frac{3}{5}$$

$$m_2 = \frac{3-1}{-5-1} = \frac{2}{-6} = \frac{-1}{3}$$

Let θ be an angle from line ℓ_1 to ℓ_2.

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$= \frac{\frac{-1}{3} - \frac{3}{5}}{1 + \left(\frac{-1}{3}\right)\left(\frac{3}{5}\right)} = \frac{\frac{-5-9}{15}}{\frac{15-3}{15}} = \frac{-14}{12}$$

$$\tan\theta = \frac{-7}{6}$$

$$\theta = \tan^{-1}\left(\frac{-7}{6}\right) = -49.4^\circ$$

$$= 180^\circ - 49.4^\circ$$

$$= 130.6^\circ \quad \text{Ans}$$

$$\text{Acute angle} = \theta = \tan^{-1}\left|\frac{-7}{6}\right| = 49.4^\circ \text{ Ans.}$$

(b) Let m_1 be the slope of line ℓ_1 and m_2 be the slope of line ℓ_2.

$$m_1 = \frac{7+1}{5-3} = \frac{8}{2} = 4$$

$$m_2 = \frac{2-4}{-8-2} = \frac{-2}{-10} = \frac{1}{5}$$

Let θ be an angle from line ℓ_1 to ℓ_2.

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$= \frac{\frac{1}{5} - 4}{1 + 4\left(\frac{1}{5}\right)} = \frac{\frac{1-20}{5}}{\frac{5+4}{5}} = \frac{-19}{9}$$

$$\theta = \tan^{-1}\left(\frac{-19}{9}\right) = -64.65^\circ$$

$= \quad 180^\circ - 64.5^\circ$

$= \quad 115.35^\circ$ Ans

Acute angle $= \theta = \tan^{-1}\left|\dfrac{-19}{9}\right| = 64.65^\circ$ Ans.

(c) Let m_1 and m_2 be the slopes of lines ℓ_1 and ℓ_2.

$$m_1 = \frac{-4+7}{6-1} = \frac{3}{5}$$

$$m_2 = \frac{-1-2}{-6+1} = \frac{-3}{-5} = \frac{3}{5}$$

Let θ be an angle from ℓ_1 to ℓ_2.

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$= \frac{\frac{3}{5} - \frac{3}{5}}{1 + \left(\frac{3}{5}\right)\left(\frac{3}{5}\right)} = \frac{0}{1 + \frac{9}{25}}$$

$$\tan\theta = 0$$

$$\theta = \tan^{-1}(0) = 0^\circ \quad \text{Ans}$$

(d) Let m_1 and m_2 be the slopes of lines ℓ_1 and ℓ_2.

$$m_1 = \frac{-5+1}{3+9} = \frac{-4}{12} = \frac{-1}{3}$$

$$m_2 = \frac{-7-7}{-6-2} = \frac{-14}{-8} = \frac{7}{4}$$

Let θ be an angle between from ℓ_1 to ℓ_2.

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$= \frac{\frac{7}{4} + \frac{1}{3}}{1 + \left(\frac{-1}{3}\right)\left(\frac{7}{4}\right)} = \frac{\frac{21+4}{12}}{\frac{12-7}{12}} = \frac{25}{5}$$

$$\tan\theta = 5$$

$$\theta = \tan^{-1}(5)$$

$$\theta = 78.69^\circ \quad \text{Ans}$$

Q.11: Find the interior angles of the triangle whose vertices are

(a) A (– 2, 11), B (– 6, – 3), C (4, – 9) (b) A (6, 1), B (2, 7), C (– 6, – 7)

(c) A (2, – 5), B (– 4, – 3), C (– 1, 5) (d) A (2, 8), B (– 5, 4), C (4, – 9)

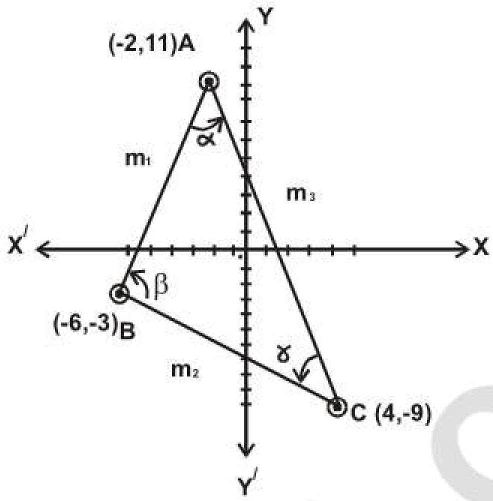
Solution:

(a)A (– 2, 11), B (– 6, – 3), C (4, – 9) (Lhr. Board 2008)

Let m_1, m_2 and m_3 be the slopes of sides AB, BC and AC respectively.

$$m_1 = \frac{-3-11}{-6+2} = \frac{-14}{-4} = \frac{7}{2}$$

$$m_2 = \frac{-9+3}{4+6} = \frac{-6}{10} = \frac{-3}{5}$$



$$m_3 = \frac{-9-11}{4+2} = \frac{-20}{6} = \frac{-10}{3}$$

Let α, β and γ be the angles of a Δ ABC.

Let θ be an angle from line ℓ_1 to ℓ_2.

$$\tan\alpha = \frac{m_3 - m_1}{1 + m_3 m_1}$$

$$= \frac{\frac{-10}{3} - \frac{7}{2}}{1 + \left(\frac{-10}{3}\right)\left(\frac{7}{2}\right)} = \frac{\frac{-20-21}{6}}{\frac{6-70}{6}} = \frac{-41}{-64} = \frac{41}{64}$$

$$\alpha = \tan^{-1}\left(\frac{41}{64}\right) = 32.64^{\circ} \quad \text{Ans.}$$

$$\tan\beta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{\frac{7}{2} + \frac{3}{5}}{1 + \left(\frac{7}{2}\right)\left(\frac{-3}{5}\right)} = \frac{\frac{35+6}{10}}{\frac{10-21}{10}} = \frac{41}{-11}$$

$$\beta = \tan^{-1}\left(\frac{-41}{11}\right)$$

$$= -74.98^\circ$$
$$= 180^\circ - 74.98^\circ$$
$$= 105.02^\circ \qquad \text{Ans.}$$

$$\tan\gamma = \frac{m_2 - m_3}{1 + m_2 m_3}$$

$$= \frac{\frac{-3}{5} + \frac{10}{3}}{1 + \left(\frac{-3}{5}\right)\left(\frac{-10}{3}\right)} = \frac{\frac{-9+50}{15}}{\frac{15+30}{15}}$$

$$\tan\gamma = \frac{41}{45}$$

$$\gamma = \tan^{-1}\left(\frac{41}{45}\right)$$

$$\gamma = 42.34^\circ \qquad \text{Ans}$$

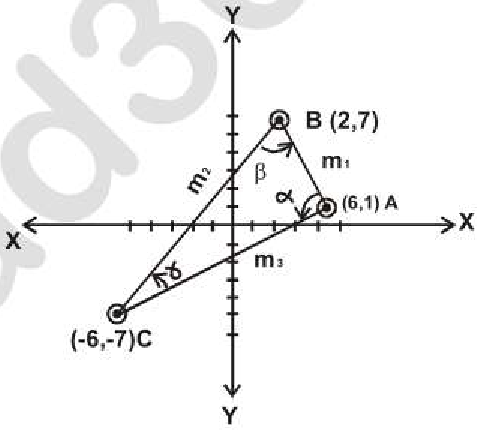
(b) A (6, 1), B (2, 7), C(– 6, – 7)

Let m_1, m_2 and m_3 be the slopes of sides AB, BC and AC respectively.

$$m_1 = \frac{7-1}{2-6} = \frac{6}{-4} = \frac{-3}{2}$$

$$m_2 = \frac{-7-7}{-6-2} = \frac{-14}{-8} = \frac{7}{4}$$

$$m_3 = \frac{-7-1}{-6-6} = \frac{-8}{-12} = \frac{2}{3}$$



Let α, β and γ be the angles of a Δ ABC.

$$\tan\alpha = \frac{m_3 - m_1}{1 + m_3 m_1}$$

$$= \frac{\frac{2}{3}+\frac{3}{2}}{1+\left(\frac{2}{3}\right)\left(\frac{-3}{2}\right)} = \frac{\frac{4+9}{6}}{1-1} = \frac{\frac{13}{6}}{0}$$

$\tan\alpha = \infty$

$\alpha = \tan^{-1}(\infty) = 90^{o}$ Ans

$$\tan\beta = \frac{m_1 - m_2}{1 + m_1m_2}$$

$$= \frac{\frac{-3}{2}-\frac{7}{4}}{1+\left(\frac{-3}{2}\right)\left(\frac{7}{4}\right)} = \frac{\frac{-12-14}{8}}{\frac{8-21}{8}} = \frac{-26}{-13} = 2$$

$\beta = \tan^{-1}(2) = 63.43^{o}$ Ans

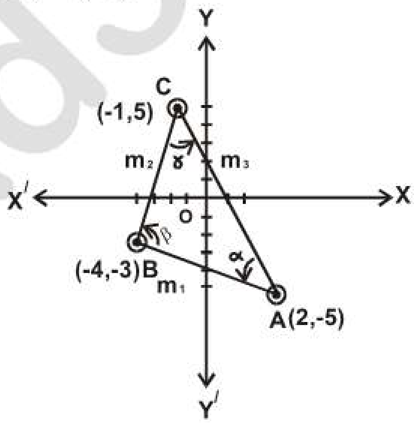
$$\tan\gamma = \frac{m_2 - m_3}{1 + m_2m_3}$$

$$= \frac{\frac{7}{4}-\frac{2}{3}}{1+\left(\frac{7}{4}\right)\left(\frac{2}{3}\right)} = \frac{\frac{21-8}{12}}{\frac{12+14}{12}} = \frac{13}{26} = \frac{1}{2}$$

$$\gamma = \tan^{-1}\left(\frac{1}{2}\right)$$

$\boxed{\gamma = 26.57^{o}}$ Ans

(c) A (2, – 5), B (– 4, – 3), C (– 1, 5)



Let m_1, m_2 and m_3 be the slopes of sides AB, BC and AC respectively.

$$m_1 = \frac{-3+5}{-4+2} = \frac{2}{-6} = \frac{-1}{3}$$

$$m_2 = \frac{5+3}{-1+4} = \frac{8}{3}$$

$$m_3 = \frac{5+5}{-1-2} = \frac{10}{-3} = \frac{-10}{3}$$

Let α, β and γ be the angles of a triangle ABC.

$$\tan\alpha = \frac{m_1 - m_3}{1 + m_1 m_3}$$

$$= \frac{\frac{-1}{3}+\frac{10}{3}}{1+\left(\frac{-1}{3}\right)\left(\frac{-10}{3}\right)} = \frac{\frac{-1+10}{3}}{\frac{9+10}{9}} = \frac{9\,(9)}{3(19)} = \frac{27}{19}$$

$$\alpha = \tan^{-1}\left(\frac{27}{19}\right) = 54.87^{\circ} \quad \text{Ans.}$$

$$\tan\beta = \frac{m_2 - m_1}{1 + m_2 m_1}$$

$$= \frac{\frac{8}{3}+\frac{1}{3}}{1+\left(\frac{8}{3}\right)\left(\frac{-1}{3}\right)} = \frac{\frac{9}{3}}{\frac{9-8}{9}} = \frac{9\times 9}{3} = 27$$

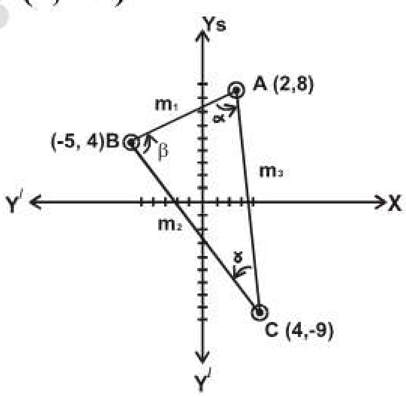
$$\beta = \tan^{-1} 27 = 87.88^{\circ} \quad \text{Ans.}$$

$$\tan\gamma = \frac{m_3 - m_2}{1 + m_3 m_2}$$

$$= \frac{\frac{-10}{3}-\frac{8}{3}}{1+\left(\frac{-10}{3}\right)\left(\frac{8}{3}\right)} = \frac{\frac{-18}{3}}{\frac{9-80}{9}} = \frac{-18\times 9}{3\times -71} = \frac{54}{71}$$

$$\gamma = \tan^{-1}\left(\frac{54}{71}\right) = 37.26^{\circ} \quad \text{Ans.}$$

(d) A (2, 8), B (– 5, 4), C (4, – 9)



Let m_1, m_2 and m_3 be the slope of sides AB, BC and AC respectively.

$$m_1 = \frac{4-8}{-5-2} = \frac{-4}{-7} = \frac{4}{7}$$

$$m_2 = \frac{-9-4}{4+5} = \frac{-13}{9}$$

$$m_3 = \frac{-9-8}{4-2} = \frac{-17}{2}$$

Let α, β and γ be the angles of ΔABC.

$$\tan\alpha = \frac{m_3 - m_1}{1 + m_3 m_1}$$

$$= \frac{\frac{-17}{2} - \frac{4}{7}}{1 + \left(\frac{-17}{2}\right)\left(\frac{4}{7}\right)} = \frac{\frac{-119-8}{14}}{\frac{14-68}{14}} = \frac{-127}{-54}$$

$$\alpha = \tan^{-1}\left(\frac{127}{54}\right) = 66.96^\circ \quad \text{Ans.}$$

$$\tan\beta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{\frac{4}{7} + \frac{13}{9}}{1 + \left(\frac{4}{7}\right)\left(\frac{-13}{9}\right)} = \frac{\frac{36+91}{63}}{\frac{63-52}{63}} = \frac{127}{11}$$

$$\beta = \tan^{-1}\left(\frac{127}{11}\right) = 85.05^\circ \quad \text{Ans.}$$

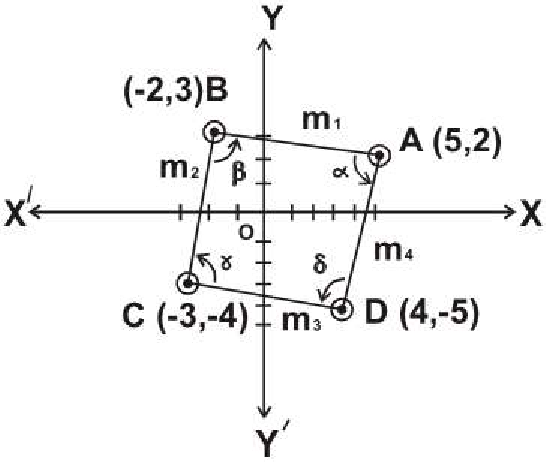
$$\tan\gamma = \frac{m_2 - m_3}{1 + m_2 m_3}$$

$$= \frac{-\frac{13}{9} + \frac{17}{2}}{1 + \left(\frac{-13}{9}\right)\left(\frac{-17}{2}\right)} = \frac{\frac{-26+153}{18}}{\frac{18+221}{18}} = \frac{127}{239}$$

$$\gamma = \tan^{-1}\left(\frac{127}{239}\right) = 27.99^\circ \quad \text{Ans}$$

Q.12 Find the interior angles of the quadrilateral whose vertices are A (5, 2), B (–2, 3), C (– 3, – 4) and D (4, – 5).

Solution:



Let m_1, m_2, m_3 and m_4 be the slopes of sides AB, BC, CD and AD respectively.

$$m_1 = \frac{3-2}{-2-5} = \frac{1}{-7} = \frac{-1}{7}$$

$$m_2 = \frac{-4-3}{-3+2} = \frac{-7}{-1} = 7$$

$$m_3 = \frac{-5+4}{4+3} = \frac{-1}{7}$$

$$m_4 = \frac{-5-2}{4-5} = \frac{-7}{-1} = 7$$

Let α, β, γ and δ be the angles of quadrilateral ABCD.

$$\tan\alpha = \frac{m_4 - m_1}{1 + m_4 m_1}$$

$$= \frac{7+\frac{1}{7}}{1+7\left(\frac{-1}{7}\right)} = \frac{\frac{49+1}{7}}{1-1} = \frac{50}{7(0)} = \frac{50}{0} = \infty$$

$$\alpha = \tan^{-1}(\infty)$$

$$\alpha = 90^\circ$$

$$\tan\beta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\frac{-1}{7} - 7}{1+\left(\frac{-1}{7}\right)(7)} = \frac{\frac{-1-49}{7}}{1-1} = \frac{-50}{7(0)}$$

$$= \infty$$

$$\beta = \tan^{-1}(\infty) = 90^\circ$$

$$\tan\gamma = \frac{m_2 - m_3}{1 + m_2m_3}$$

$$= \frac{7 + \frac{1}{7}}{1 + (7)\left(\frac{-1}{7}\right)} = \frac{\frac{49+1}{7}}{1-1} = \frac{50}{7(0)}$$

$$\tan\gamma = \infty$$

$$\gamma = \tan^{-1}(\infty) = 90^\circ$$

$$\tan\delta = \frac{m_3 - m_4}{1 + m_3m_4}$$

$$= \frac{\frac{-1}{7} - 7}{1 + \left(\frac{-1}{7}\right)(7)} = \frac{\frac{-1-49}{7}}{1-1} = \frac{-50}{7(0)}$$

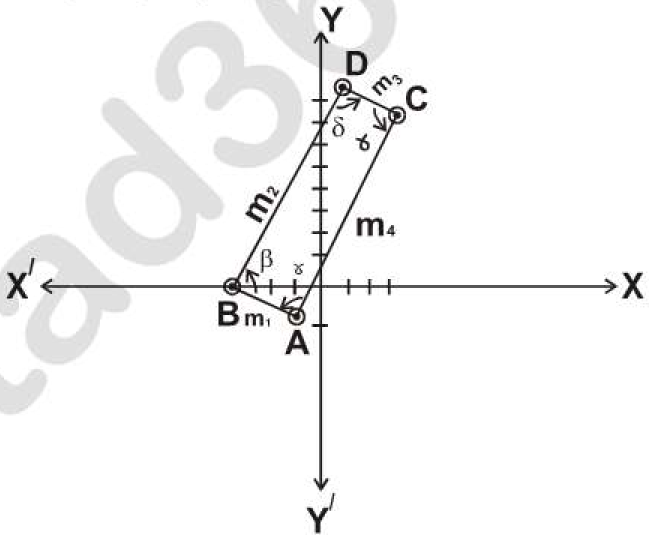
$$\tan\delta = \infty$$

$$\delta = \tan^{-1}(\infty) = 90^\circ$$

Q.13: Show that the points A (– 1, – 1), B (– 3, 0), C (3, 7), D (1, 8) are the vertices of a rectangle. Find its interior angles.

Solution:

A (– 1, – 1), B (– 3, 0), C (3, 7), D (1, 8)



Let m_1, m_2, m_3 and m_4 be the slopes of sides AB, BD, DC and AC respectively

$$m_1 = \frac{0+1}{-3+1} = \frac{1}{-2} = \frac{-1}{2}$$

$$m_2 = \frac{8-0}{1+3} = \frac{8}{4} = 2$$

$$m_3 = \frac{7-8}{3-1} = \frac{-1}{2}$$

$$m_4 = \frac{7+1}{3+1} = \frac{8}{4} = 2$$

$$\tan\alpha = \frac{m_1 - m_4}{1 + m_1 m_4}$$

$$= \frac{\frac{-1}{2} - 2}{1 + \left(\frac{-1}{2}\right)(2)} = \frac{\frac{-1-4}{2}}{1-1} = \frac{\frac{-5}{2}}{0}$$

$$\tan\alpha = \infty$$

$$\alpha = \tan^{-1}(\infty) = 90^\circ$$

$$\tan\beta = \frac{m_2 - m_1}{1 + m_2 m_1}$$

$$= \frac{2 + \frac{1}{2}}{1 + (2)\left(\frac{-1}{2}\right)} = \frac{\frac{4+1}{2}}{1-1} = \frac{\frac{5}{2}}{0}$$

$$\tan\beta = \infty$$

$$\beta = \tan^{-1}(\infty) = 90^\circ$$

$$\tan\delta = \frac{m_3 - m_2}{1 + m_3 m_2}$$

$$= \frac{\frac{-1}{2} - 2}{1 + \left(\frac{-1}{2}\right)(2)} = \frac{\frac{-1-4}{2}}{1-1} = \frac{\frac{5}{2}}{0}$$

$$\tan\delta = \infty$$

$$\delta = \tan^{-1}(\infty) = 90^\circ$$

$$\tan\gamma = \frac{m_4 - m_3}{1 + m_4 m_3}$$

$$= \frac{2 + \frac{1}{2}}{1 + (2)\left(\frac{-1}{2}\right)} = \frac{\frac{4+1}{2}}{1-1} = \frac{\frac{5}{2}}{0}$$

$\tan\gamma = \infty$

$\gamma = \tan^{-1}(\infty) = 90^\circ$

Q.14 Find the area of a region bounded by the triangle whose sides are $7x - y - 10 = 0$; $10x + y - 41 = 0$; $3x + 2y + 3 = 0$.

Solution:

$$7x - y - 10 = 0 \quad .... (1)$$
$$10x + y - 41 = 0 \quad .... (2)$$
$$3x + 2y + 3 = 0 \quad .... (3)$$

For point of intersection solving equation (1) and equation (2)

$$\frac{x}{(-1)(-41) - 1(-10)} = \frac{-y}{7(-41) - 10(-10)} = \frac{1}{7(1) - 10(-1)}$$

$$\frac{x}{41 + 10} = \frac{-y}{-287 + 100} = \frac{1}{7 + 10}$$

$$\frac{x}{51} = \frac{-y}{-187} = \frac{1}{17}$$

$\Rightarrow \quad \frac{x}{51} = \frac{1}{17} \quad$ and $\quad \frac{-y}{-187} = \frac{1}{17}$

$x = \frac{51}{17} \qquad y = \frac{187}{17}$

$x = 3 \qquad y = 11$

$\therefore$ (3, 11) is the point of intersection of equation (1) and equation (2).

For point of intersection solving equation (2) and equation (3)

$$\frac{x}{1(3) - 2(-41)} = \frac{-y}{3(10) - 3(-41)} = \frac{1}{10(2) - 3(1)}$$

$$\frac{x}{3 + 82} = \frac{-y}{30 + 123} = \frac{1}{20 - 3}$$

$$\frac{x}{85} = \frac{-y}{153} = \frac{1}{17}$$

$\Rightarrow \quad \frac{x}{85} = \frac{1}{17} \quad$ and $\quad \frac{-y}{153} = \frac{1}{17}$

$x = \frac{85}{17} \qquad y = \frac{-153}{17}$

$x = 5 \qquad y = -9$

$\therefore$ (5, – 9) is the point of intersection of equation (2) and equation (3).

For point of intersection solving equation (1) and equation (3).

$$\frac{x}{3(-1)-2(-10)} = \frac{-y}{3(7)-3(-10)} = \frac{1}{7(2)-3(-1)}$$

$$\frac{x}{-3+20} = \frac{-y}{21+30} = \frac{1}{14+3}$$

$$\frac{x}{17} = \frac{-y}{51} = \frac{1}{17}$$

$$\Rightarrow \quad \frac{x}{17} = \frac{1}{17} \quad \text{and} \quad \frac{-y}{51} = \frac{1}{17}$$

$$x = \frac{17}{17} \qquad y = \frac{-51}{17}$$

$$x = 1 \qquad y = -3$$

$\therefore$ (1, – 3) is the point of intersection of equation (1) and equation (2).

$$\text{Area of } \Delta\, ABC = \frac{1}{2}\begin{vmatrix} 3 & 11 & 1 \\ 5 & -9 & 1 \\ 1 & -3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\left[3\begin{vmatrix} -9 & 1 \\ -3 & 1 \end{vmatrix} - 11\begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} + 1\begin{vmatrix} 5 & -9 \\ 1 & -3 \end{vmatrix}\right]$$

$$= \frac{1}{2}\left[3(-9+3) - 11(5-1) + 1(-15+9)\right]$$

$$= \frac{1}{2}\left[3(-6) - 11(4) + 1(-6)\right]$$

$$= \frac{1}{2}\left[-18-44-6\right] = \frac{-68}{2} = -34$$

$= 34$ sq. unit Neglecting negative sign. Ans.

Q.15 The vertices of a triangle are A (– 2, 3), B (– 4, 1) and C (3, 5). Find the centre of the circum circle of the triangle.

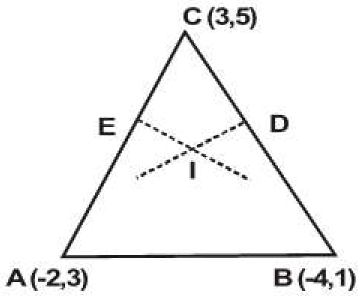
Solution:

A (– 2, 3), B (– 4, 1), C (3, 5)

Let I be the center of the circum circle.

Since D and E are the mid points of sides $\overline{BC}$ and $\overline{AC}$ respectively.

Let $\overline{CE}$ and $\overline{CD}$ be the perpendicular bisectors of sides $\overline{AC}$ and $\overline{BC}$ respectively.



Coordinates of D $= \left(\dfrac{3-4}{2}\ ,\ \dfrac{5+1}{2}\right)$

Coordinates of D $= \left(\dfrac{-1}{2}\ ,\ \dfrac{6}{2}\right)$

Coordinates of D $= \left(\dfrac{-1}{2}\ ,\ 3\right)$

Coordinates of E $= \left(\dfrac{3-2}{2}\ ,\ \dfrac{5+3}{2}\right)$

Coordinates of E $= \left(\dfrac{1}{2}\ ,\ \dfrac{8}{2}\right)$

Coordinates of E $= \left(\dfrac{1}{2}\ ,\ 4\right)$

Slope of $\overline{AC}$ $= \dfrac{5-3}{3+2}$

$= \dfrac{2}{5}$

Slope of $\overline{BC}$ $= \dfrac{5-1}{3+4} = \dfrac{4}{7}$

Since perpendicular bisectors are $\perp$ to the sides

$\therefore$ Slope of $\overline{CE}$ $= \dfrac{-1}{\frac{2}{5}} = \dfrac{-5}{2}$

Slope of $\overline{CD}$ $= \dfrac{-1}{\frac{4}{7}} = \dfrac{-7}{4}$

Equation of perpendicular bisector $\overline{CE}$ is

$y - y_1 = m\,(x - x_1)$

$y - 4 = \dfrac{-5}{2}\left(x - \dfrac{1}{2}\right)$

$$2(y-4) = -5\left(\frac{2x-1}{2}\right)$$
$$4(y-4) = -10x+5$$
$$4y-16+10x-5 = 0$$
$$10x+4y-21 = 0 \qquad \text{.... (1)}$$

Equation of perpendicular bisector $\overline{CD}$ is

$$y-y_1 = m(x-x_1)$$
$$y-3 = \frac{-7}{4}\left(x+\frac{1}{2}\right)$$
$$y-3 = \frac{-7}{4}\left(\frac{2x+1}{2}\right)$$
$$8(y-3) = -14x-7$$
$$8y-24+14x+7 = 0$$
$$14x+8y-17 = 0 \qquad \text{..... (2)}$$

Since I be the point of intersection of equation (1) and equation (2)

Solving eq. (1) and eq. (2) for point for intersection.

$$\frac{x}{4(-17)-8(-21)} = \frac{-y}{(-17)(10)-14(-21)} = \frac{1}{10(8)-4(14)}$$
$$\frac{x}{-68+168} = \frac{-y}{-170+294} = \frac{1}{80-56}$$
$$\frac{x}{100} = \frac{-y}{124} = \frac{1}{24}$$

$$\Rightarrow \quad \frac{x}{100} = \frac{1}{24} \quad \text{and} \quad \frac{-y}{124} = \frac{1}{24}$$
$$x = \frac{100}{24} \qquad y = \frac{-124}{24}$$
$$x = \frac{25}{6} \qquad y = \frac{-31}{6}$$

$\therefore$ $I\left(\frac{25}{6}, \frac{-31}{6}\right)$ is the centre of circum circle.

Q.16 Express the given system of equations in matrix form. Find in each case whether the lines are concurrent.

(a) $x+3y-2 = 0$; $2x-y+4 = 0$; $x-11y+14 = 0$

(b) $2x+3y+4 = 0$; $x-2y-3 = 0$; $3x+y-8 = 0$

(c) $3x-4y-2 = 0$; $x+2y-4 = 0$; $3x-2y+5 = 0$

Solution:

(a)

$$x + 3y - 2 = 0$$
$$2x - y + 4 = 0$$
$$x - 11y + 14 = 0$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Taking $\begin{vmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{vmatrix}$

$$= 1\begin{vmatrix} -1 & 4 \\ -11 & 14 \end{vmatrix} - 3\begin{vmatrix} 2 & 4 \\ 1 & 14 \end{vmatrix} + (-2)\begin{vmatrix} 2 & -1 \\ 1 & -11 \end{vmatrix}$$

$$= 1(-14 + 44) - 3(28 - 4) - 2(-22 + 1)$$

$$= 30 - 3(24) - 2(-21) = 30 - 72 + 42 = 0$$

$\therefore$ The given lines are concurrent.

(b)

$$2x + 3y + 4 = 0$$
$$x - 2y - 3 = 0$$
$$3x + y - 8 = 0$$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 2 & 3 & 4 \\ 1 & -2 & -3 \\ 3 & 1 & -8 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Taking $\begin{vmatrix} 2 & 3 & 4 \\ 1 & -2 & -3 \\ 3 & 1 & -8 \end{vmatrix}$

$$= 2\begin{vmatrix} -2 & -3 \\ 1 & -8 \end{vmatrix} - 3\begin{vmatrix} 1 & -3 \\ 3 & -8 \end{vmatrix} + 4\begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix}$$

$$= 2(16 + 3) - 3(-8 + 9) + 4(1 + 6)$$

$= \quad 2(19) - 3(1) + 4(7)$
$= \quad 38 - 3 + 28$
$= \quad 63 \neq 0$

$\therefore$ The given lines are not concurrent.

(c)

$3x - 4y - 2 = 0$
$x + 2y - 4 = 0$
$3x - 2y + 5 = 0$

This system of equations can be written in matrix form as

$$\begin{bmatrix} 3 & -4 & -2 \\ 1 & 2 & -4 \\ 3 & -2 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Taking $\begin{vmatrix} 3 & -4 & -2 \\ 1 & 2 & -4 \\ 3 & -2 & 5 \end{vmatrix}$

$= \quad 3\begin{vmatrix} 2 & -4 \\ -2 & 5 \end{vmatrix} - (-4)\begin{vmatrix} 1 & -4 \\ 3 & 5 \end{vmatrix} + (-2)\begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix}$
$= \quad 3(10 - 8) + 4(5 + 12) - 2(-2 - 6)$
$= \quad 3(2) + 4(17) - 2(-8)$
$= \quad 6 + 68 + 16$
$= \quad 90 \neq 0$

$\therefore$ The given lines are not concurrent.

Q.17: Find a system of linear equations corresponding to the given matrix form. Check whether the lines represented by the system are concurrent.

(a) $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 1 \\ 0 & -1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ **(b)** $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Solution:

(a) $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & 1 \\ 0 & -1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$x - 1 = 0$$
$$2x + 1 = 0$$
$$-y + 2 = 0$$

Taking $\begin{vmatrix} 1 & 0 & -1 \\ 2 & 0 & 1 \\ 0 & -1 & 2 \end{vmatrix}$

$= 1\begin{vmatrix} 0 & 1 \\ -1 & 2 \end{vmatrix} - 0\begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} + (-1)\begin{vmatrix} 2 & 0 \\ 0 & -1 \end{vmatrix}$

$= 1(0 + 1) - 0 - 1(-2 - 0)$

$= 1 + 2$

$= 3 \neq 0$

$\therefore$ The lines are not concurrent.

(b) $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$x + y + 2 = 0$$
$$2x + 4y - 3 = 0$$
$$3x + 6y - 5 = 0$$

Taking $\begin{vmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{vmatrix}$

$= 1\begin{vmatrix} 4 & -3 \\ 6 & -5 \end{vmatrix} - 1\begin{vmatrix} 2 & -3 \\ 3 & -5 \end{vmatrix} + 2\begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix}$

$= 1(-20 + 18) - 1(-10 + 9) + 2(12 - 12)$

$= 1(-2) - 1(-1) + 2(0)$

$= -2 + 1$

$= -1 \neq 0$

$\therefore$ The lines are not concurrent.

EXERCISE 4.5

Q:1 Find the lines represented by each of the following and also find measure of the angle between them (Problems 1 – 6).

1. $10x^2 - 23xy - 5y^2 = 0$
2. $3x^2 + 7xy + 2y^2 = 0$ **(Lhr. Board 2009 (S))**
3. $9x^2 + 24xy + 16y^2 = 0$
4. $2x^2 + 3xy - 5y^2 = 0$
5. $6x^2 - 19xy + 15y^2 = 0$
6. $x^2 + 2xy \sec\alpha + y^2 = 0$

Solution:

$10x^2 - 23xy - 5y^2 = 0$ **(Guj. Board 2007)**

$10x^2 - 25xy + 2xy - 5y^2 = 0$

$5x(2x - 5y) + y(2x - 5y) = 0$

Either

$2x - 5y = 0$ or $5x + y = 0$

Comparing the given equation with $ax^2 + 2hxy + by^2 = 0$

$a = 10,\quad 2h = -23,\quad b = -5$

$h = \dfrac{-23}{2}$

Let θ be an angle between the lines

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan\theta = \frac{2\sqrt{\left(\frac{-23}{2}\right)^2 - (10)(-5)}}{10 - 5}$$

$$\tan\theta = \frac{2\sqrt{\frac{529}{4} + 50}}{5}$$

$$\tan\theta = \frac{2\sqrt{\frac{529 + 200}{4}}}{5}$$

$$\tan\theta = \frac{2\sqrt{\frac{729}{4}}}{5}$$

$$\tan\theta = \frac{2\left(\frac{27}{2}\right)}{5}$$

$$\tan\theta = \frac{27}{5}$$

$$\theta = \tan^{-1}\left(\frac{27}{5}\right)$$

$$\boxed{\theta = 79.51^\circ} \quad \text{Ans.}$$

2. $3x^2 + 7xy + 2y^2 = 0$ (Guj. Board 2006) (Lhr. Board 2006)

$$3x^2 + 6xy + xy + 2y^2 = 0$$

$$3x(x + 2y) + y(x + 2y) = 0$$

$$(x + 2y)(3x + y) = 0$$

Either

$x + 2y = 0$ or $3x + y = 0$

Comparing the given equation with $ax^2 + 2hxy + by^2 = 0$

$a = 3$, $2h = 7$, $b = 2$

$$h = \frac{7}{2}$$

Let θ be an angle between the lines.

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan\theta = \frac{2\sqrt{\left(\frac{7}{2}\right)^2 - (3)(2)}}{3 + 2}$$

$$\tan\theta = \frac{2\sqrt{\frac{49}{4} - 6}}{5}$$

$$\tan\theta = \frac{2\sqrt{\frac{25}{4}}}{5}$$

$$\tan\theta = \frac{2\left(\frac{5}{2}\right)}{5}$$

$$\tan\theta = \frac{5}{5}$$

$$\tan\theta = 1$$

$$\theta = \tan^{-1}(1)$$

$$\boxed{\theta = 45^\circ} \quad \text{Ans}$$

3. $9x^2 + 24xy + 16y^2 = 0$

$$9x^2 + 12xy + 12xy + 16y^2 = 0$$

$$3x(3x + 4y) + 4y(3x + 4y) = 0$$

$$(3x + 4y)(3x + 4y) = 0$$

$$3x + 4y = 0$$

is the equation of two coincident lines passing through origin.

4. $2x^2 + 3xy - 5y^2 = 0$

$$2x^2 + 5xy - 2xy - 5y^2 = 0$$

$$x(2x + 5y) - y(2x + 5y) = 0$$

$$(2x + 5y)(x - y) = 0$$

Either

$2x + 5y = 0$ or $x - y = 0$

Comparing the given equation with $ax^2 + 2hxy + by^2 = 0$

$a = 2$, $2h = 3$, $b = -5$

$$h = \frac{3}{2}$$

Let θ be an angle between the lines

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan\theta = \frac{2\sqrt{\left(\frac{3}{2}\right)^2 - (2)(-5)}}{2 - 5}$$

$$\tan\theta = \frac{2\sqrt{\frac{9}{4} + 10}}{-3} = \frac{2\sqrt{\frac{9 + 40}{4}}}{-3}$$

$$\tan\theta = \frac{-2\sqrt{\frac{49}{4}}}{3}$$

$$\tan\theta = \frac{-2\left(\frac{7}{2}\right)}{3}$$

$$\tan\theta = \frac{-7}{3}$$

$$-\tan\theta = \frac{7}{3}$$

$$\tan(180^\circ - \theta) = \frac{7}{3}$$

$$180 - \theta = \tan^{-1}\left(\frac{7}{3}\right)$$

$$180 - \theta = 66.80^\circ$$

$$\theta = 180^\circ - 66.80^\circ$$

$$\boxed{\theta = 113.2^\circ} \quad \text{Ans.}$$

5. $6x^2 - 19xy + 15y^2 = 0$

$$6x^2 - 10xy - 9xy + 15y^2 = 0$$

$$2x(3x - 5y) - 3y(3x - 5y) = 0$$

$$(3x - 5y)(2x - 3y) = 0$$

Either

$3x - 5y = 0$ or $2x - 3y = 0$

Comparing the given equation with $ax^2 + 2hxy + by^2 = 0$

$a = 6$, $2h = -19$, $b = 15$

$$h = \frac{-19}{2}$$

Let 'θ' be an angle between the lines

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan\theta = \frac{2\sqrt{\left(\frac{-19}{2}\right)^2 - (6)(15)}}{6 + 15}$$

$$\tan\theta = \frac{2\sqrt{\frac{361}{4} - 90}}{21}$$

$$\tan\theta = \frac{2\sqrt{\frac{361 - 360}{4}}}{21}$$

$$\tan\theta = \frac{2\sqrt{\frac{1}{4}}}{21}$$

$$\tan\theta = \frac{2\left(\frac{1}{2}\right)}{21}$$

$$\tan\theta = \frac{1}{21}$$

$$\theta = \tan^{-1}\left(\frac{1}{21}\right)$$

$$\boxed{\theta = 2.73^\circ} \quad \text{Ans}$$

6. $x^2 + 2xy\sec\alpha + y^2 = 0$

Dividing both sides by x^2

$$\frac{y^2}{x^2} + \frac{2xy\sec\alpha}{x^2} + \frac{x^2}{x^2} = \frac{0}{x^2}$$

$$\left(\frac{y}{x}\right)^2 + 2\left(\frac{y}{x}\right)\sec\alpha + 1 = 0$$

Equation is quadratic in $\frac{y}{x}$

So,

$a = 1$, $b = 2\sec\alpha$, $c = 1$

$$\frac{y}{x} = \frac{-2\sec\alpha \pm \sqrt{(2\sec\alpha)^2 - 4(1)(1)}}{2(1)}$$

$$\frac{y}{x} = \frac{-2\sec\alpha \pm \sqrt{\sec^2\alpha - 1}}{2}$$

$$\frac{y}{x} = -\sec\alpha \pm \tan\alpha$$

$$\frac{y}{x} = -\frac{1}{\cos\alpha} \pm \frac{\sin\alpha}{\cos\alpha}$$

$$\frac{y}{x} = \frac{-1 \pm \sin\alpha}{\cos\alpha}$$

$$\boxed{x(1 - \sin\alpha) + y\cos\alpha = 0}$$

$$\boxed{x(1 + \sin\alpha) + y\cos\alpha = 0}$$

Comparing the given equation with $ax^2 + 2hxy + by^2 = 0$

$a = 1,\quad 2h = 2\sec\alpha\ ,\quad b = 1$

$h = \sec\alpha$

Let θ be an angle between the lines.

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a+b}$$

$$\tan\theta = \frac{2\sqrt{\sec^2\alpha - (1)(1)}}{1+1}$$

$$\tan\theta = \frac{2\sqrt{\sec^2\alpha - 1}}{2}$$

$$\tan\theta = \sqrt{\tan^2\alpha}$$

$$\tan\theta = \tan\alpha$$

$$\boxed{\theta = \alpha}$$ Ans.

Q.7 Find a joint equation of the lines through the origin and perpendicular to the lines $x^2 - 2xy\tan\alpha - y^2 = 0$

Solution:

$$x^2 - 2xy\tan\alpha - y^2 = 0$$

Compare it with

$$ax^2 + 2hxy + by^2 = 0$$

$a = 1, \quad 2h = -2\tan\alpha, \quad b = -1$

$h = -\tan\alpha$

Let m_1, m_2 be the slopes of given lines.

$$m_1 + m_2 = \frac{-2h}{b}$$

$$= \frac{-2(-\tan\alpha)}{-1} = -2\tan\alpha$$

$$m_1 m_2 = \frac{a}{b}$$

$$= \frac{1}{-1} = -1$$

Slopes of lines perpendicular to given lines are $\frac{-1}{m_1}$ and $\frac{-1}{m_2}$

$y = \frac{-1}{m_1}x$ and $y = -\frac{1}{m_2}x$

$m_1 y = -x$ and $m_2 y = -x$

$x + m_1 y = 0$ and $x + m_2 y = 0$

So the joint eq. is

$$(x + m_1 y)(x + m_2 y) = 0$$

$$x^2 + m_2 xy + m_1 xy + m_1 m_2 y^2 = 0$$

$x^2 + (m_1 + m_2)\,xy + m_1m_2y^2 = 0$

$x^2 + (-2\tan\alpha)\,xy + (-1)y^2 = 0$

$\boxed{x^2 - 2\tan\alpha\ xy - y^2 = 0}$ Ans

Q.8 Find a joint equation of the lines through the origin and perpendicular to the lines $ax^2 + 2hxy + by^2 = 0$

Solution:

$$ax^2 + 2hxy + by^2 = 0$$

Let m_1, m_2 be the slopes of given lines.

$$m_1 + m_2 = \frac{-2h}{b}$$

$$m_1 m_2 = \frac{a}{b}$$

Slopes of lines perpendicular to given lines are $\frac{-1}{m_1}$ and $\frac{-1}{m_2}$ then their equations are

$y = \frac{-1}{m_1}x$ and $y = \frac{-1}{m_2}x$

$m_1y = -x$ and $m_2y = -x$

$x + m_1y = 0$ and $x + m_2y = 0$

So the joint equation is

$(x + m_1y)(x + m_2y) = 0$

$x^2 + m_2yx + m_1yx + m_1m_2y^2 = 0$

$x^2 + (m_1 + m_2)\,xy + m_1m_2y^2 = 0$

$x^2 - \frac{2h}{b}xy + \frac{a}{b}y^2 = 0$

$\frac{bx^2 - 2hxy + ay^2}{b} = 0$

$\boxed{bx^2 - 2hxy + ay^2 = 0}$ Ans.

Q.9 Find the area of the region bounded by

$10x^2 - xy - 21y^2 = 0$ and $x + y + 1 = 0$

Solution:

$10x^2 - xy - 21y^2 = 0$ ..... (1)

$x + y + 1 = 0$ ..... (2)

From equation (1)

$10x^2 - xy - 21y^2 = 0$

$10x^2 - 15xy + 14xy - 21y^2 = 0$

$$5x(2x-3y)+7y(2x-3y) = 0$$
$$(2x-3y)(5x+7y) = 0$$
$$2x-3y = 0 \quad ..... (3) \qquad \text{or} \qquad 5x+7y = 0 \quad ...... (4)$$

Let A (x, y) be the point of intersection of equation (3) and equation (4) since equation (3) and equation (4) passing through origin so the point of intersection of (3) and (4) is A (0, 0).

Let B (x, y) be the point of intersection of equation (2) and (3).

Equation (2) × 3 + Equation (3), we get

$$\begin{array}{rl} 3x+3y+3 &= 0 \\ 2x-3y &= 0 \\ \hline 5x+3 &= 0 \end{array}$$
$$5x = -3$$
$$x = \frac{-3}{5}$$

Put $x = \frac{-3}{5}$ in eq. (3)

$$2\left(\frac{-3}{5}\right)-3y = 0$$
$$\frac{-6}{5} = 3y$$
$$y = \frac{-6}{15} = \frac{-2}{5}$$

$\therefore$ $B(x,y) = B\left(\frac{-3}{5}, \frac{-2}{5}\right)$

Let C (x, y) be the point of intersection of equation (2) and equation (4)

Equation (2) × 5 – Equation (4), we get

$$\begin{array}{rl} 5x+5y+5 &= 0 \\ -5x \pm 7y &= 0 \\ \hline -2y+5 &= 0 \end{array}$$
$$-2y = -5$$
$$y = \frac{5}{2}$$

Put $y = \frac{5}{2}$ in equation (4)

$$5x+7\left(\frac{5}{2}\right) = 0$$

$$5x + \frac{35}{2} = 0$$

$$5x = \frac{-35}{2}$$

$$x = \frac{-35}{10} = \frac{-7}{2}$$

$$\therefore \quad C(x, y) = C\left(\frac{-7}{2}, \frac{5}{2}\right)$$

Area of region ABC is

$$= \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ \frac{-3}{5} & \frac{-2}{5} & 1 \\ \frac{-7}{2} & \frac{5}{2} & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} \frac{-3}{5} & \frac{-2}{5} \\ \frac{-7}{2} & \frac{5}{2} \end{vmatrix}$$

$$= \frac{1}{2}\left(\frac{-15}{10} - \frac{14}{10}\right) = \frac{1}{2}\left(\frac{-15-14}{10}\right)$$

$$= \frac{1}{2}\left(\frac{-29}{10}\right) = \frac{29}{20}$$ Sq. unit (Neglecting – ve sign) Ans